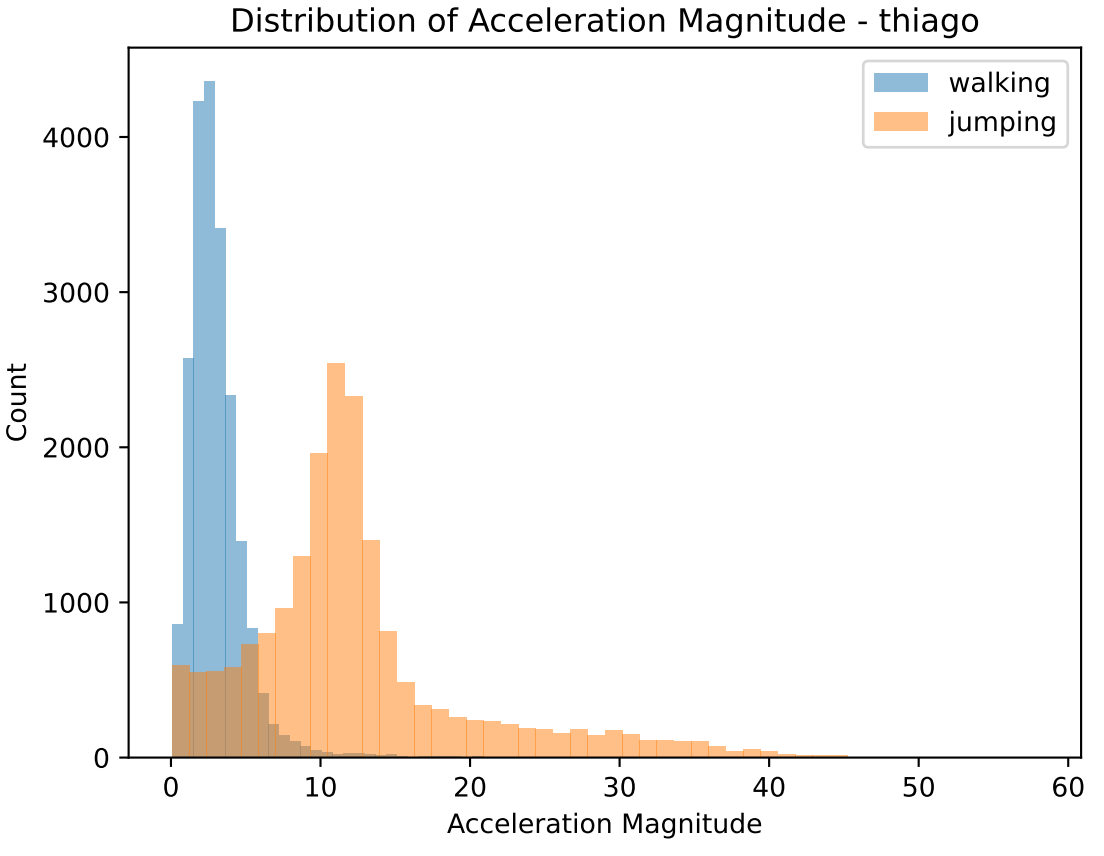
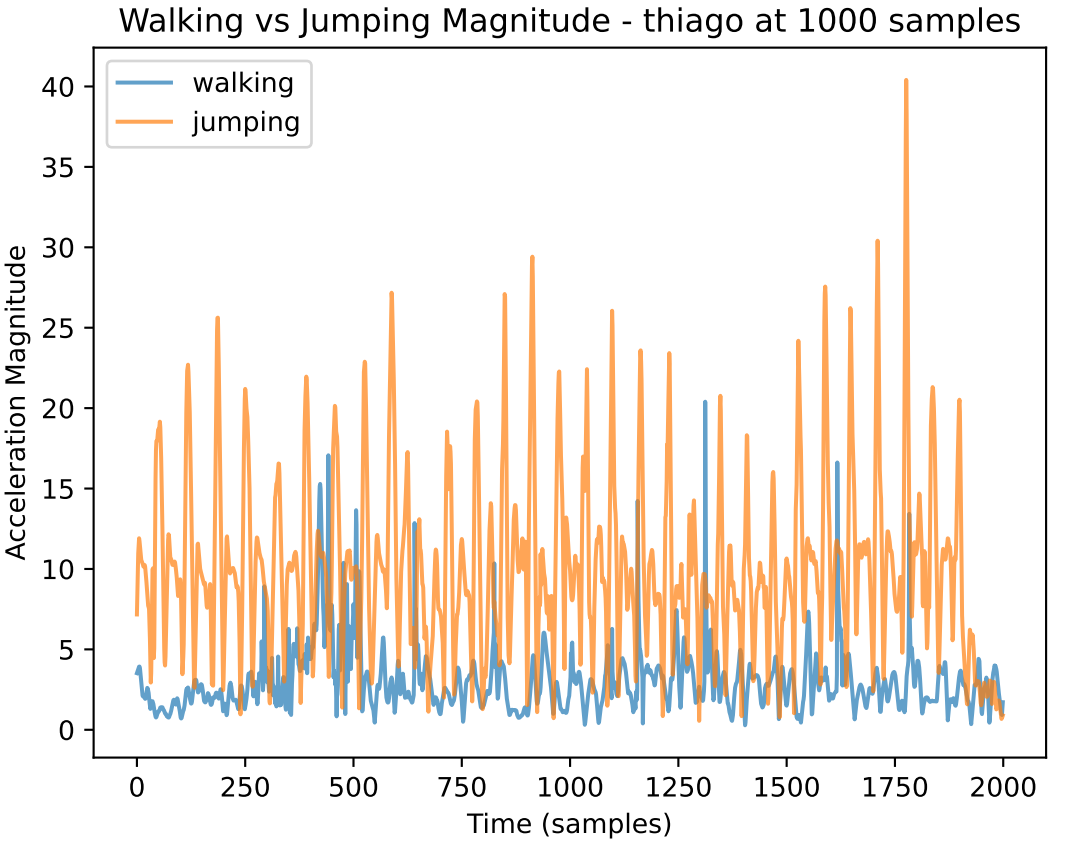
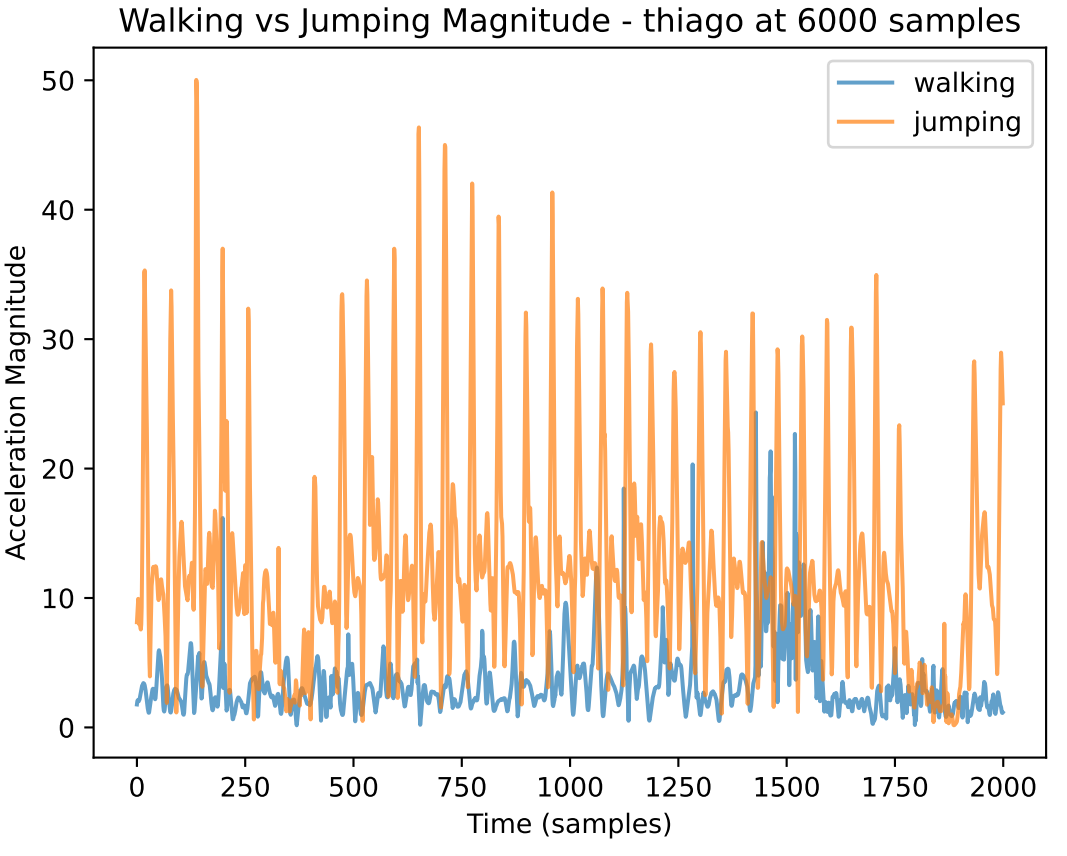
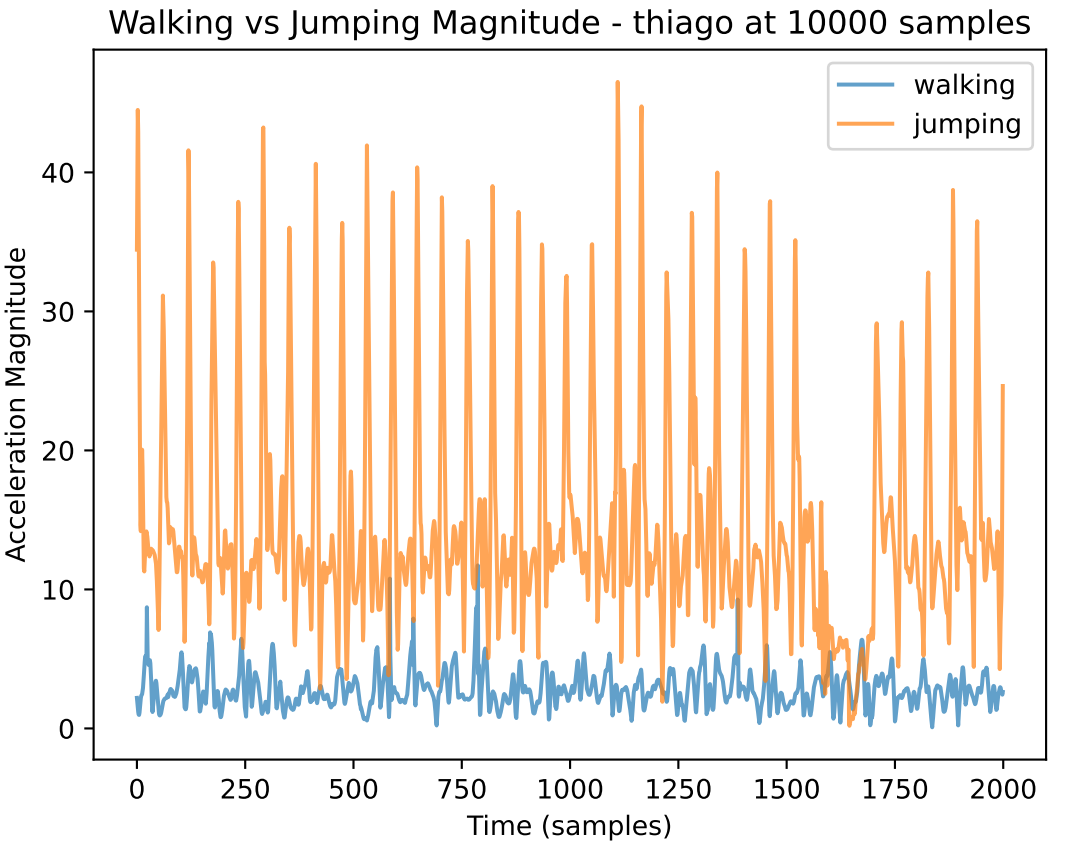
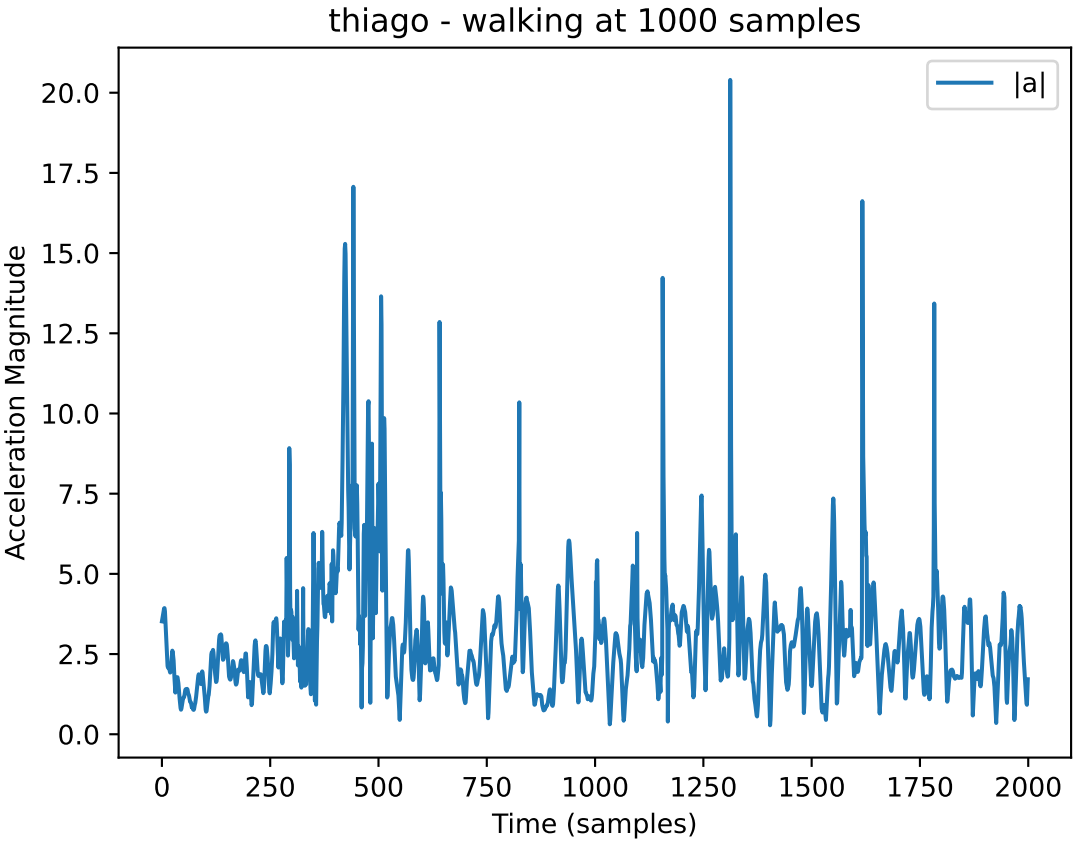

Distribution of Acceleration Magnitude - thiago
walking
jumping
Count
0
1000
2000
3000
4000
0
10
20
30
40
50
60
Acceleration Magnitude

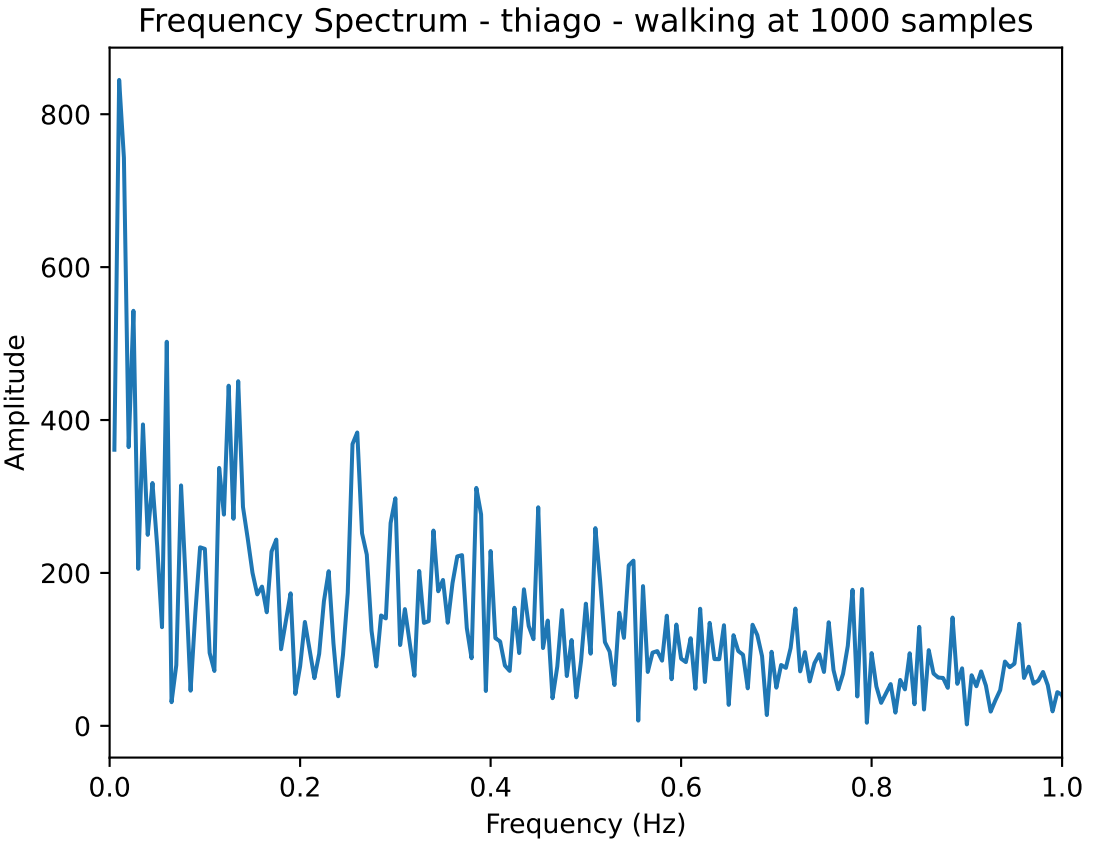

Walking vs Jumping Magnitude - thiago at 1000 samples
walking
jumping
Acceleration Magnitude
Time (samples)
0
5
10
15
20
25
30
35
40
0
250
500
750
1000
1250
1500
1750
2000

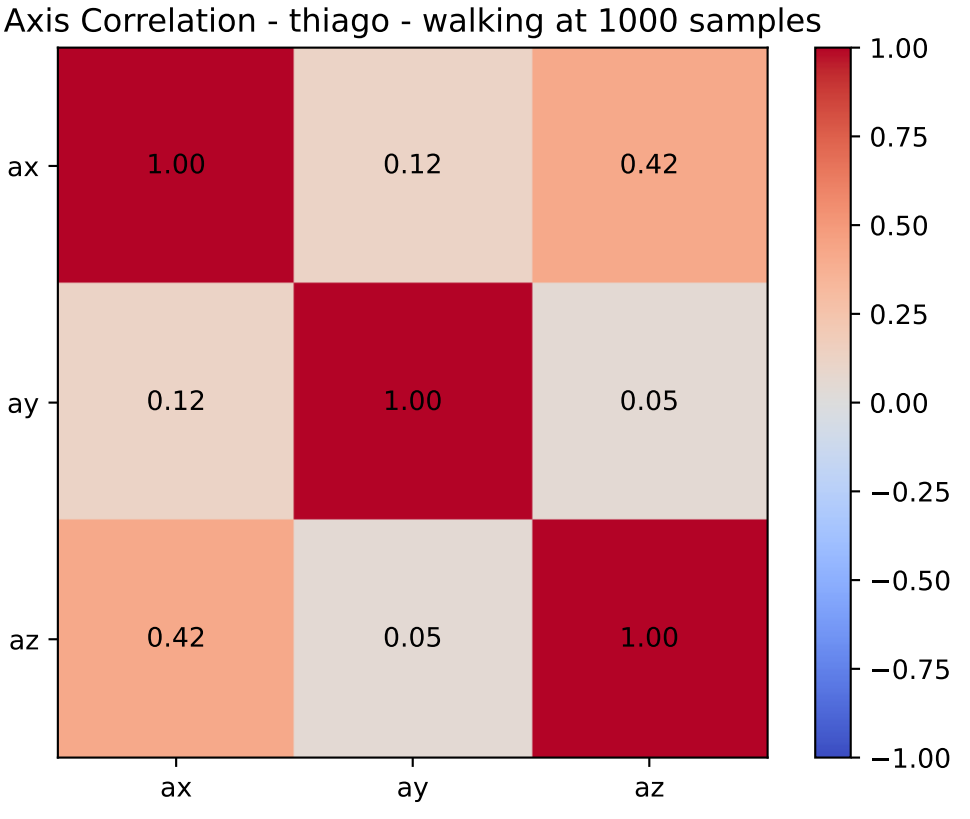

Walking vs Jumping Magnitude - thiago at 6000 samples
walking
jumping
Acceleration Magnitude
Time (samples)
0
10
20
30
40
50
0
250
500
750
1000
1250
1500
1750
2000

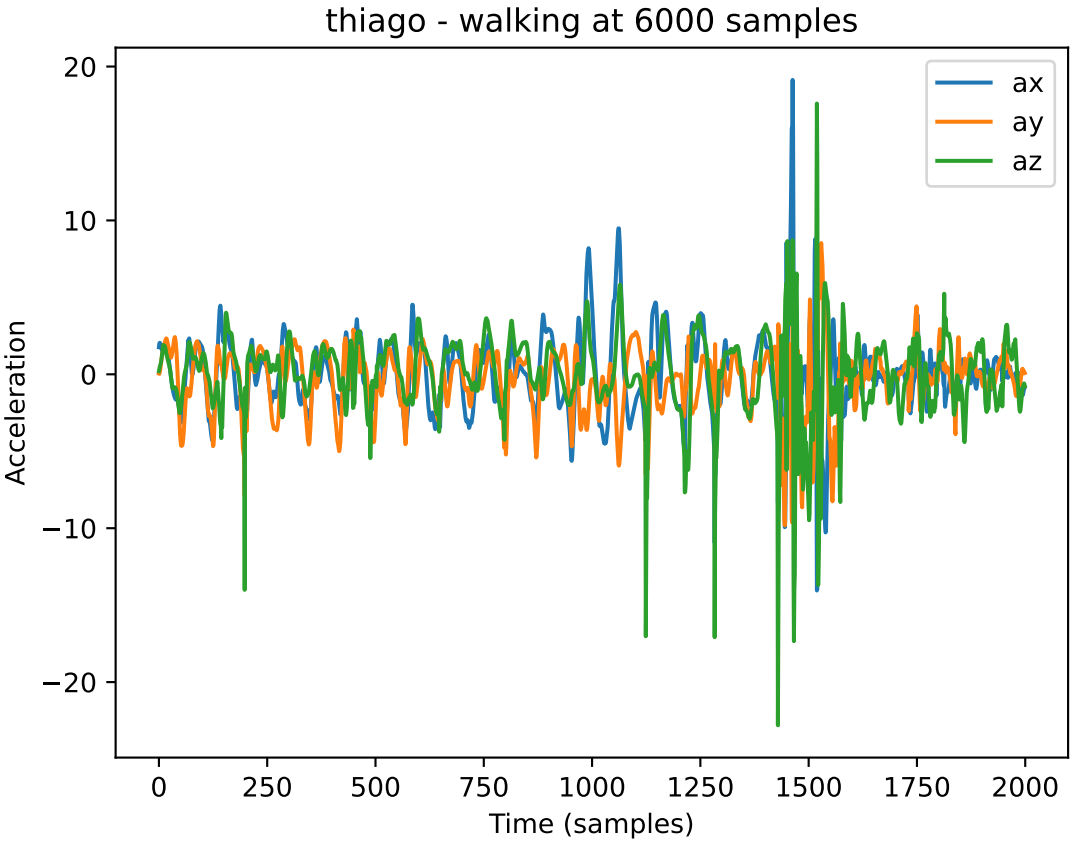

Walking vs Jumping Magnitude - thiago at 10000 samples
walking
jumping
Acceleration Magnitude
0
10
20
30
40
0
250
500
750
1000
1250
1500
1750
2000
Time (samples)

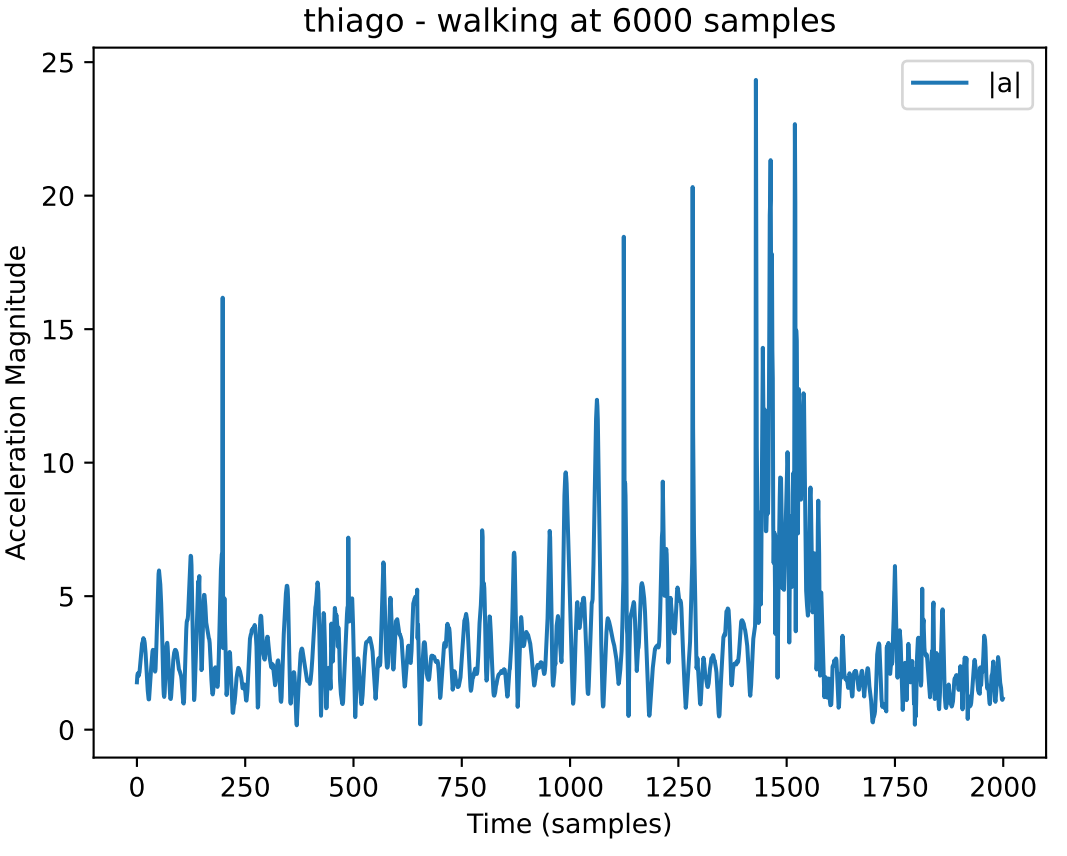
thiago - walking at 1000 samples
ax
ay
az
Acceleration
Time (samples)
15
10
5
0
−5
−10
−15
0
250
500
750
1000
1250
1500
1750
2000

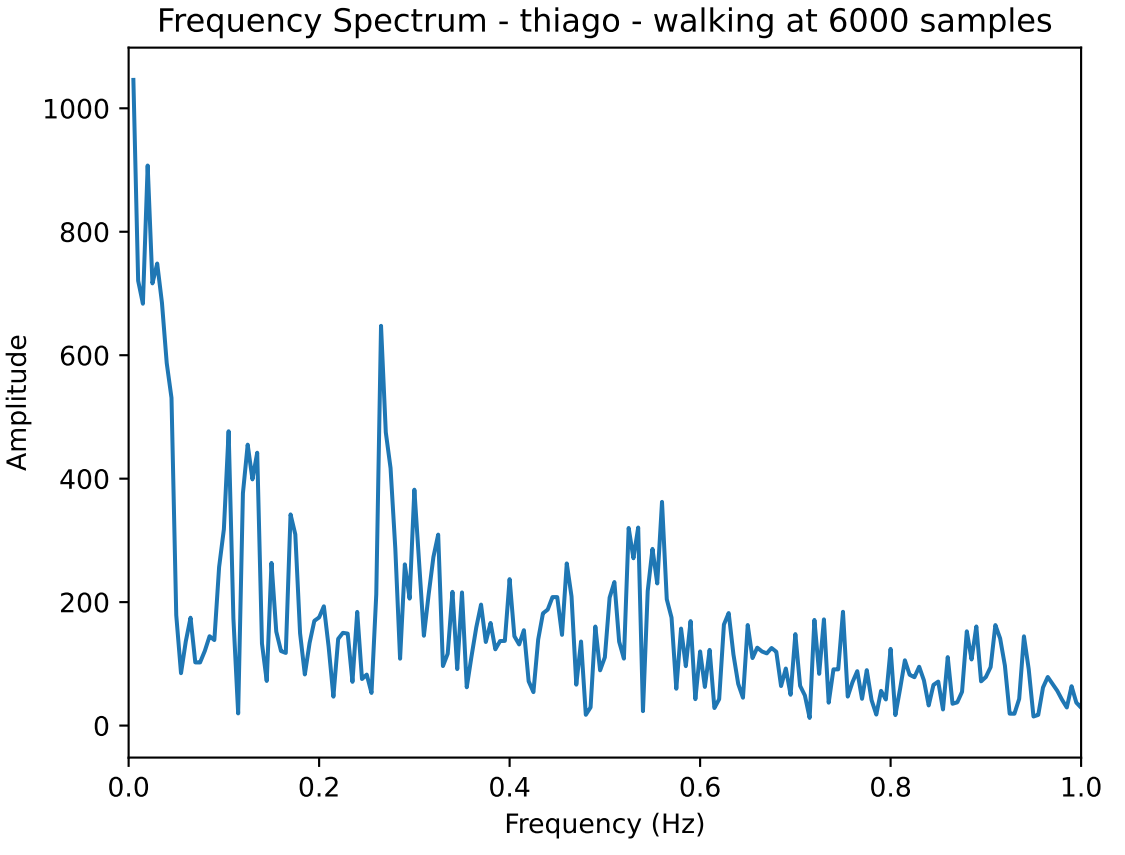

thiago - walking at 1000 samples
|a|
Acceleration Magnitude
Time (samples)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
0
250
500
750
1000
1250
1500
1750
2000

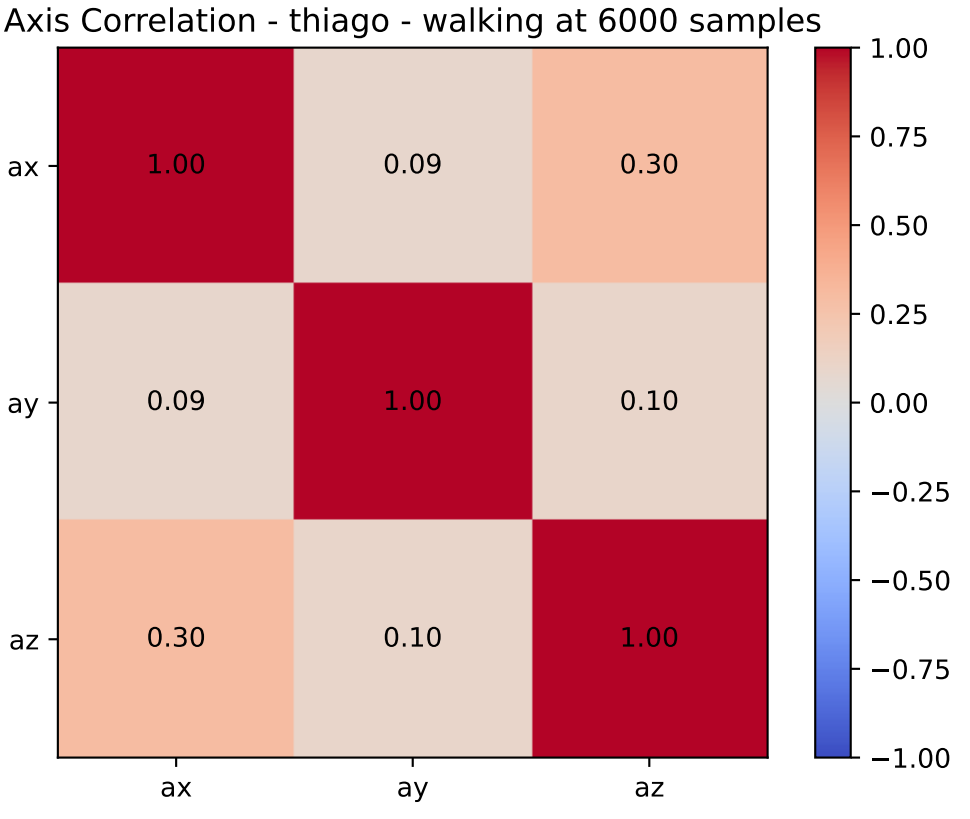

Frequency Spectrum - thiago - walking at 1000 samples
Amplitude
800
600
400
200
0
0.0
0.2
0.4
0.6
0.8
1.0
Frequency (Hz)

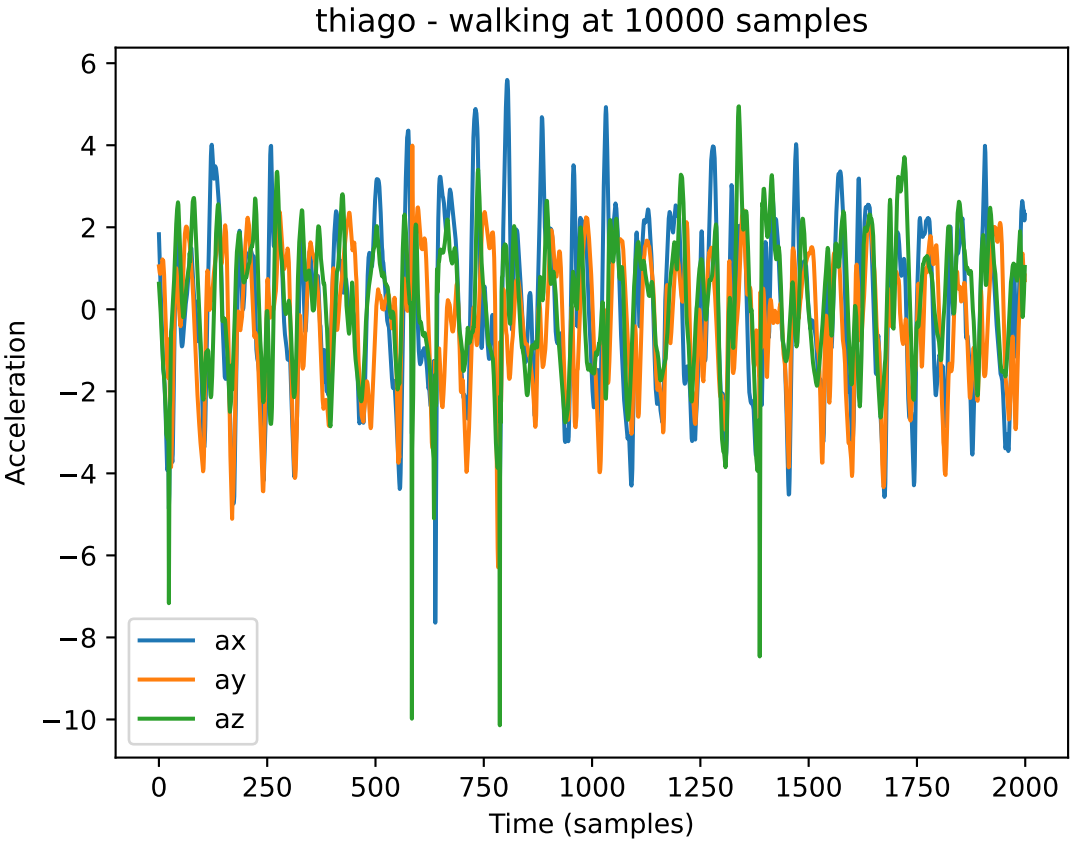

Axis Correlation - thiago - walking at 1000 samples
ax
ay
az
1.00
0.12
0.42
0.12
1.00
0.05
0.42
0.05
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

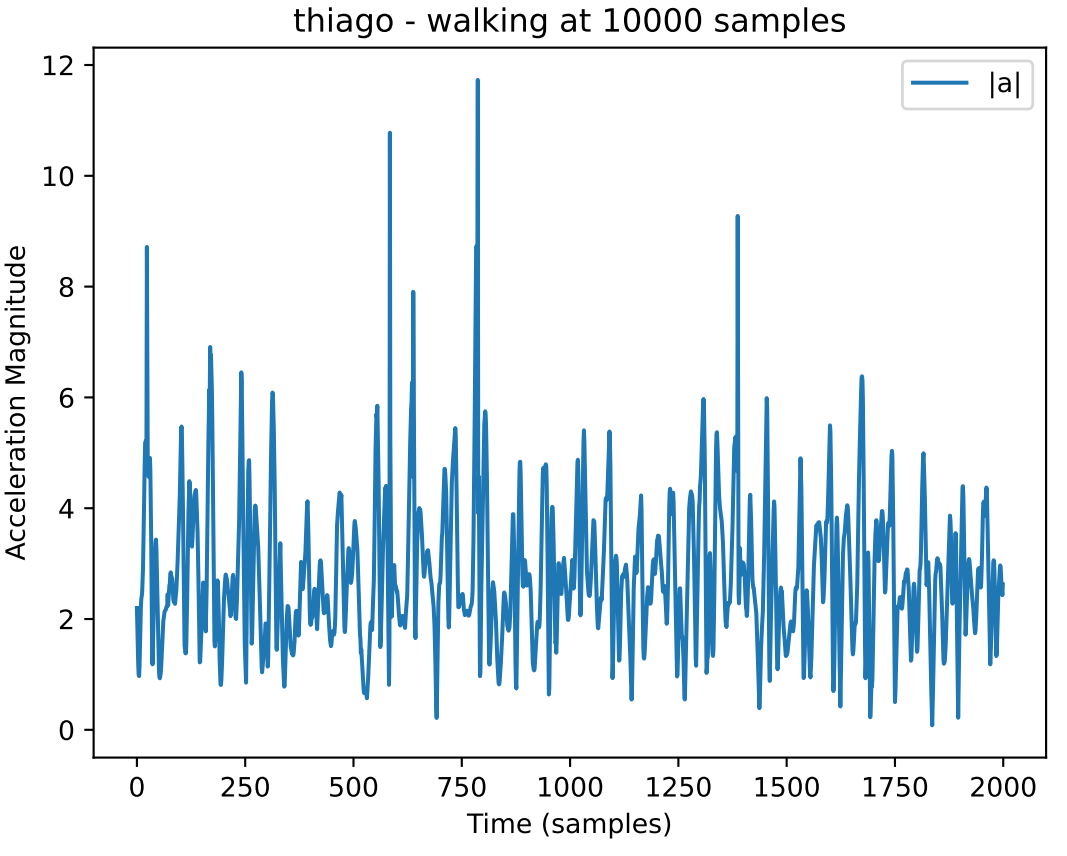

thiago - walking at 6000 samples
ax
ay
az
Acceleration
Time (samples)
20
10
0
−10
−20
0
250
500
750
1000
1250
1500
1750
2000

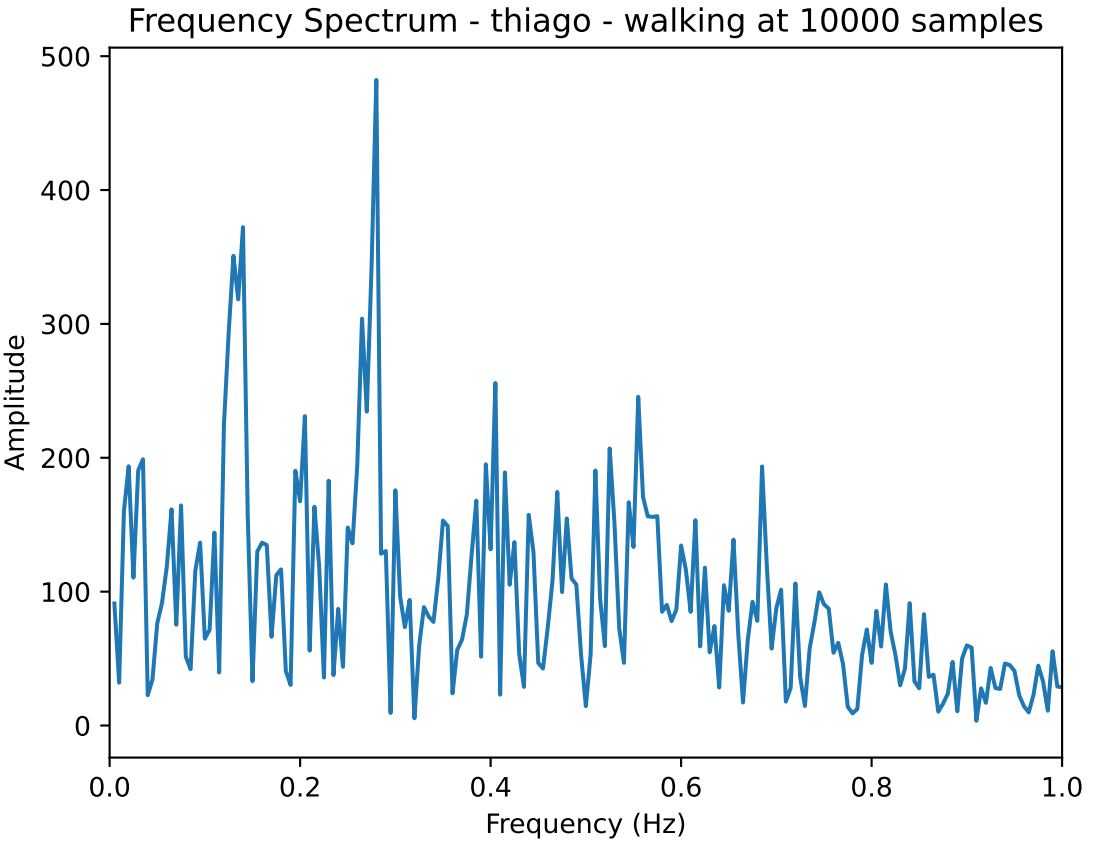

thiago - walking at 6000 samples
|a|
Acceleration Magnitude
Time (samples)
0
5
10
15
20
25
0
250
500
750
1000
1250
1500
1750
2000

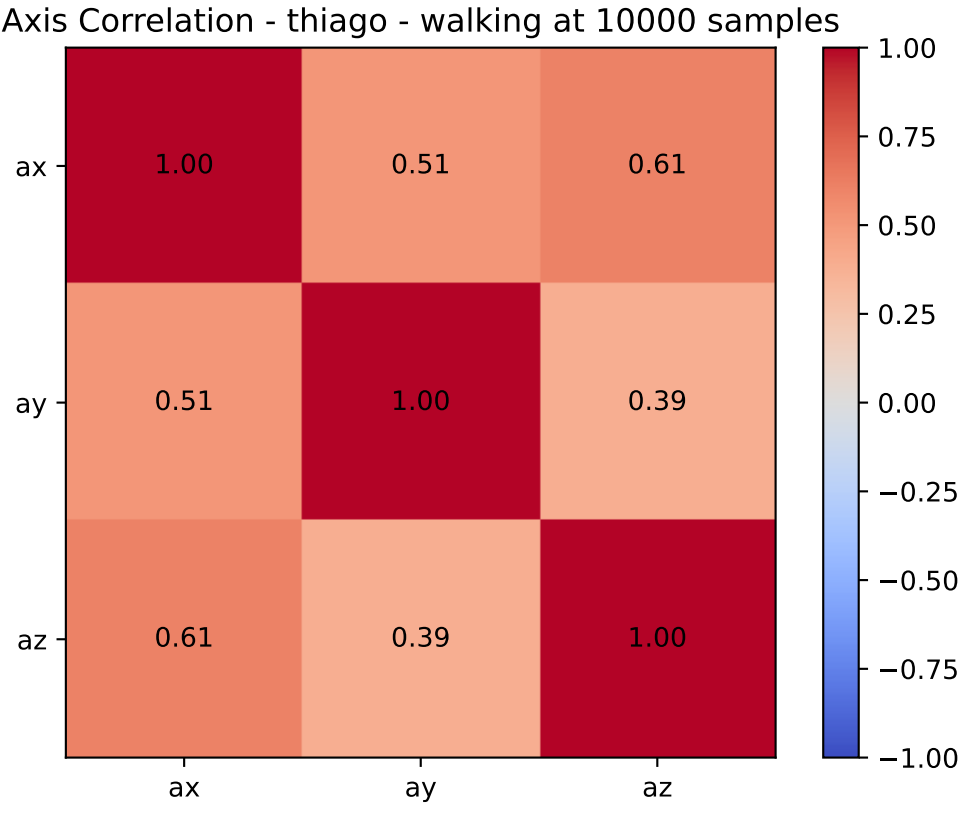

Frequency Spectrum - thiago - walking at 6000 samples
Amplitude
1000
800
600
400
200
0
0.0
0.2
0.4
0.6
0.8
1.0
Frequency (Hz)

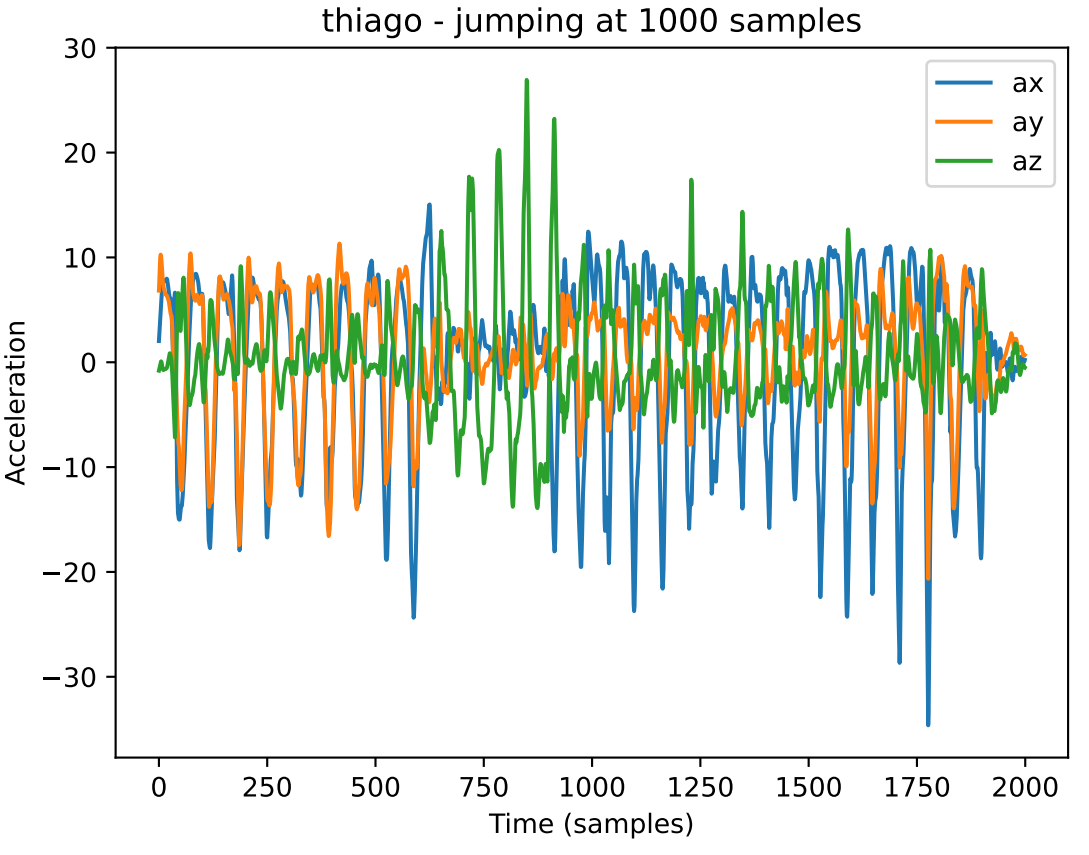

Axis Correlation - thiago - walking at 6000 samples
ax
ay
az
1.00
0.09
0.30
0.09
1.00
0.10
0.30
0.10
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

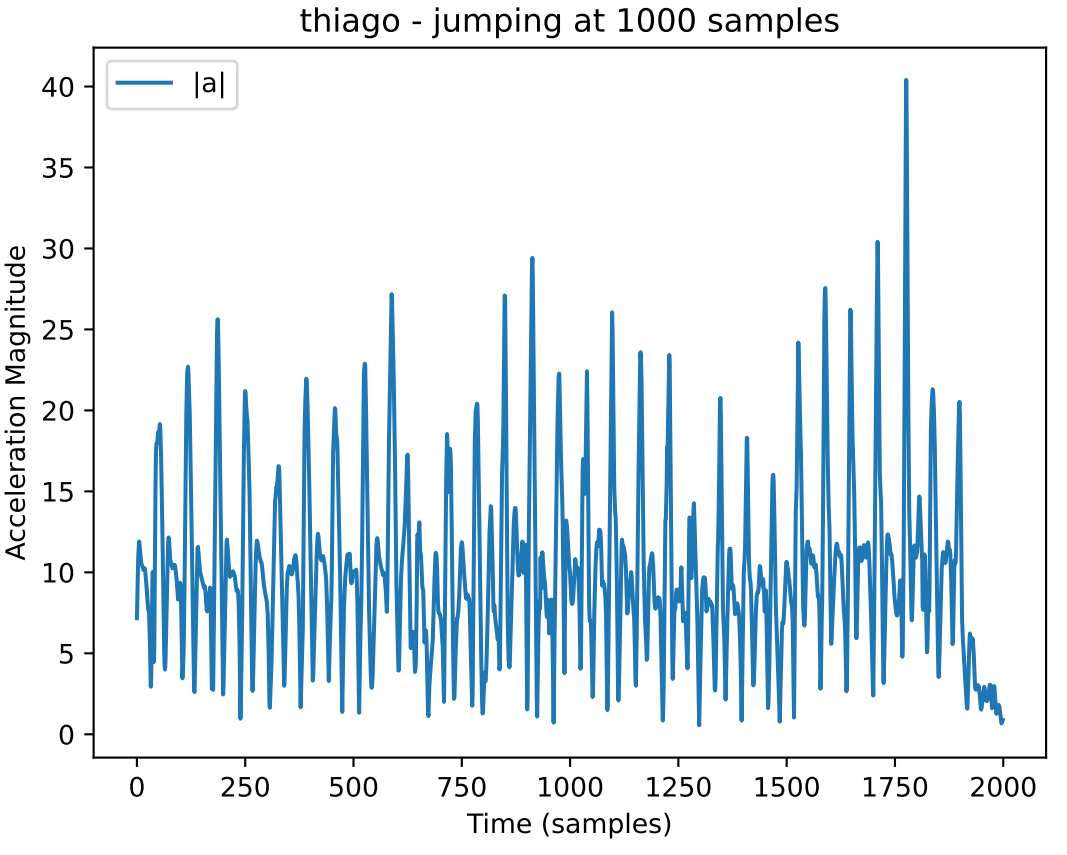

thiago - walking at 10000 samples
Acceleration
Time (samples)
ax
ay
az
6
4
2
0
−2
−4
−6
−8
−10
0
250
500
750
1000
1250
1500
1750
2000

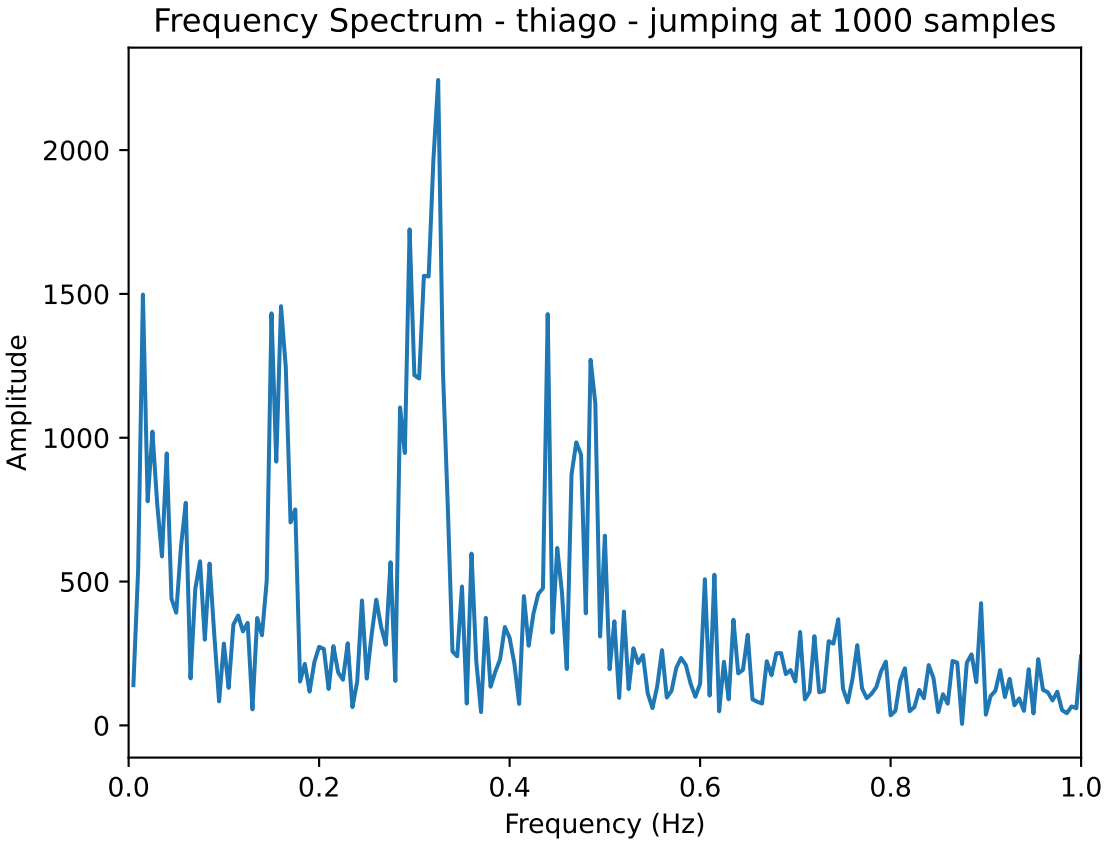

thiago - walking at 10000 samples
|a|
Acceleration Magnitude
Time (samples)
0
2
4
6
8
10
12
0
250
500
750
1000
1250
1500
1750
2000

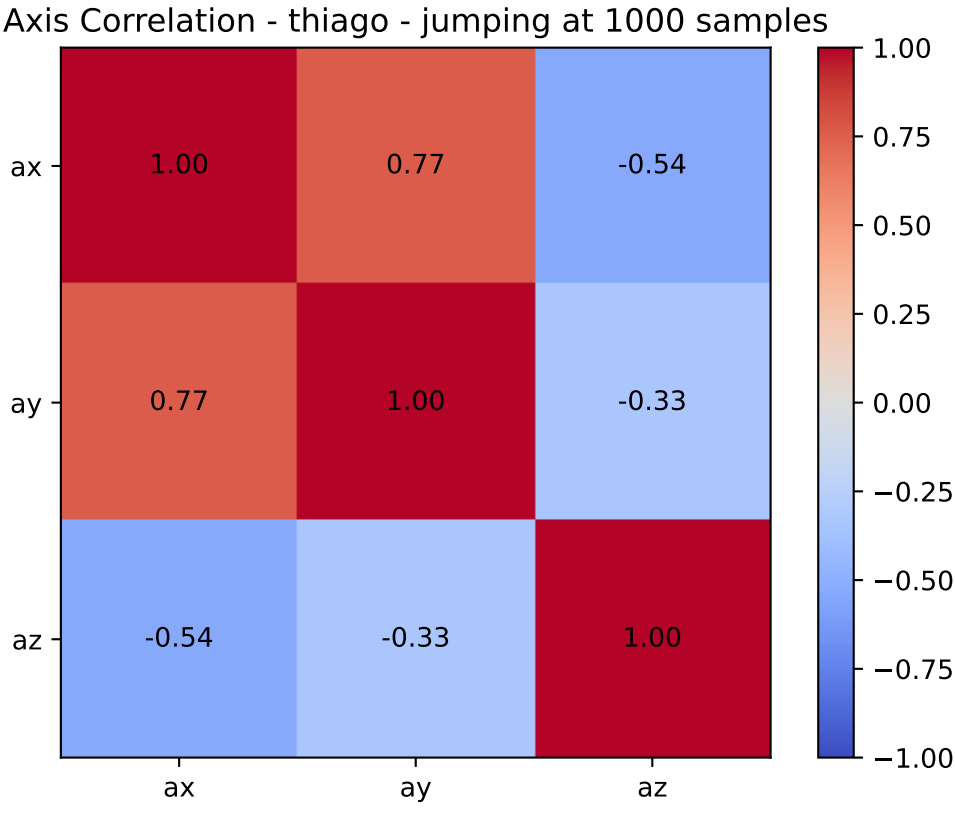

Frequency Spectrum - thiago - walking at 10000 samples
Amplitude
Frequency (Hz)
500
400
300
200
100
0
0.0
0.2
0.4
0.6
0.8
1.0

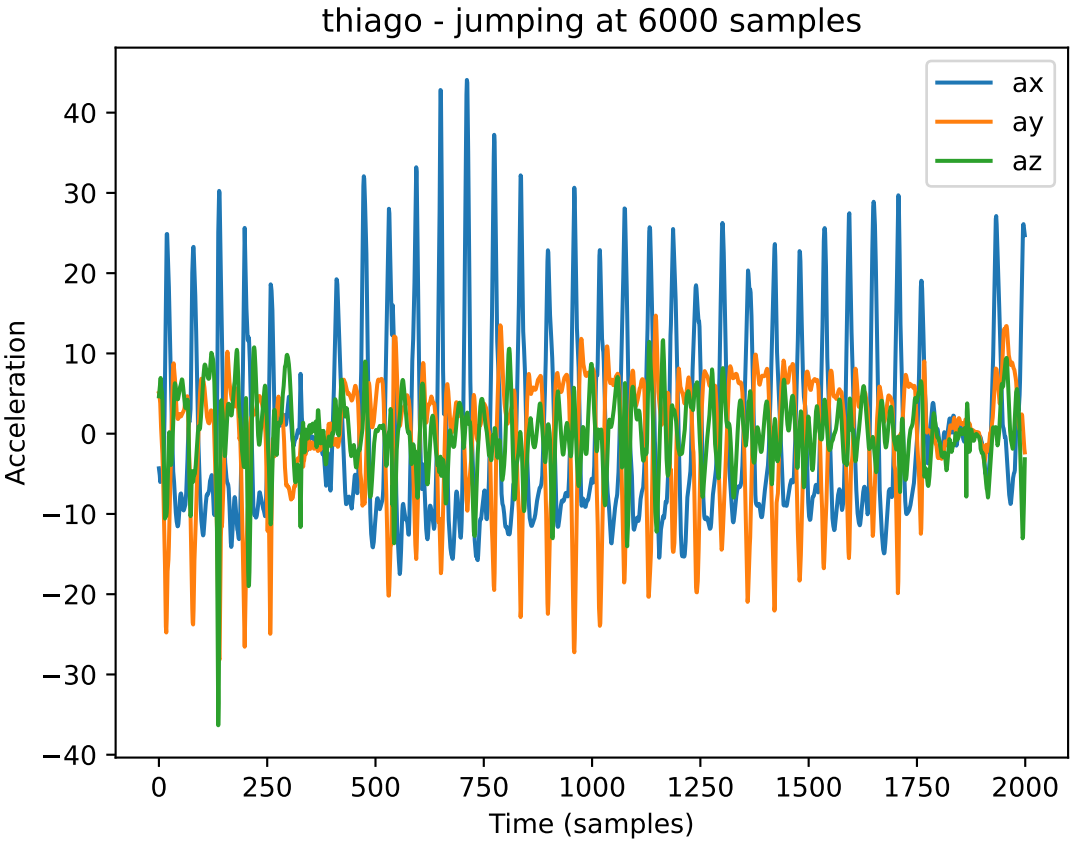

Axis Correlation - thiago - walking at 10000 samples
ax
ay
az
1.00
0.51
0.61
0.51
1.00
0.39
0.61
0.39
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

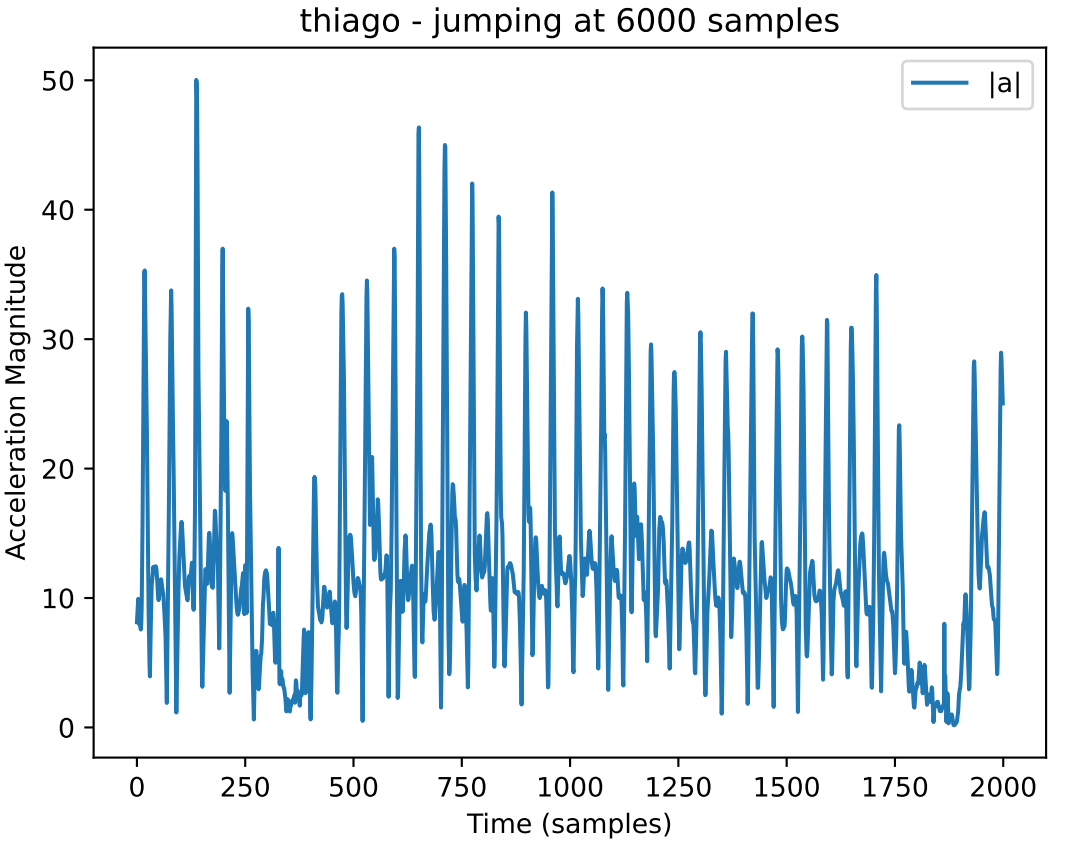

thiago - jumping at 1000 samples
ax
ay
az
Acceleration
Time (samples)
30
20
10
0
−10
−20
−30
0
250
500
750
1000
1250
1500
1750
2000

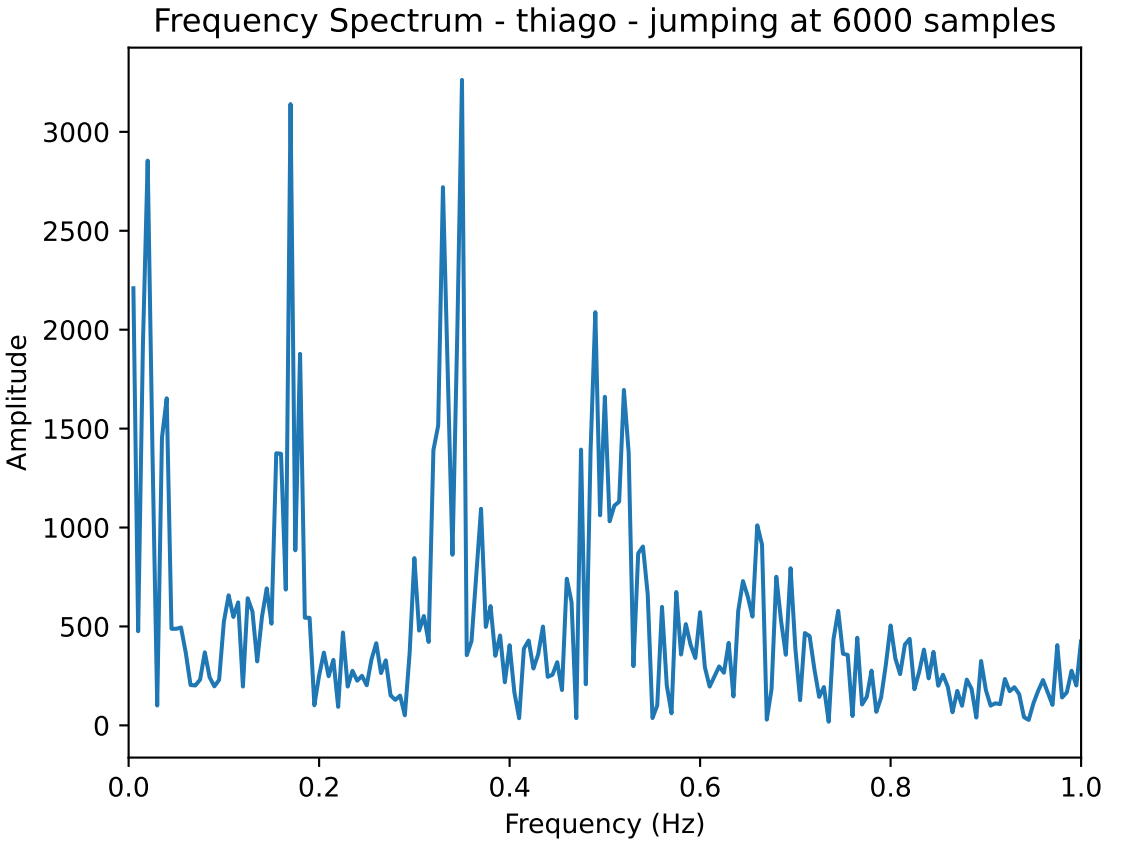

thiago - jumping at 1000 samples
|a|
Acceleration Magnitude
Time (samples)
0
5
10
15
20
25
30
35
40
0
250
500
750
1000
1250
1500
1750
2000

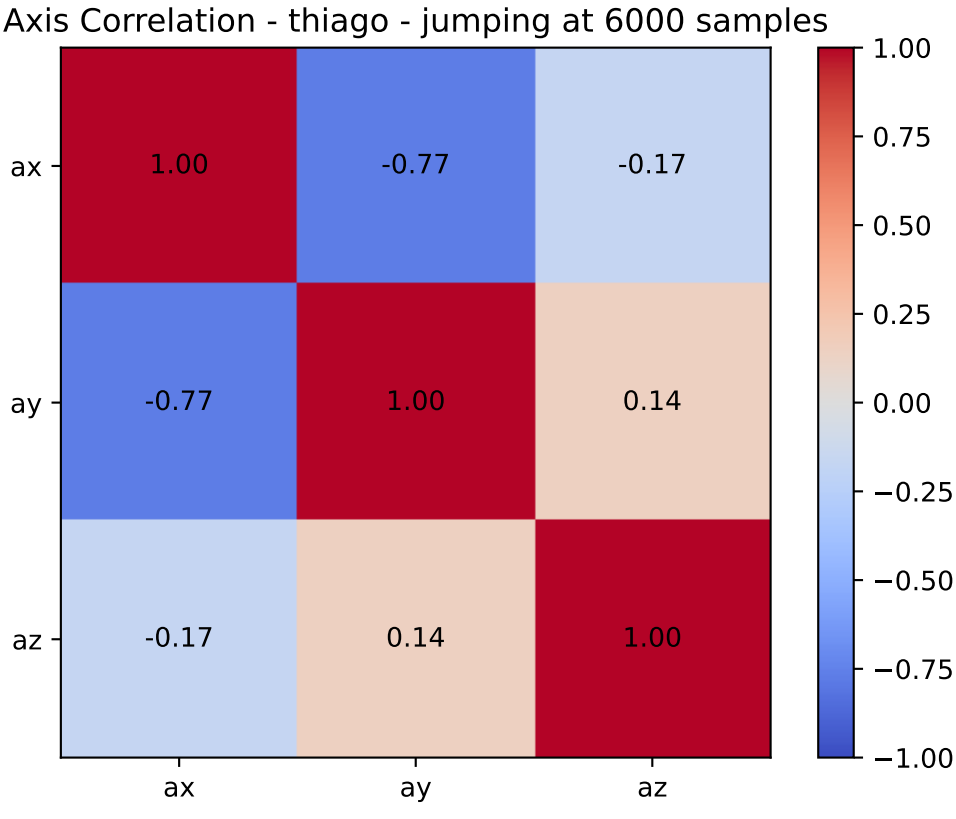

Frequency Spectrum - thiago - jumping at 1000 samples
Amplitude
2000
1500
1000
500
0
0.0
0.2
0.4
0.6
0.8
1.0
Frequency (Hz)

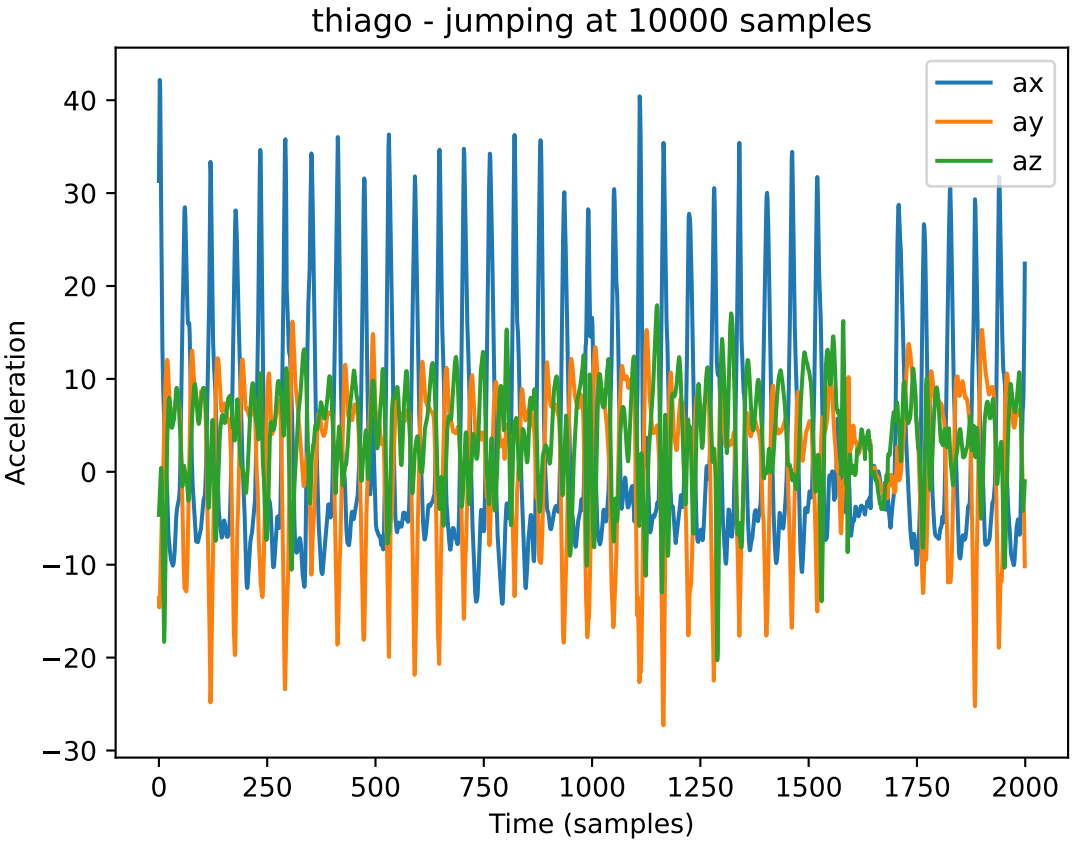

Axis Correlation - thiago - jumping at 1000 samples
ax
ay
az
1.00
0.77
-0.54
0.77
1.00
-0.33
-0.54
-0.33
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

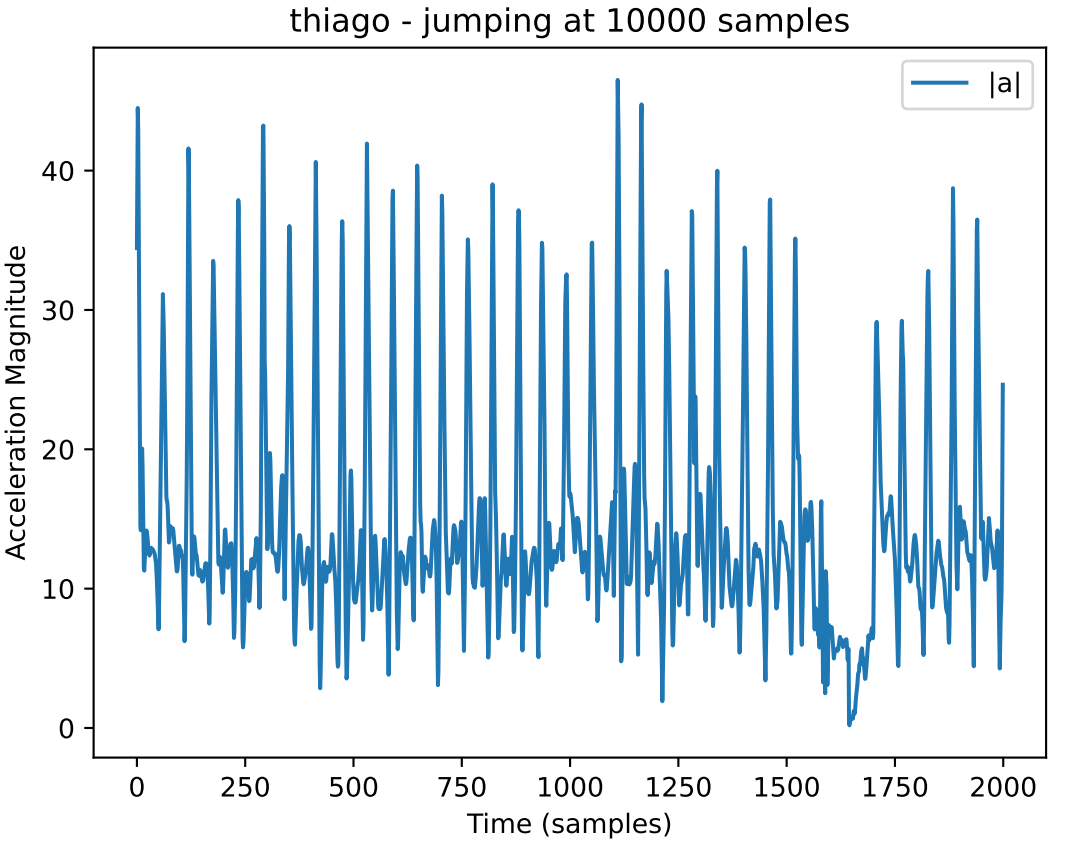

thiago - jumping at 6000 samples
ax
ay
az
Acceleration
Time (samples)
40
30
20
10
0
−10
−20
−30
−40
0
250
500
750
1000
1250
1500
1750
2000

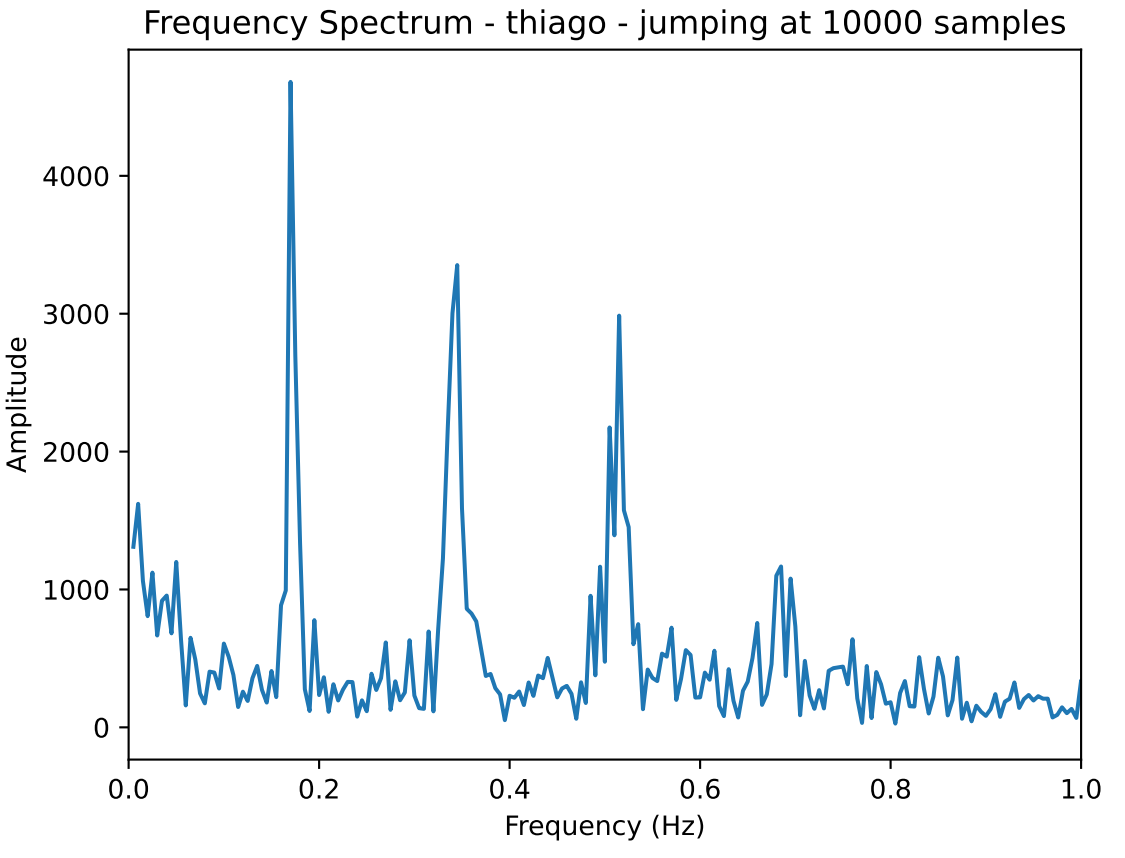

thiago - jumping at 6000 samples
|a|
Acceleration Magnitude
Time (samples)
0
10
20
30
40
50
0
250
500
750
1000
1250
1500
1750
2000

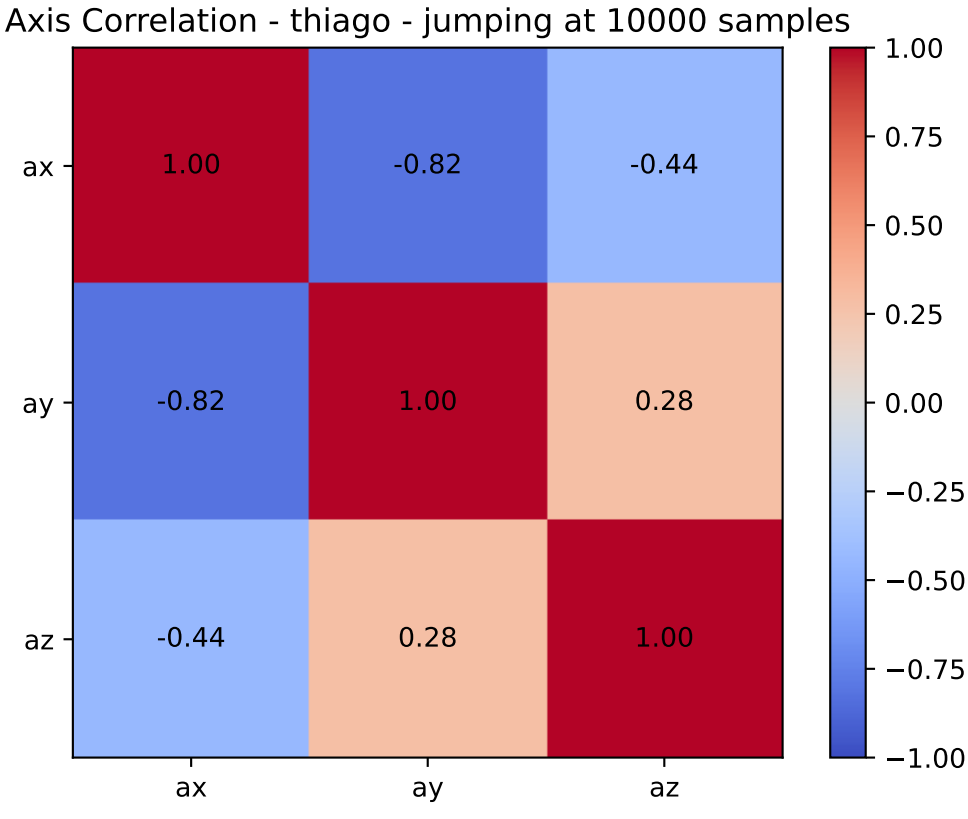

Frequency Spectrum - thiago - jumping at 6000 samples
Amplitude
3000
2500
2000
1500
1000
500
0
0.0
0.2
0.4
0.6
0.8
1.0
Frequency (Hz)

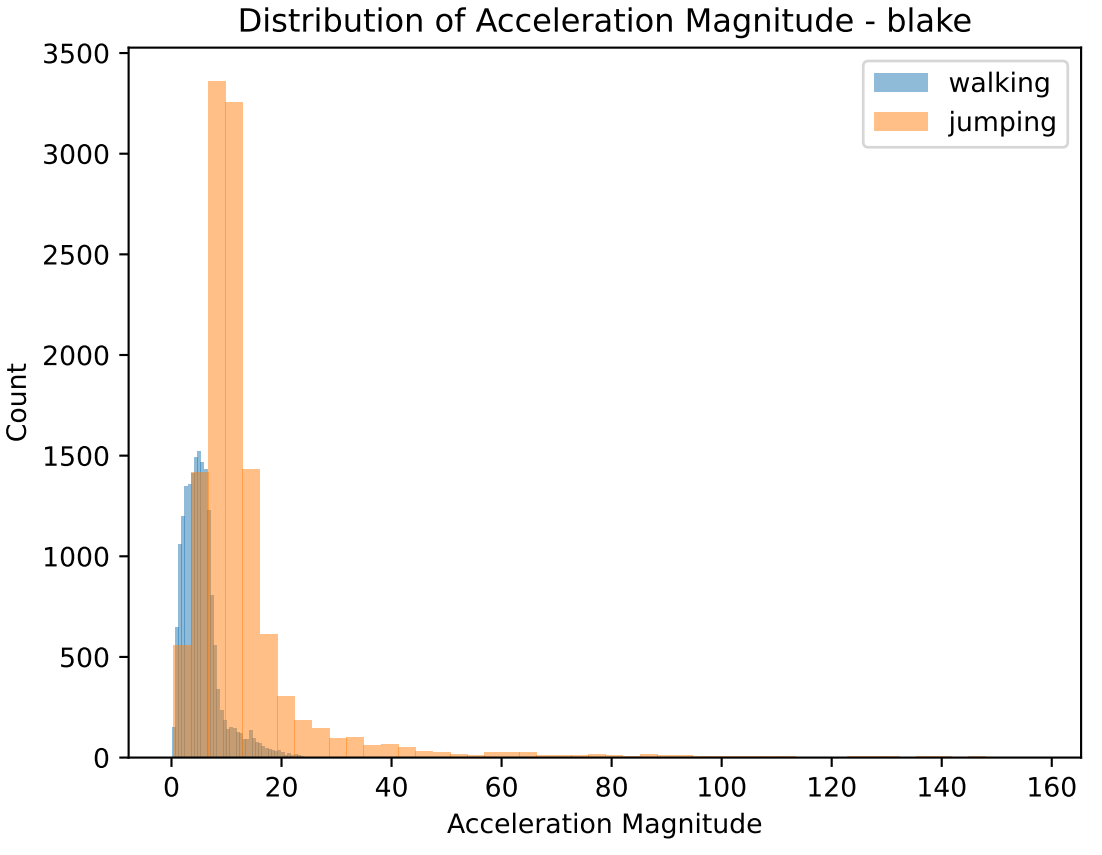

Axis Correlation - thiago - jumping at 6000 samples
ax
ay
az
1.00
-0.77
-0.17
-0.77
1.00
0.14
-0.17
0.14
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

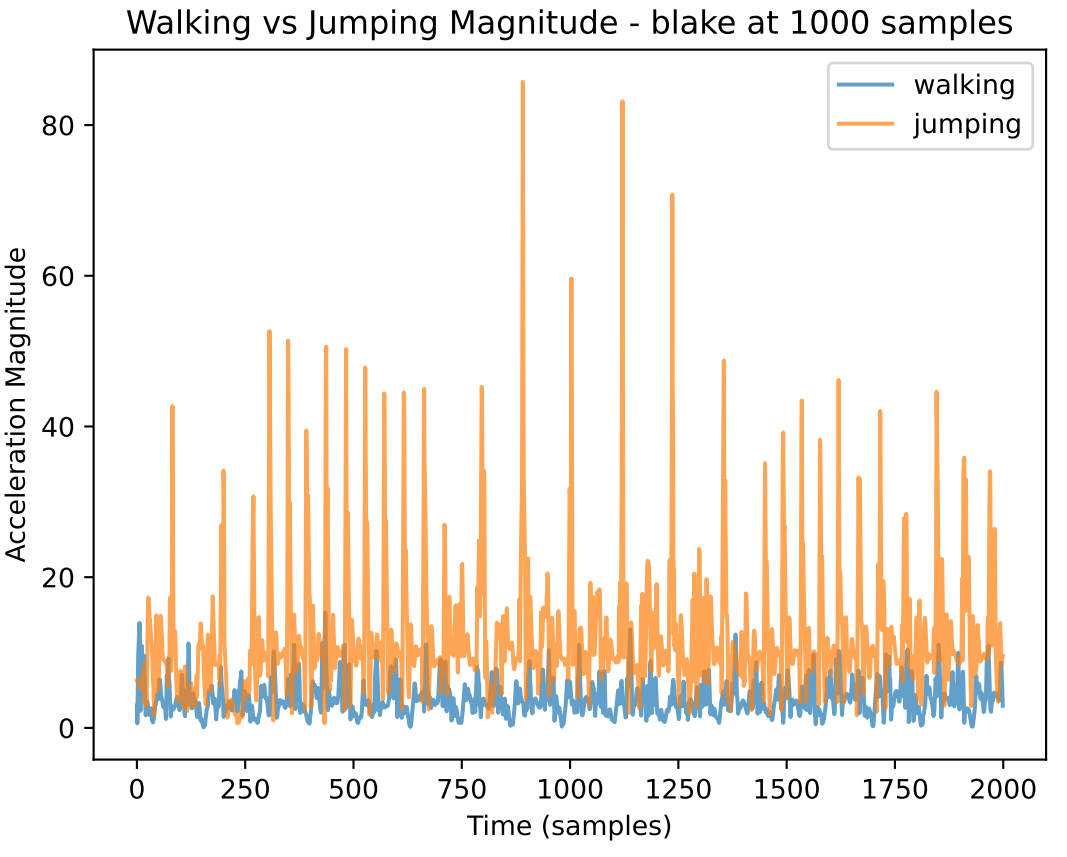

thiago - jumping at 10000 samples
ax
ay
az
Acceleration
Time (samples)
40
30
20
10
0
−10
−20
−30
0
250
500
750
1000
1250
1500
1750
2000

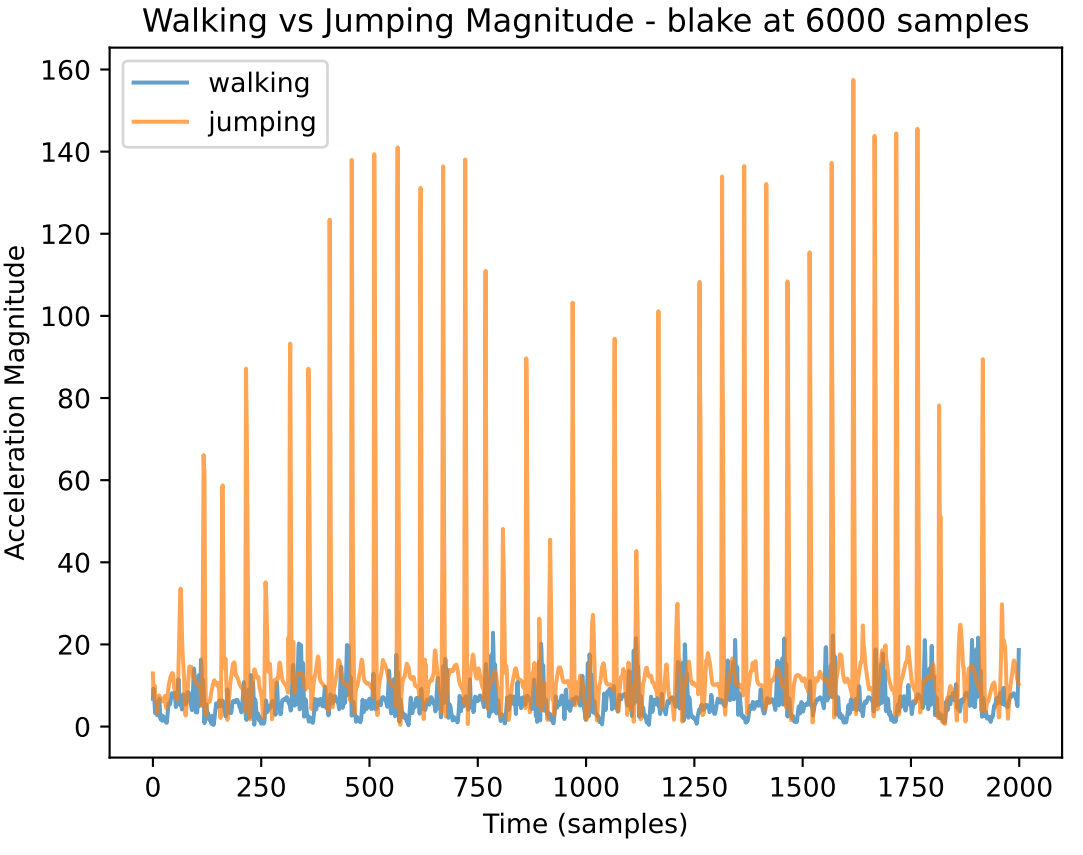

thiago - jumping at 10000 samples
|a|
Acceleration Magnitude
0
10
20
30
40
0
250
500
750
1000
1250
1500
1750
2000
Time (samples)

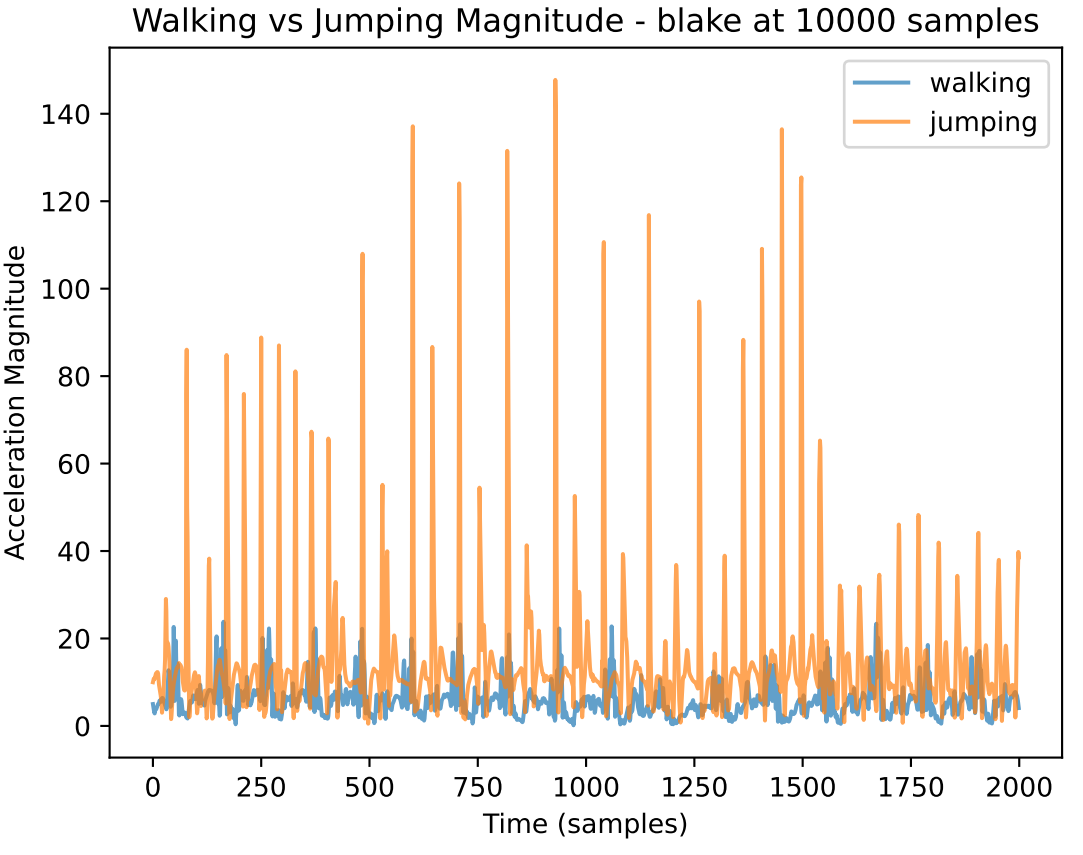

Frequency Spectrum - thiago - jumping at 10000 samples
Amplitude
4000
3000
2000
1000
0
0.0
0.2
0.4
0.6
0.8
1.0
Frequency (Hz)

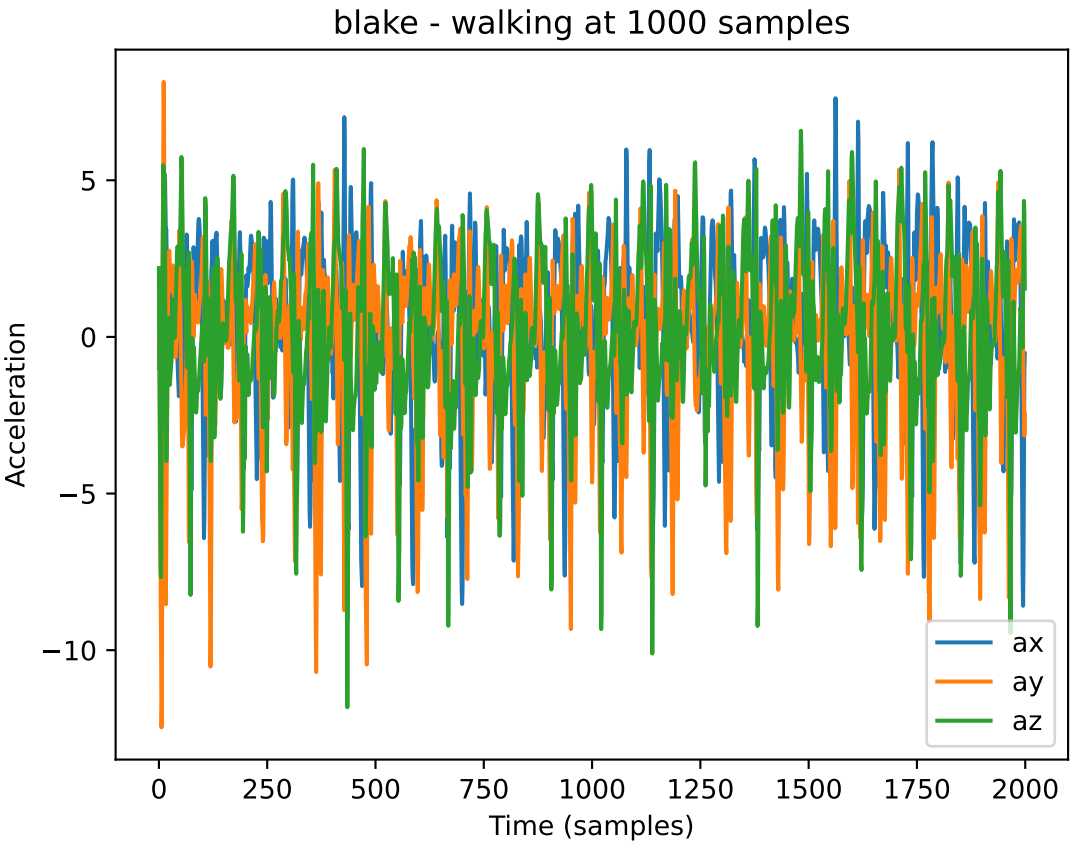

Axis Correlation - thiago - jumping at 10000 samples
ax
ay
az
1.00
-0.82
-0.44
-0.82
1.00
0.28
-0.44
0.28
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

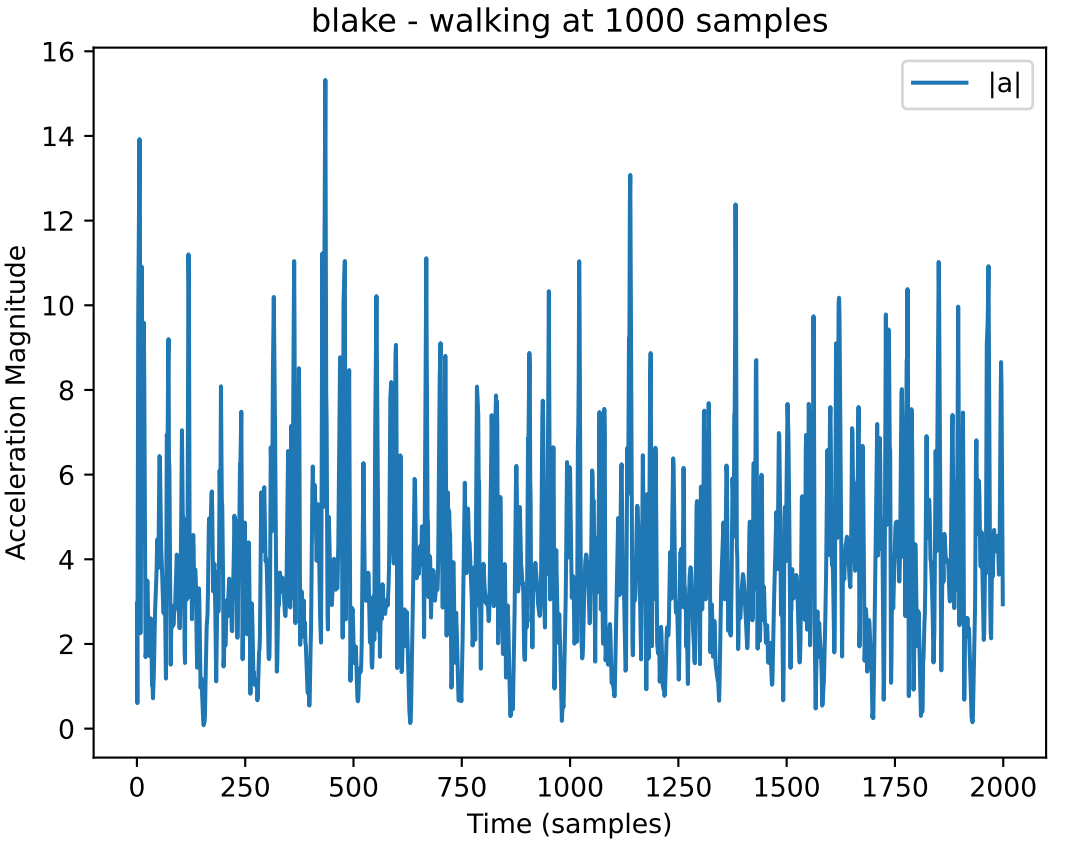

Distribution of Acceleration Magnitude - blake
walking
jumping
Count
Acceleration Magnitude
0
500
1000
1500
2000
2500
3000
3500
0
20
40
60
80
100
120
140
160

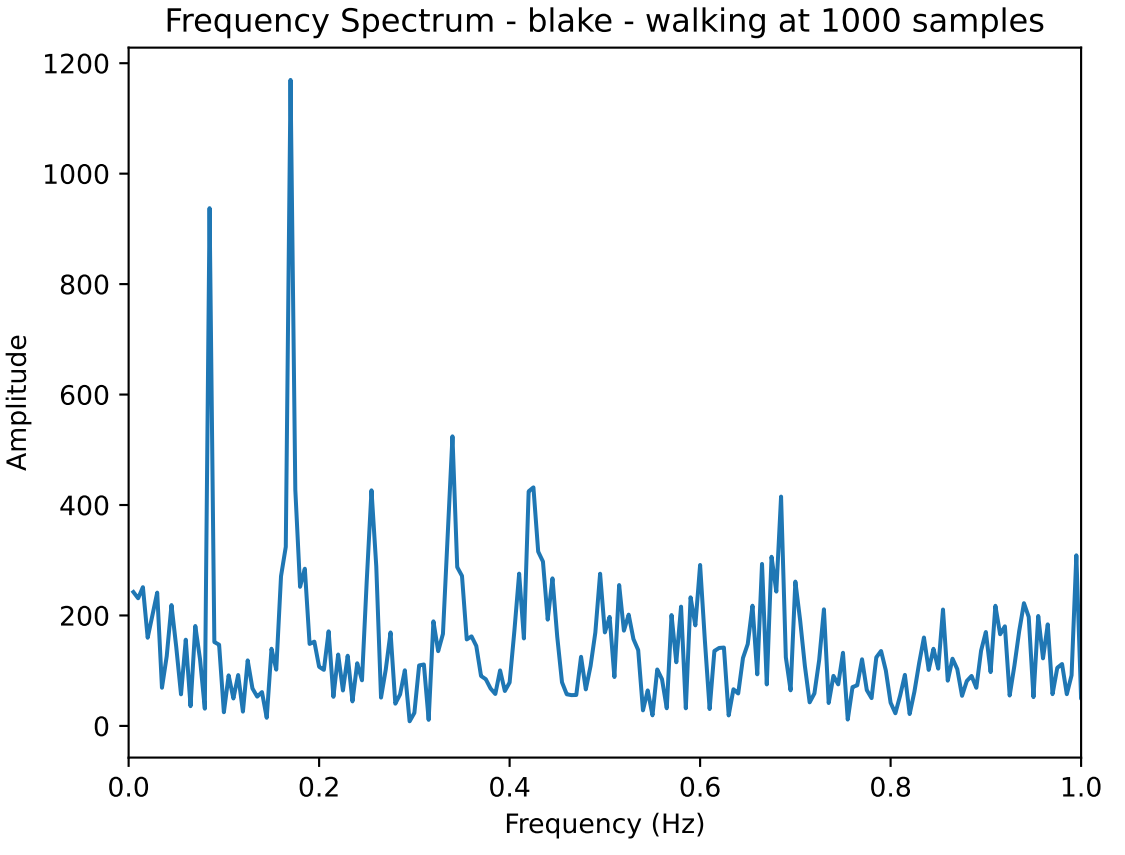

Walking vs Jumping Magnitude - blake at 1000 samples
walking
jumping
Acceleration Magnitude
0
20
40
60
80
0
250
500
750
1000
1250
1500
1750
2000
Time (samples)

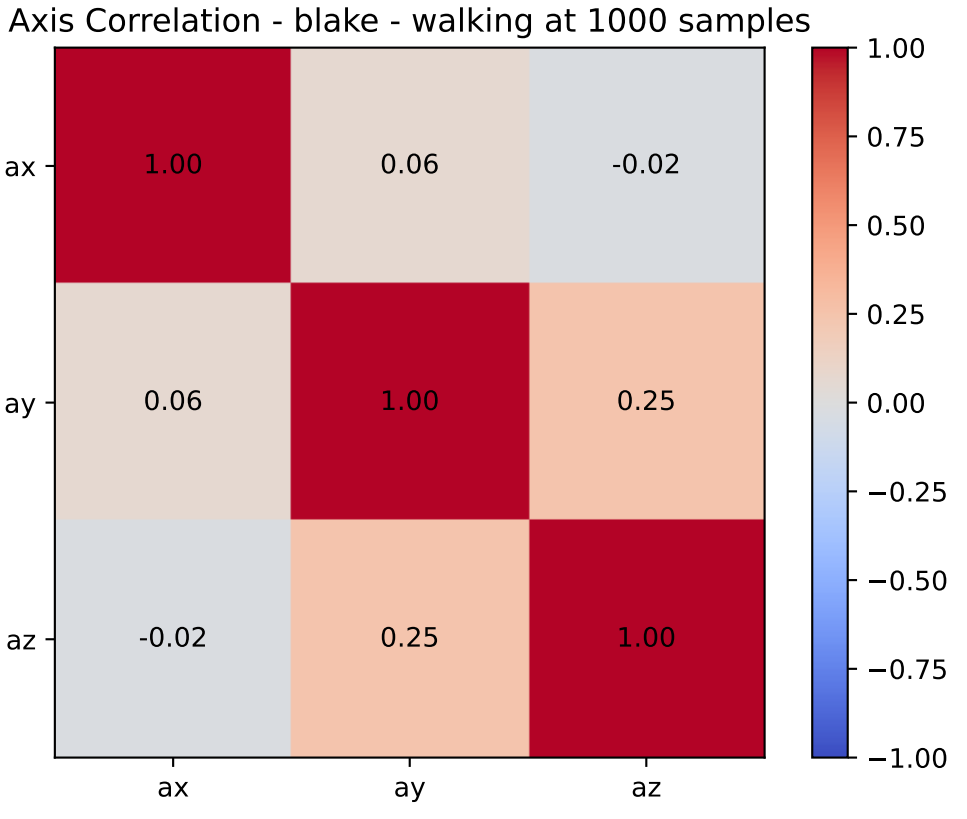

Walking vs Jumping Magnitude - blake at 6000 samples
walking
jumping
Acceleration Magnitude
Time (samples)
0
20
40
60
80
100
120
140
160
0
250
500
750
1000
1250
1500
1750
2000

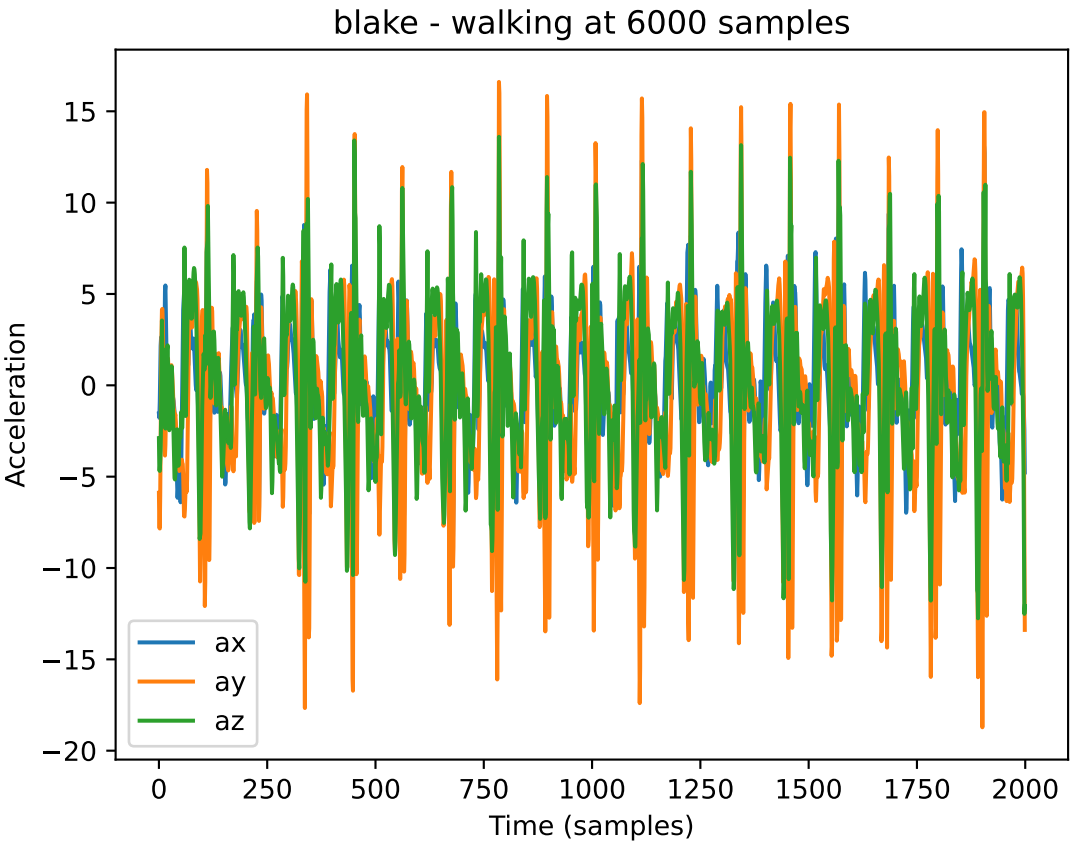

Walking vs Jumping Magnitude - blake at 10000 samples
walking
jumping
Acceleration Magnitude
Time (samples)
0
20
40
60
80
100
120
140
0
250
500
750
1000
1250
1500
1750
2000

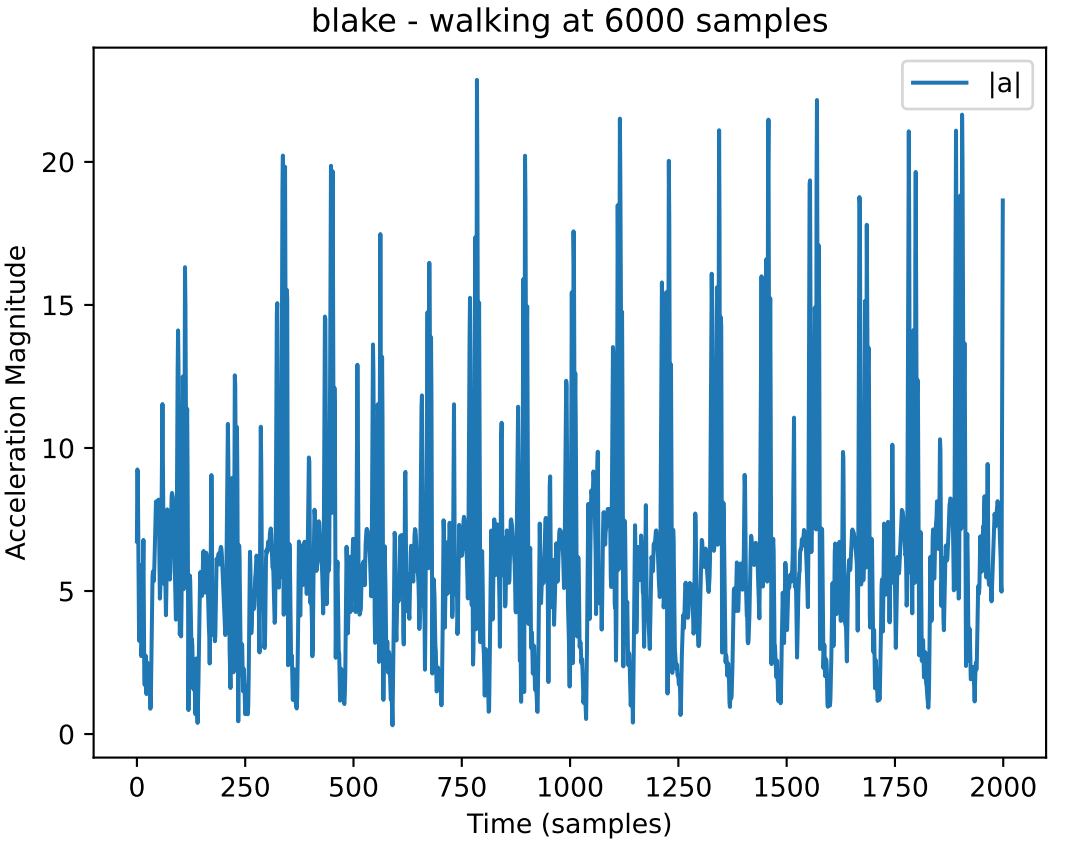

blake - walking at 1000 samples
Acceleration
Time (samples)
5
0
−5
−10
0
250
500
750
1000
1250
1500
1750
2000
ax
ay
az

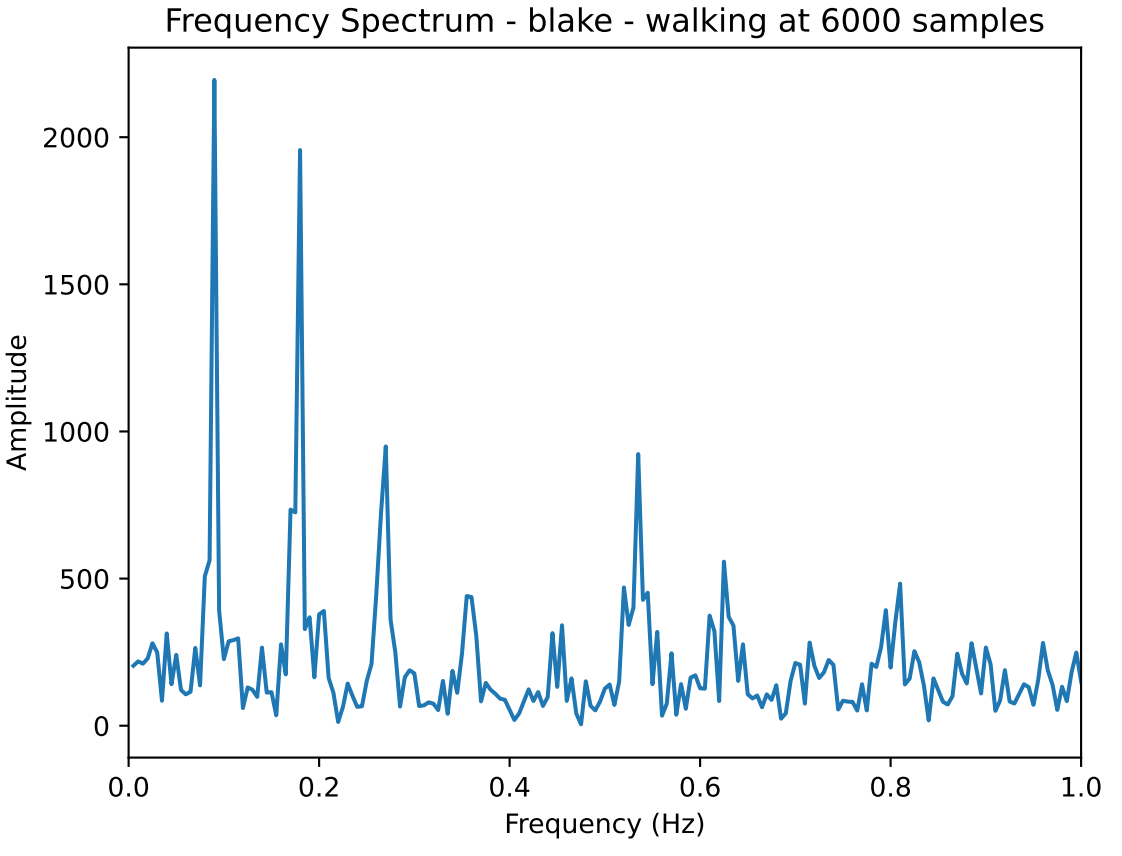

blake - walking at 1000 samples
|a|
Acceleration Magnitude
Time (samples)
0
2
4
6
8
10
12
14
16
0
250
500
750
1000
1250
1500
1750
2000

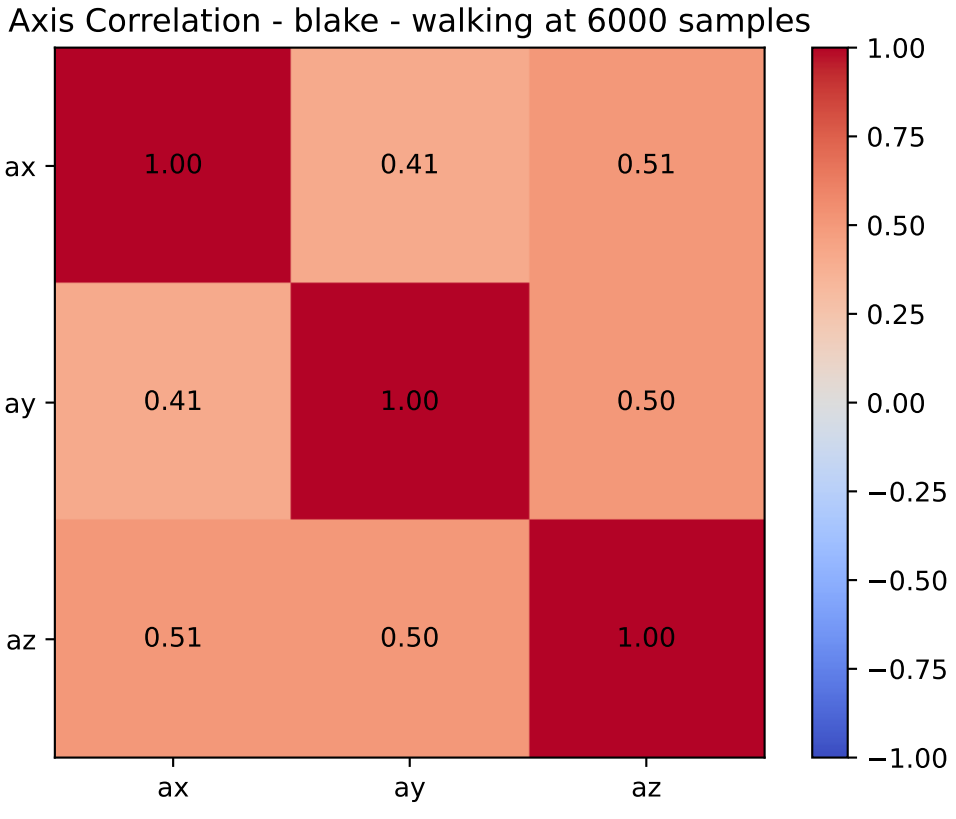

Frequency Spectrum - blake - walking at 1000 samples
Amplitude
Frequency (Hz)
0
200
400
600
800
1000
1200
0.0
0.2
0.4
0.6
0.8
1.0

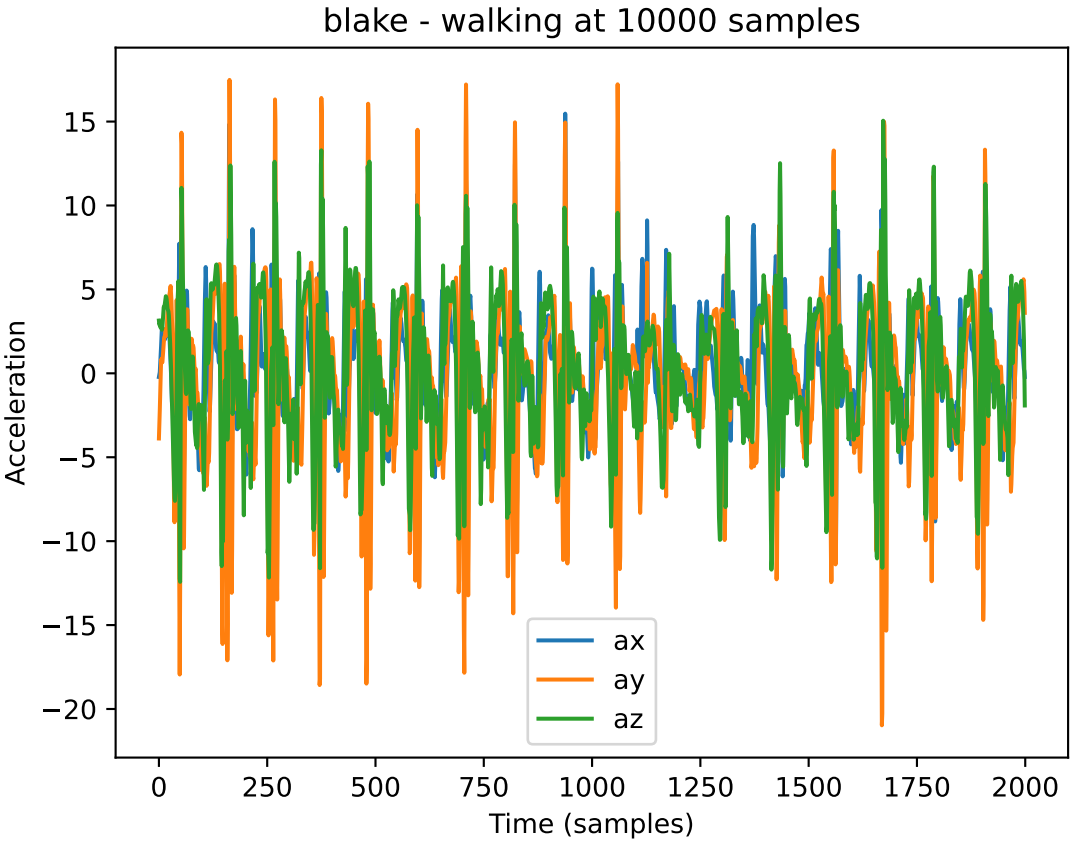

Axis Correlation - blake - walking at 1000 samples
ax
ay
az
1.00
0.06
-0.02
0.06
1.00
0.25
-0.02
0.25
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

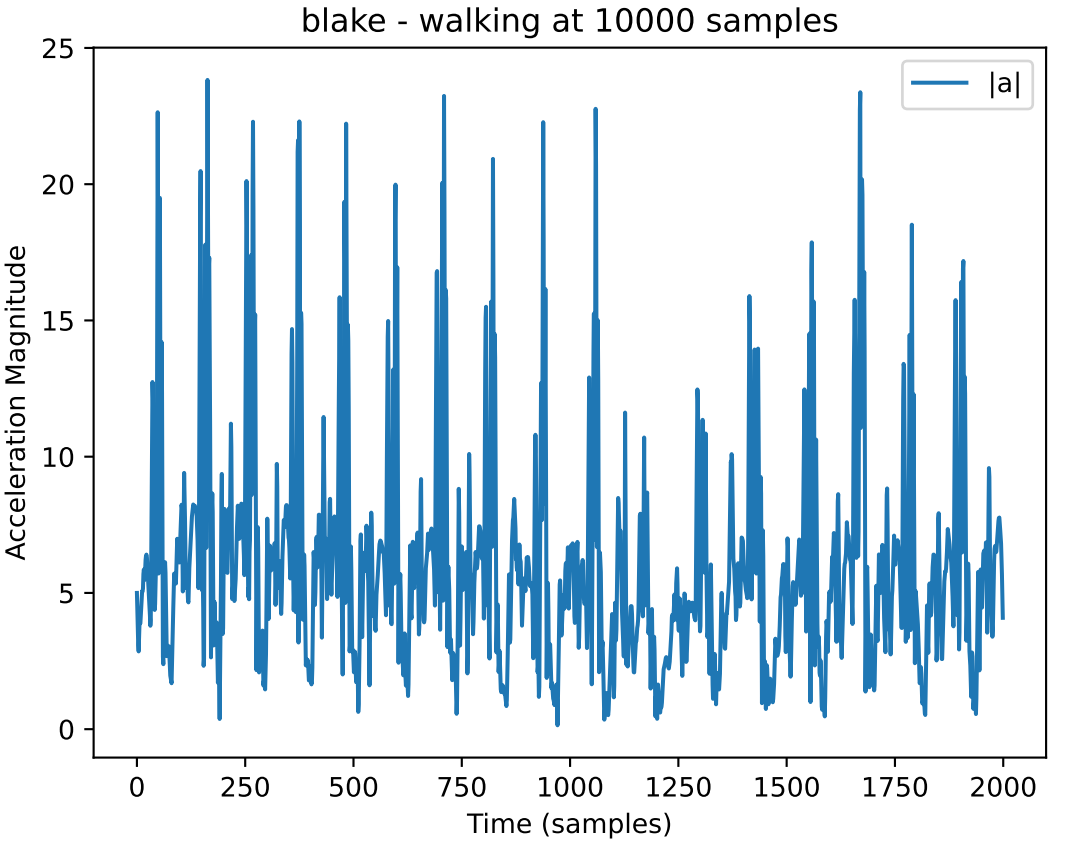

blake - walking at 6000 samples
Acceleration
Time (samples)
ax
ay
az
15
10
5
0
−5
−10
−15
−20
0
250
500
750
1000
1250
1500
1750
2000

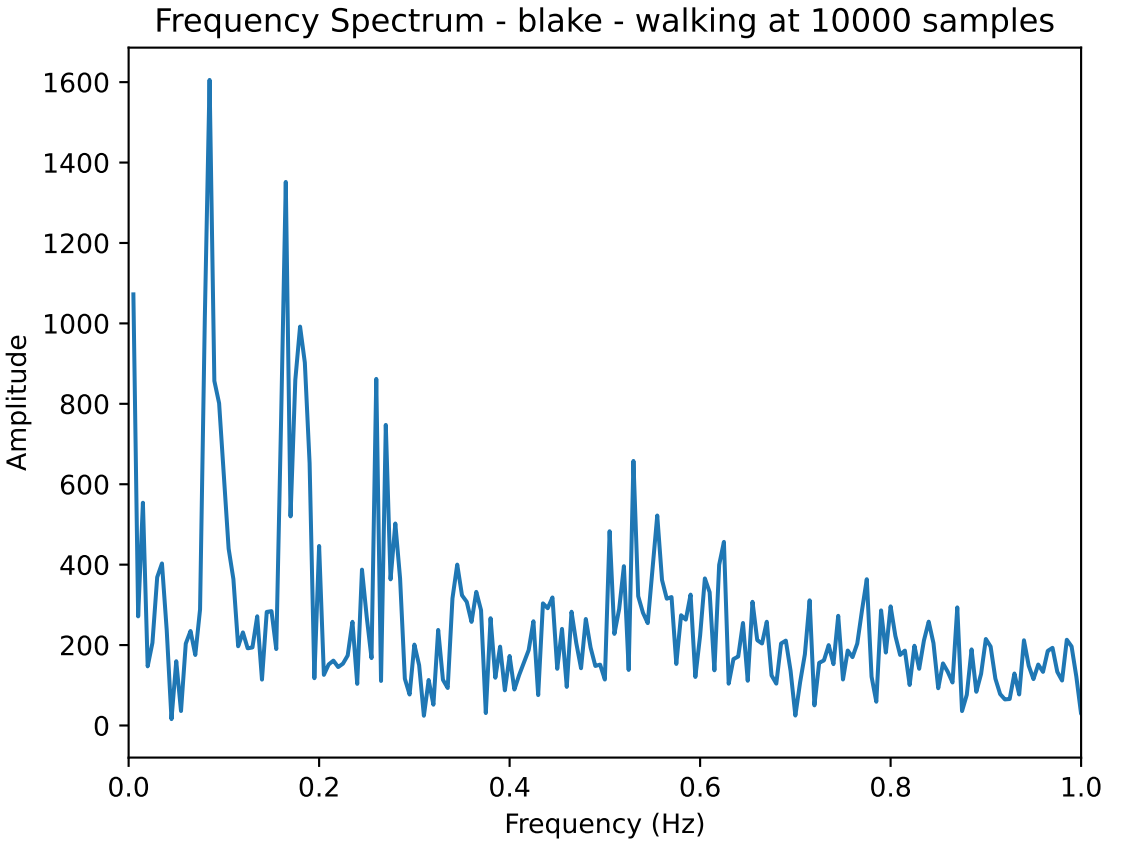

blake - walking at 6000 samples
|a|
Acceleration Magnitude
0
5
10
15
20
0
250
500
750
1000
1250
1500
1750
2000
Time (samples)

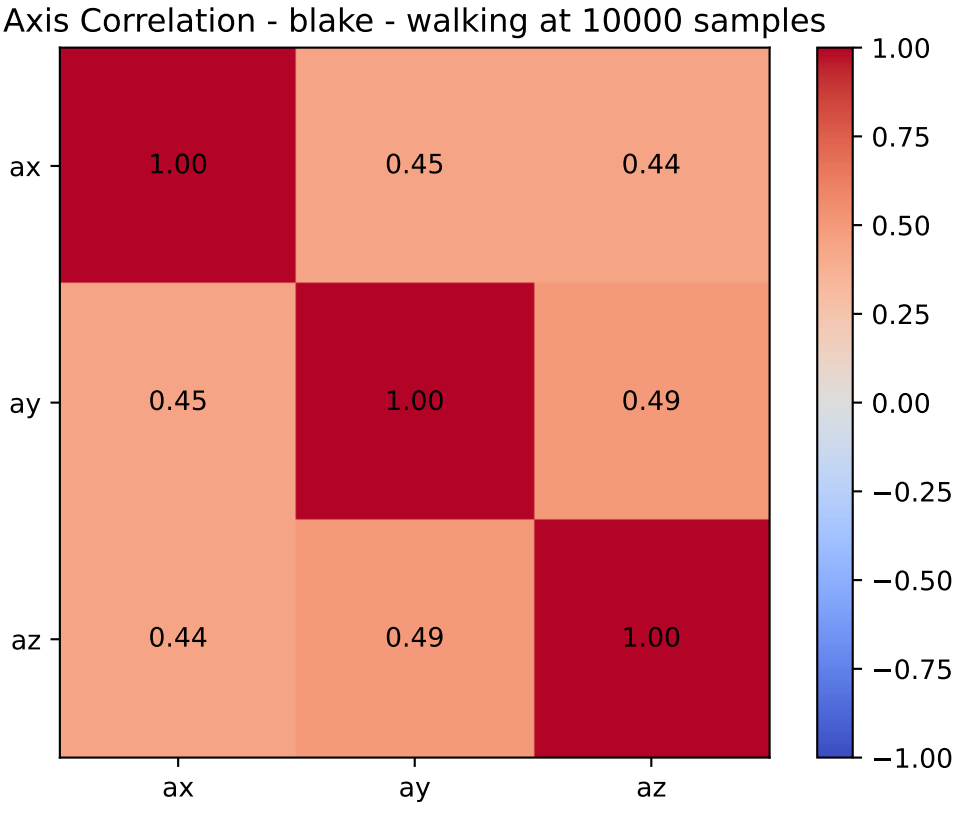

Frequency Spectrum - blake - walking at 6000 samples
Amplitude
2000
1500
1000
500
0
0.0
0.2
0.4
0.6
0.8
1.0
Frequency (Hz)

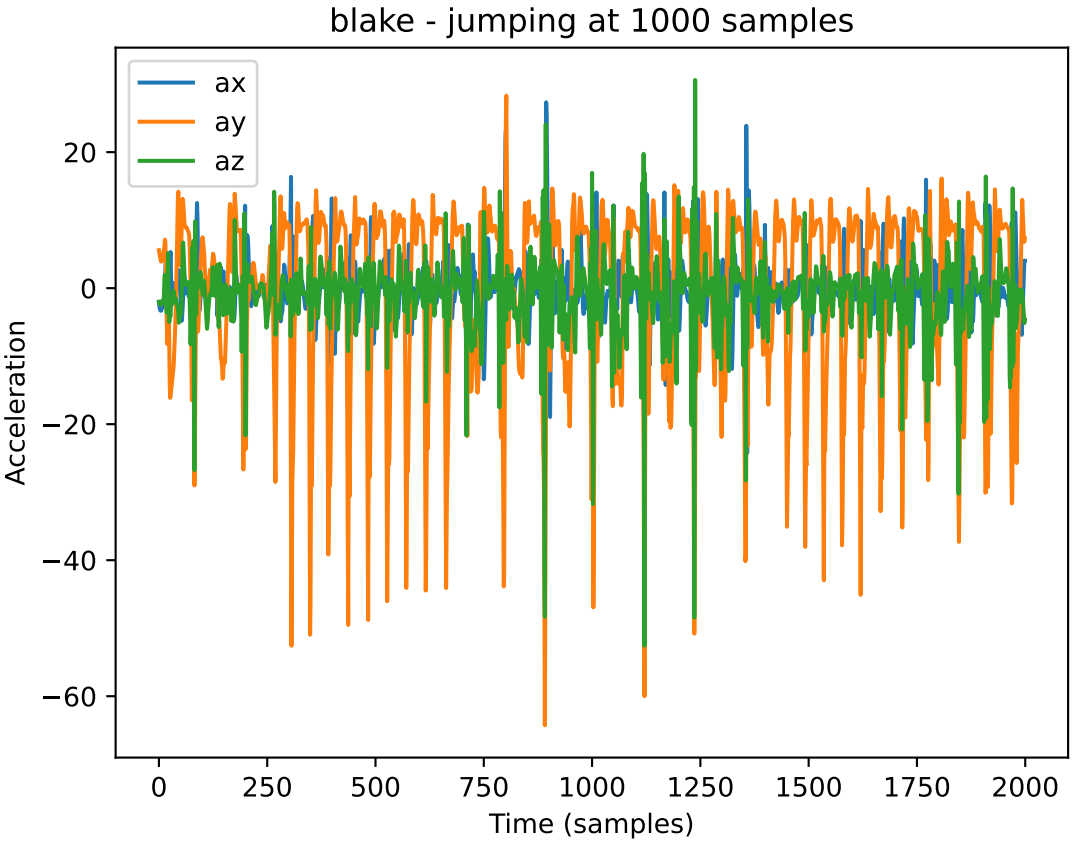

Axis Correlation - blake - walking at 6000 samples
ax
ay
az
1.00
0.41
0.51
0.41
1.00
0.50
0.51
0.50
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

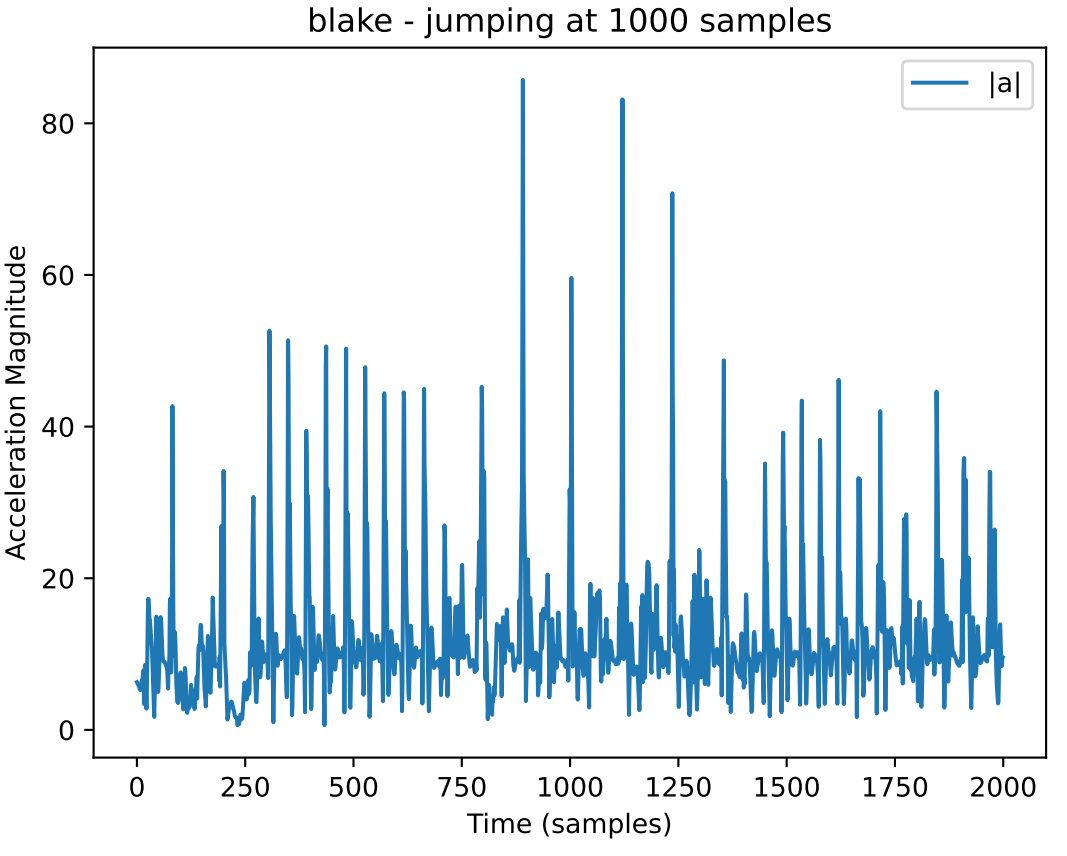

blake - walking at 10000 samples
Acceleration
Time (samples)
ax
ay
az
15
10
5
0
−5
−10
−15
−20
0
250
500
750
1000
1250
1500
1750
2000

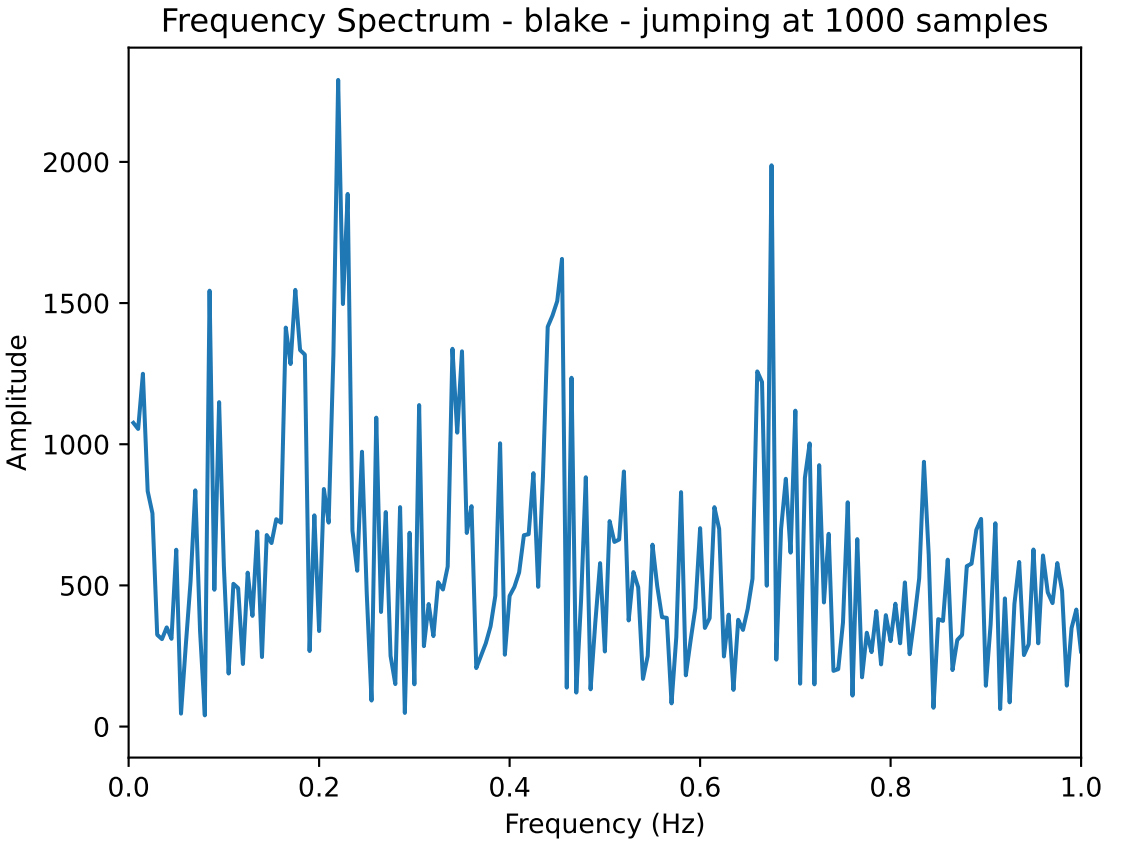

blake - walking at 10000 samples
|a|
Acceleration Magnitude
Time (samples)
0
5
10
15
20
25
0
250
500
750
1000
1250
1500
1750
2000

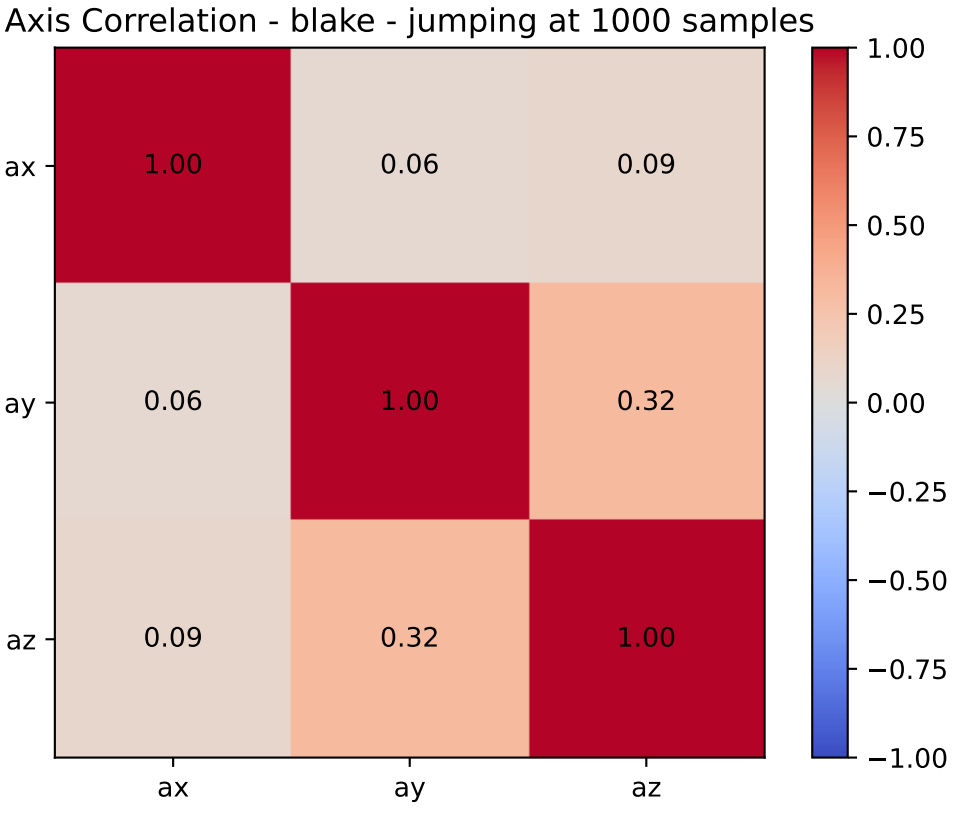

Frequency Spectrum - blake - walking at 10000 samples
Amplitude
Frequency (Hz)
0
200
400
600
800
1000
1200
1400
1600
0.0
0.2
0.4
0.6
0.8
1.0

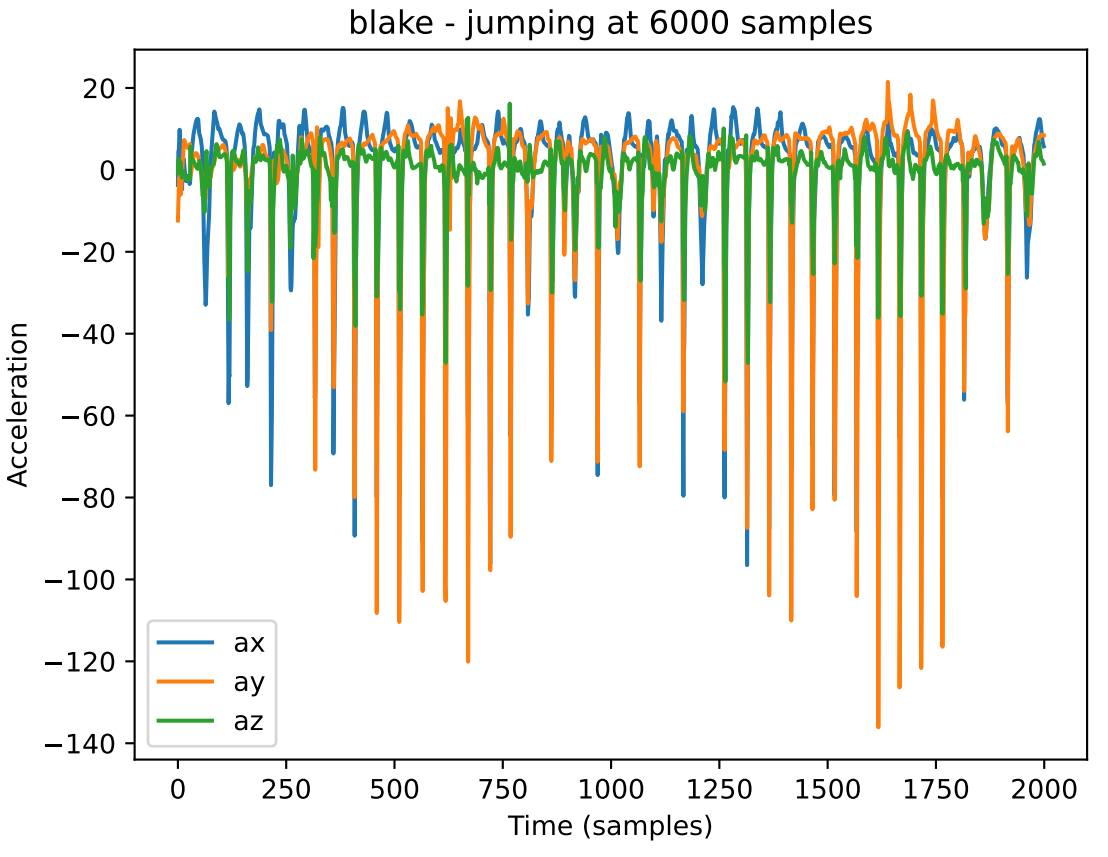

Axis Correlation - blake - walking at 10000 samples
ax
ay
az
1.00
0.45
0.44
0.45
1.00
0.49
0.44
0.49
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

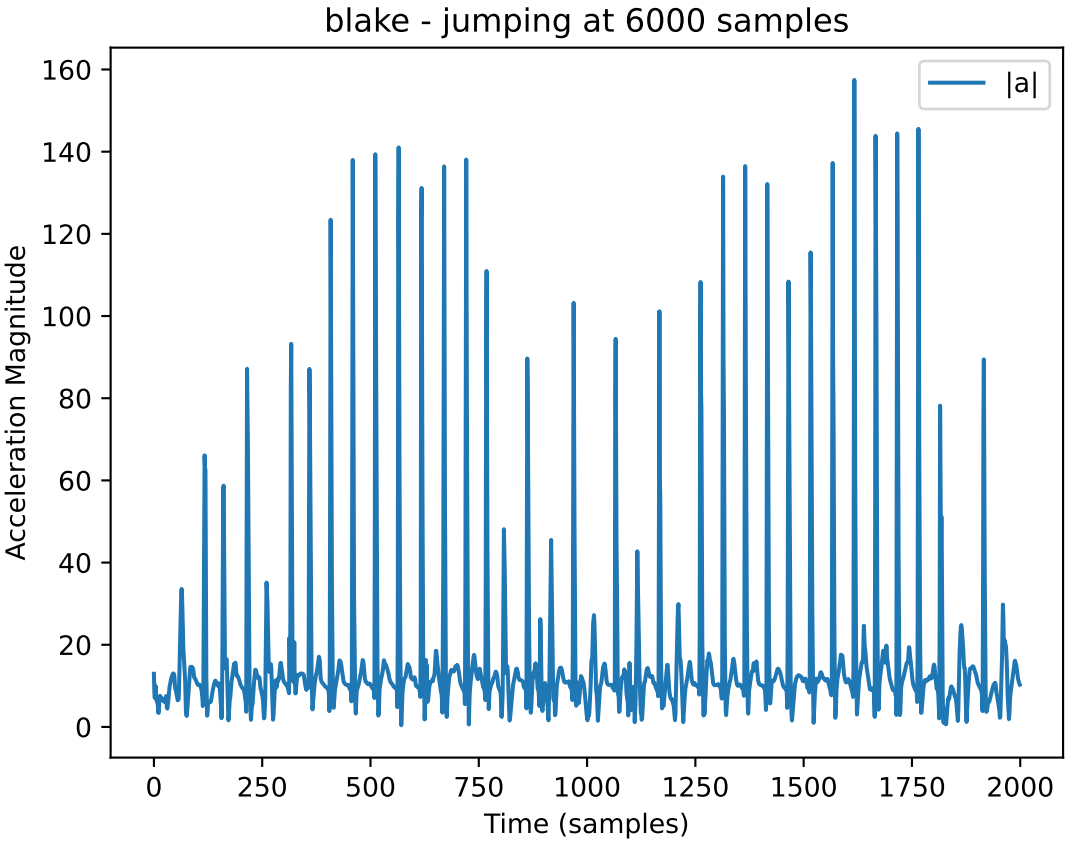

blake - jumping at 1000 samples
ax
ay
az
20
0
−20
−40
−60
Acceleration
0
250
500
750
1000
1250
1500
1750
2000
Time (samples)

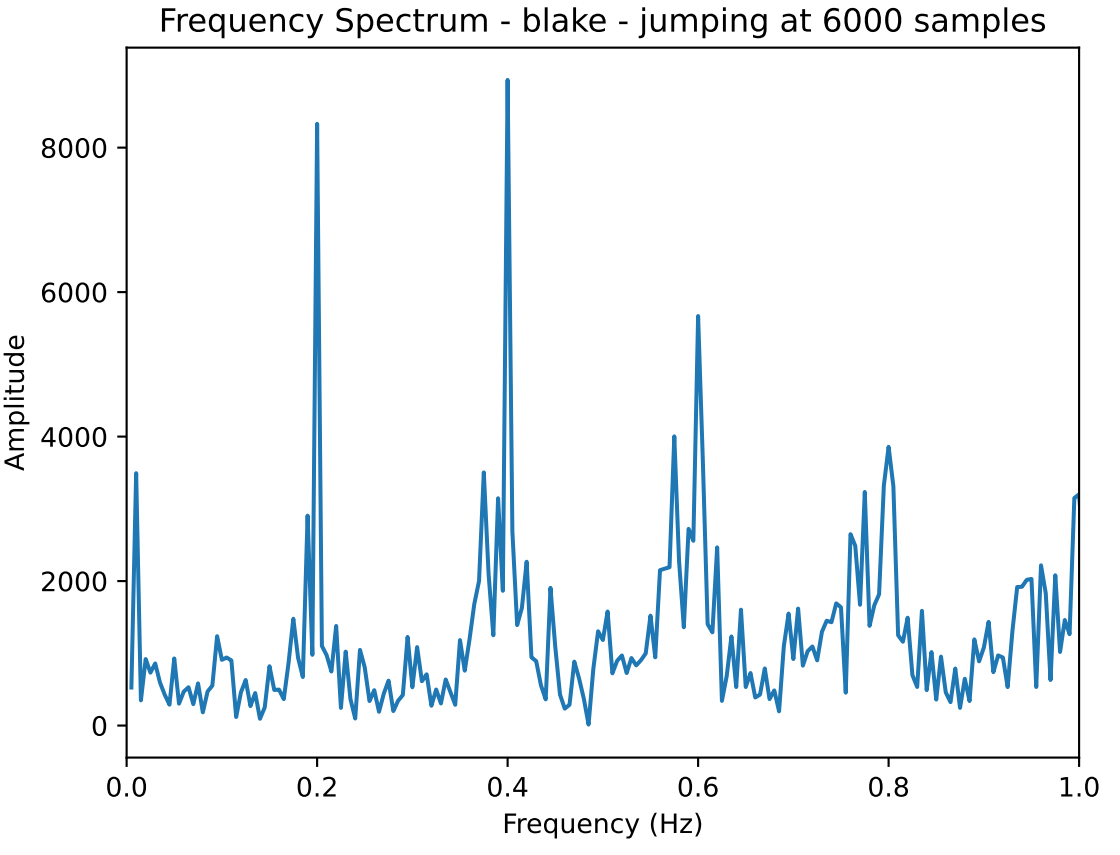

blake - jumping at 1000 samples
|a|
Acceleration Magnitude
0
20
40
60
80
0
250
500
750
1000
1250
1500
1750
2000
Time (samples)

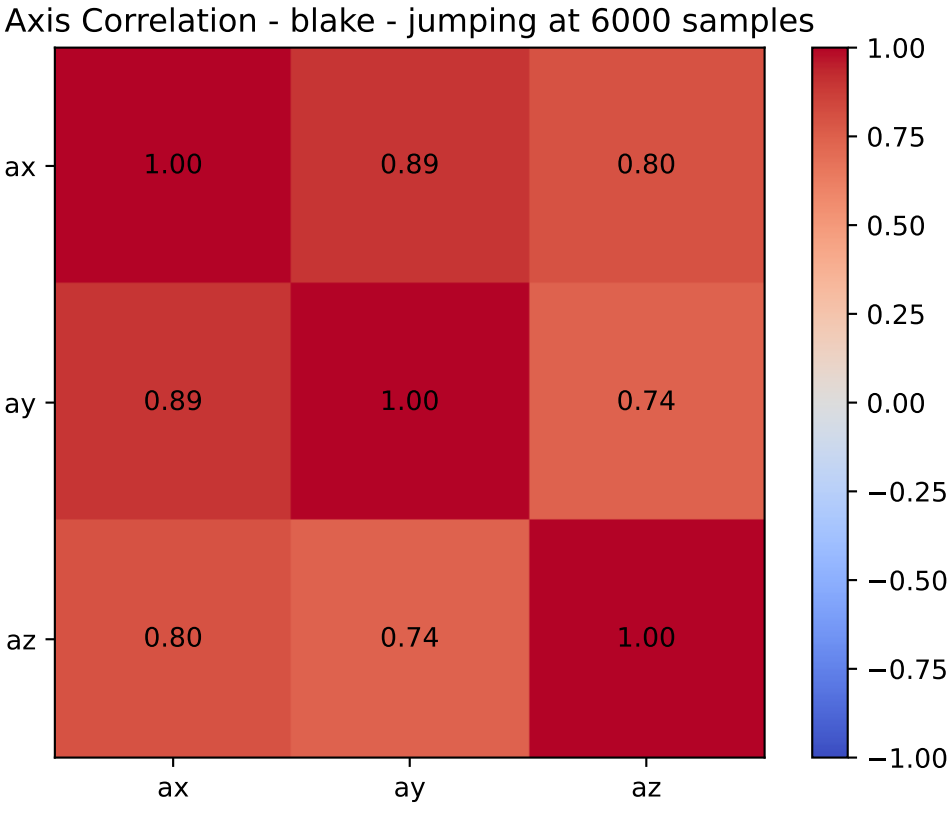

Frequency Spectrum - blake - jumping at 1000 samples
Amplitude
2000
1500
1000
500
0
0.0
0.2
0.4
0.6
0.8
1.0
Frequency (Hz)

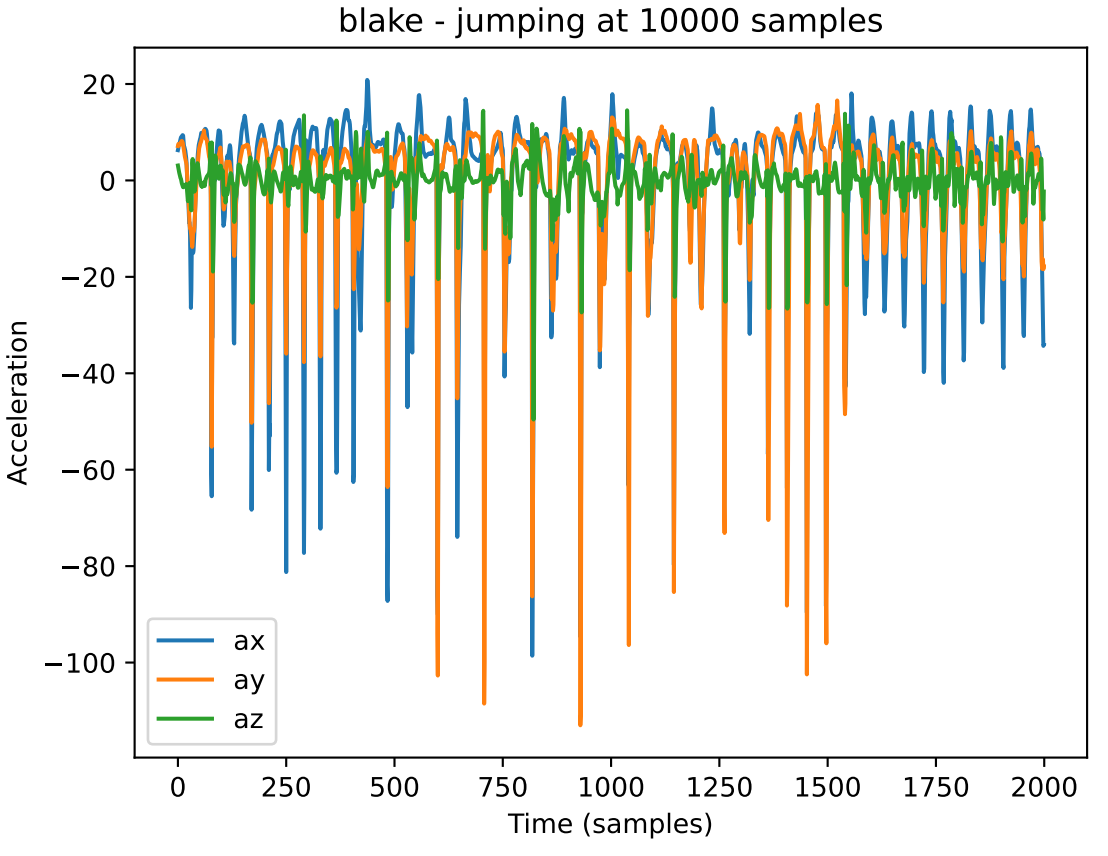

Axis Correlation - blake - jumping at 1000 samples
ax
ay
az
1.00
0.06
0.09
0.06
1.00
0.32
0.09
0.32
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

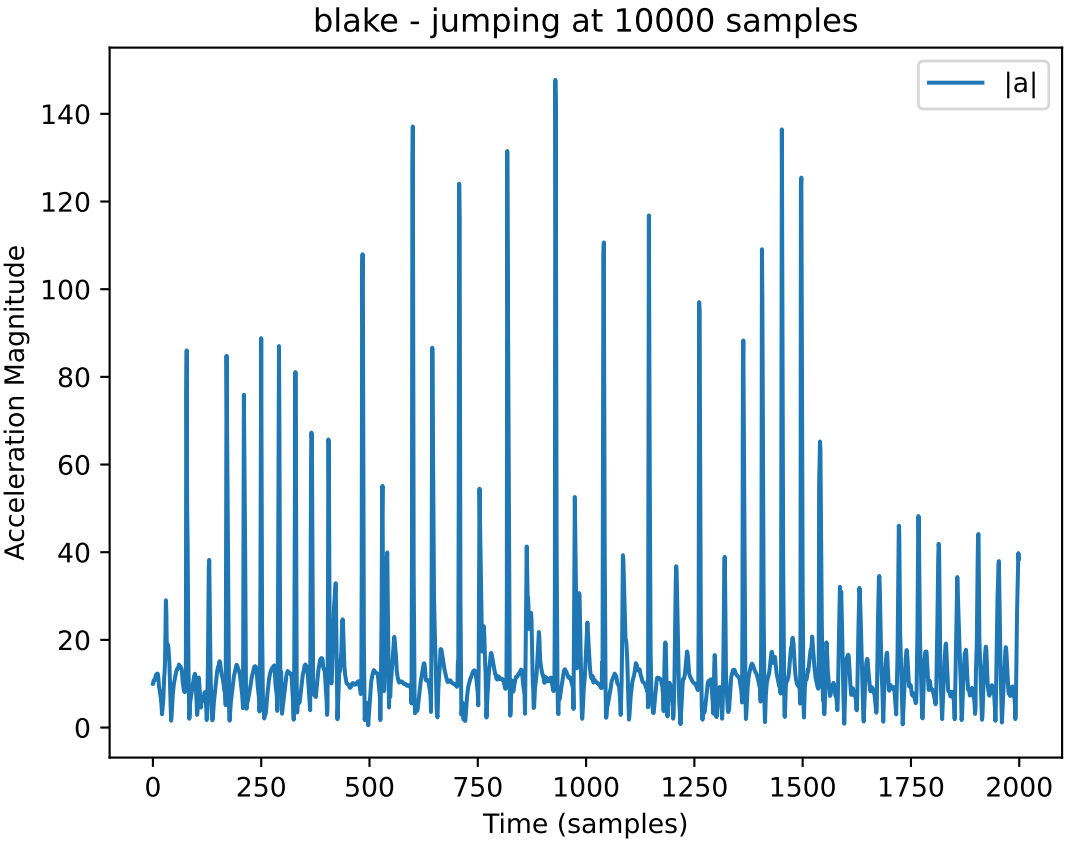

blake - jumping at 6000 samples
Acceleration
Time (samples)
20
0
−20
−40
−60
−80
−100
−120
−140
0
250
500
750
1000
1250
1500
1750
2000
ax
ay
az

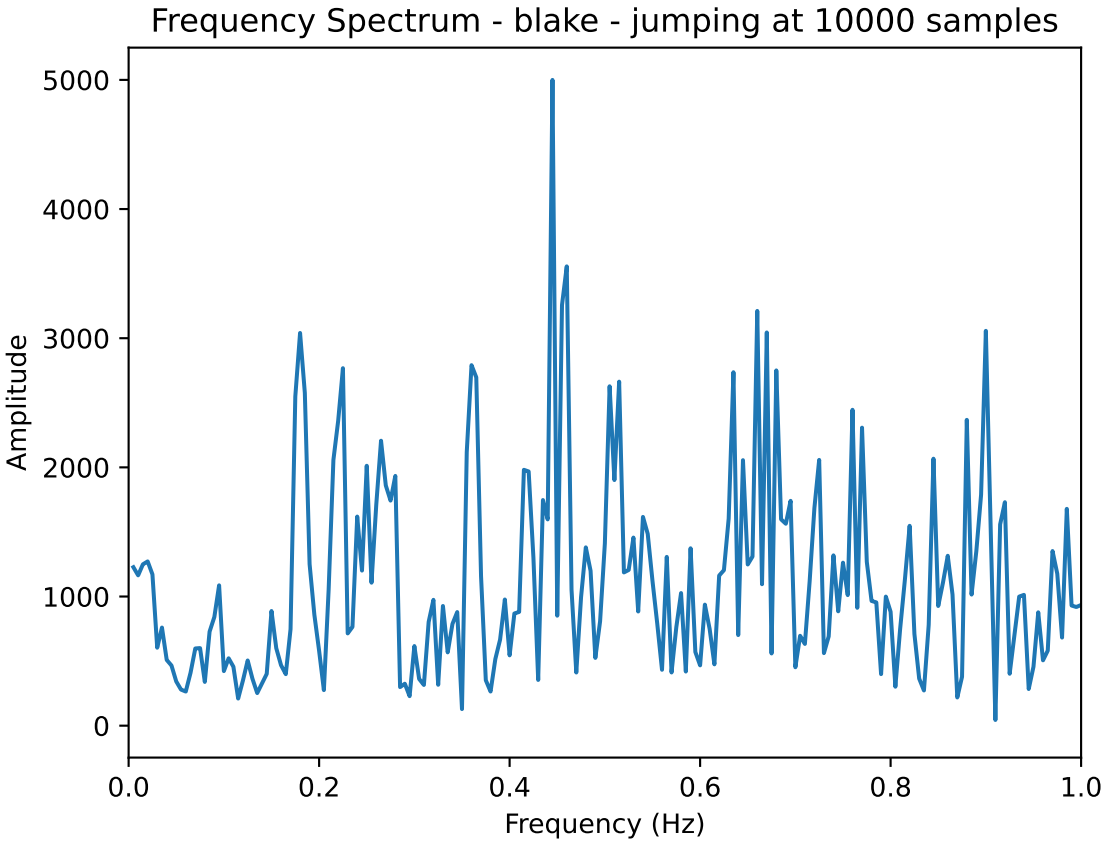

blake - jumping at 6000 samples
|a|
Acceleration Magnitude
Time (samples)
0
20
40
60
80
100
120
140
160
0
250
500
750
1000
1250
1500
1750
2000

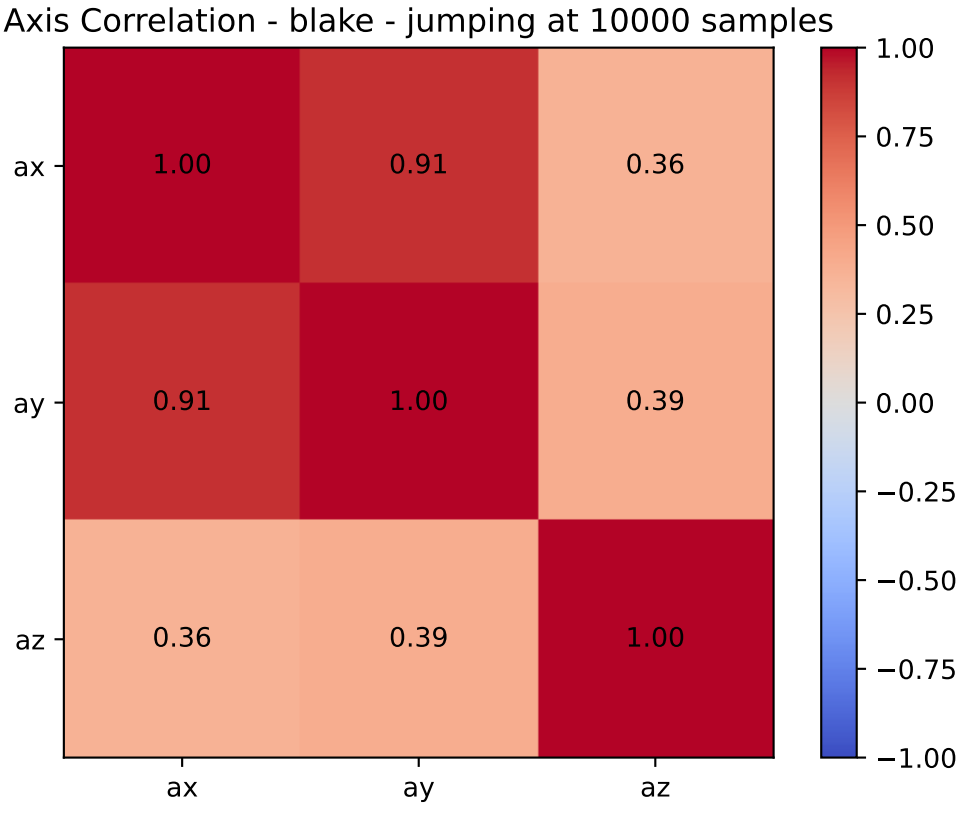

Frequency Spectrum - blake - jumping at 6000 samples
Amplitude
8000
6000
4000
2000
0
0.0
0.2
0.4
0.6
0.8
1.0
Frequency (Hz)

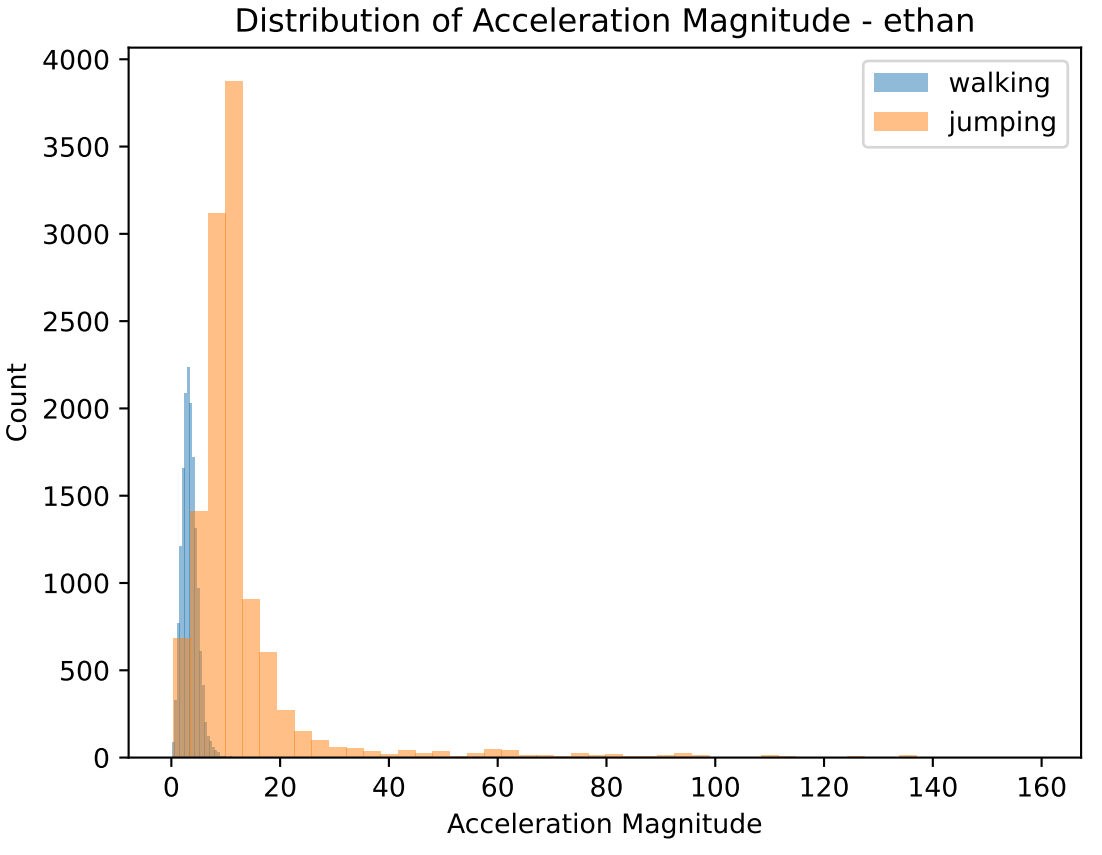

Axis Correlation - blake - jumping at 6000 samples
ax
ay
az
1.00
0.89
0.80
0.89
1.00
0.74
0.80
0.74
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

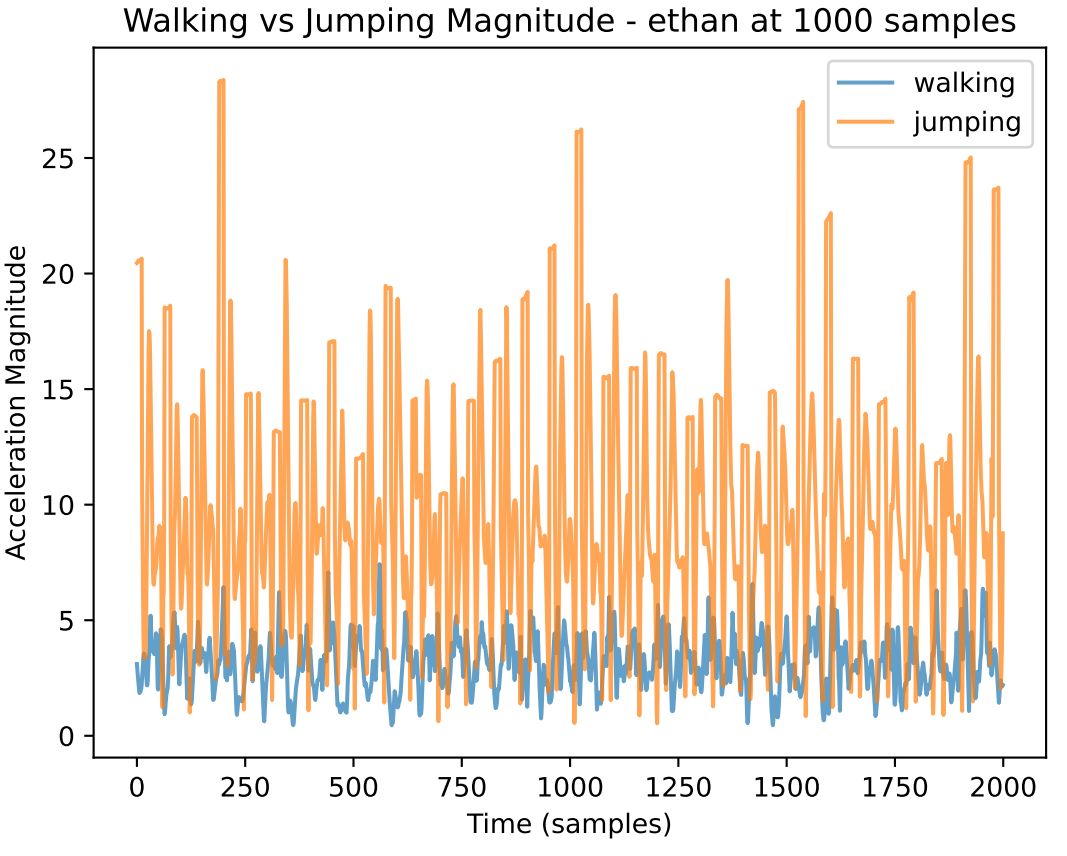

blake - jumping at 10000 samples
Acceleration
Time (samples)
ax
ay
az
20
0
−20
−40
−60
−80
−100
0
250
500
750
1000
1250
1500
1750
2000

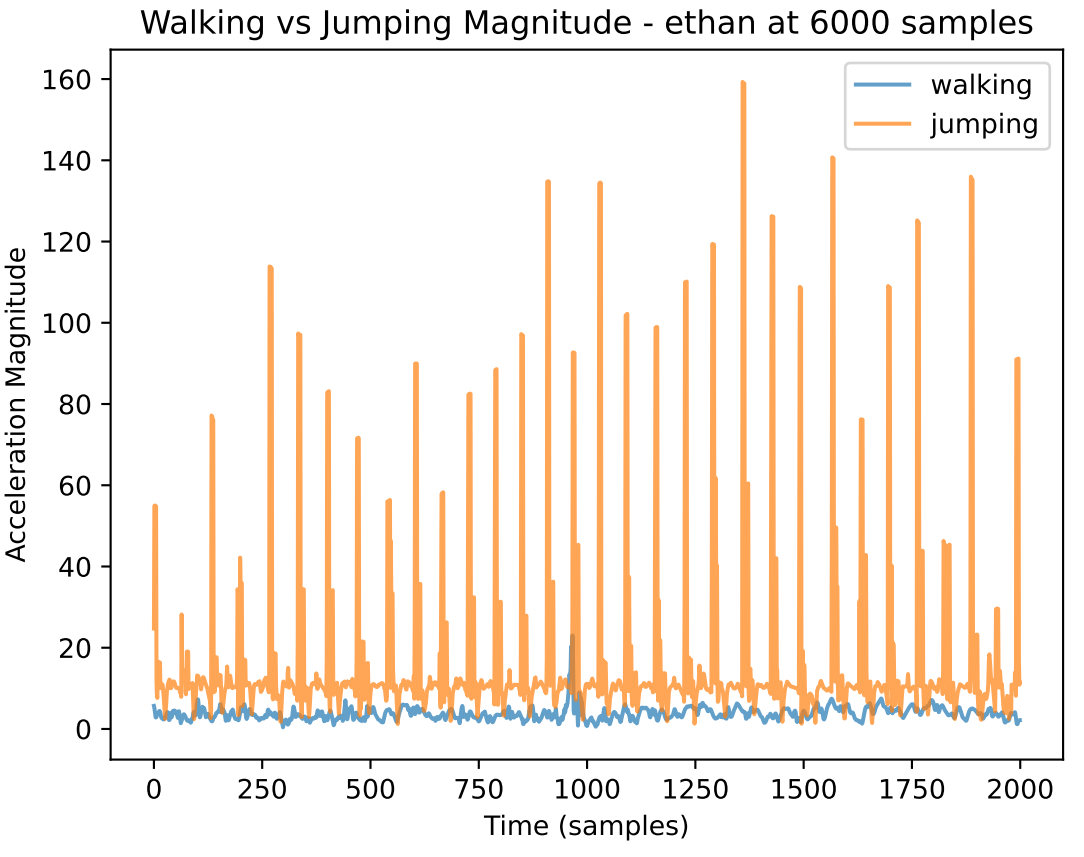

blake - jumping at 10000 samples
|a|
Acceleration Magnitude
Time (samples)
0
20
40
60
80
100
120
140
0
250
500
750
1000
1250
1500
1750
2000

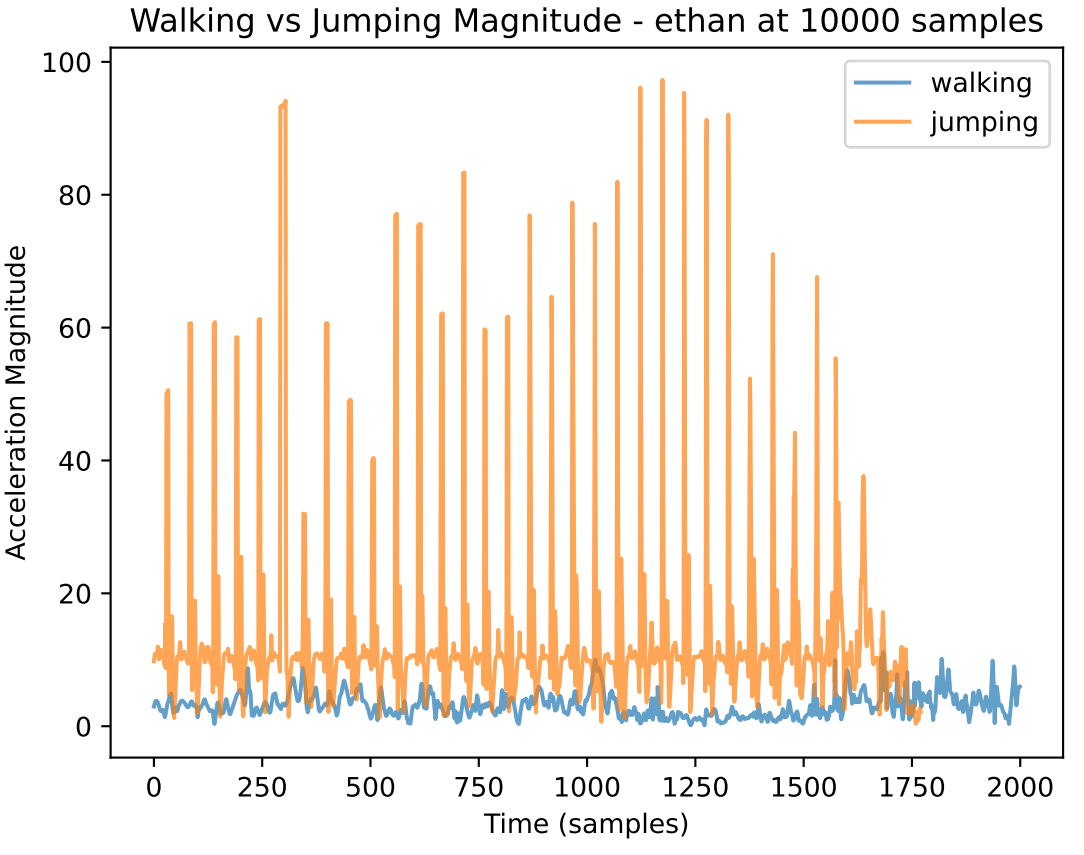

Frequency Spectrum - blake - jumping at 10000 samples
Amplitude
5000
4000
3000
2000
1000
0
0.0
0.2
0.4
0.6
0.8
1.0
Frequency (Hz)

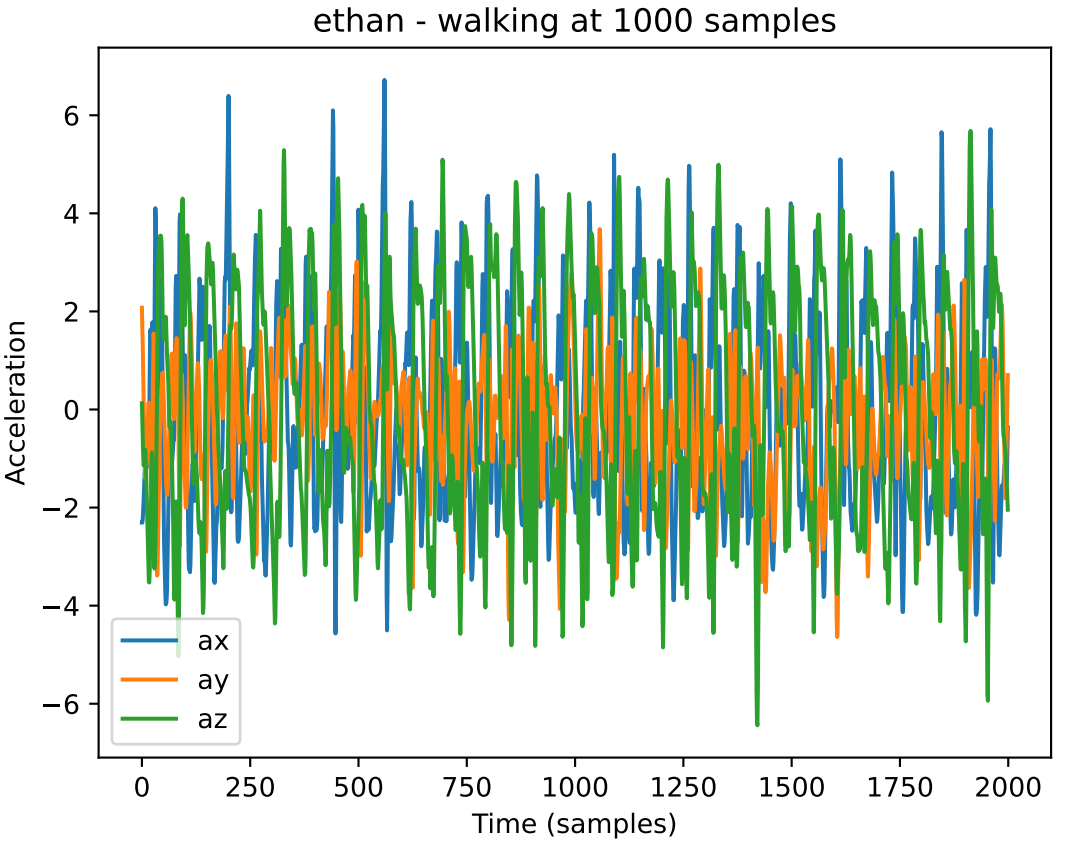

Axis Correlation - blake - jumping at 10000 samples
ax
ay
az
1.00
0.91
0.36
0.91
1.00
0.39
0.36
0.39
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

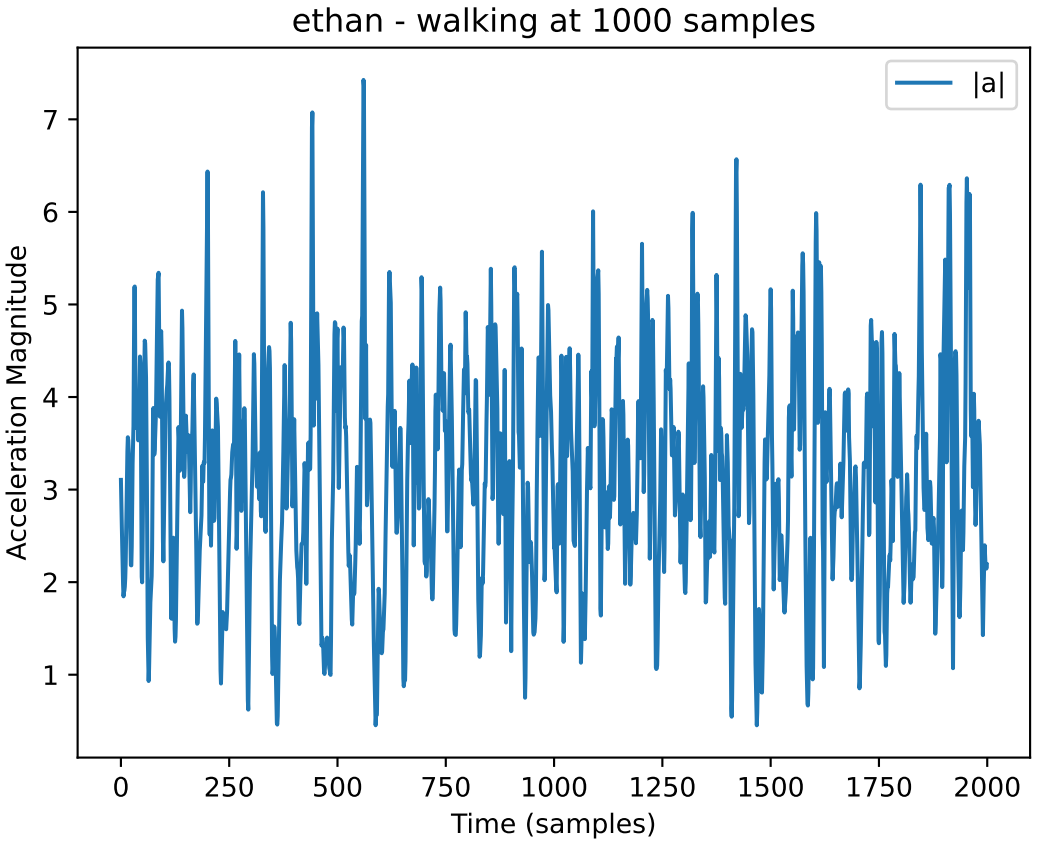

Distribution of Acceleration Magnitude - ethan
walking
jumping
Count
Acceleration Magnitude
0
500
1000
1500
2000
2500
3000
3500
4000
0
20
40
60
80
100
120
140
160

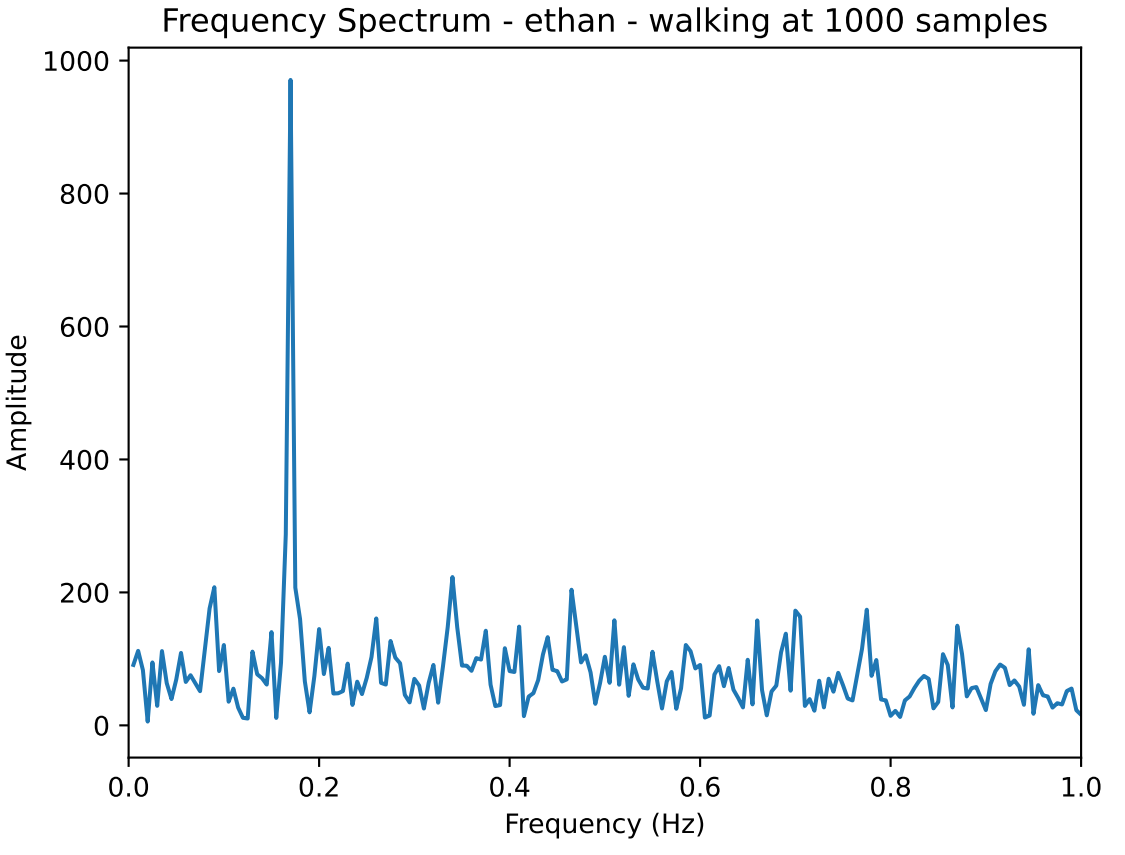

Walking vs Jumping Magnitude - ethan at 1000 samples
walking
jumping
Acceleration Magnitude
Time (samples)
0
5
10
15
20
25
0
250
500
750
1000
1250
1500
1750
2000

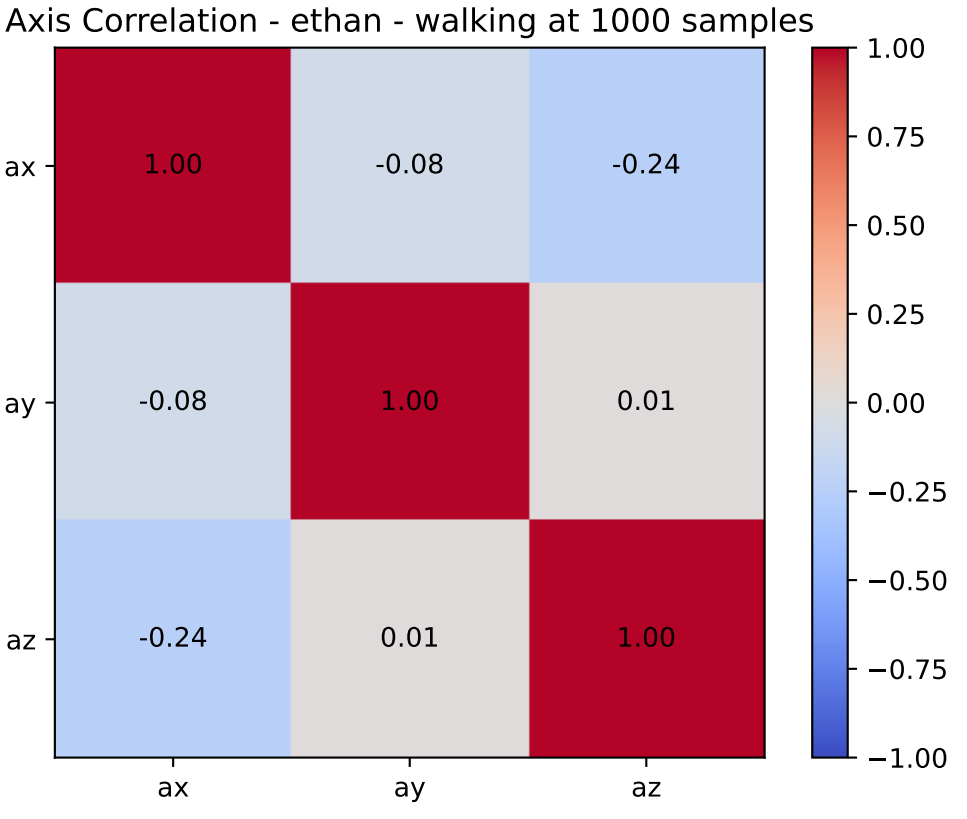

Walking vs Jumping Magnitude - ethan at 6000 samples
walking
jumping
Acceleration Magnitude
Time (samples)
0
20
40
60
80
100
120
140
160
0
250
500
750
1000
1250
1500
1750
2000

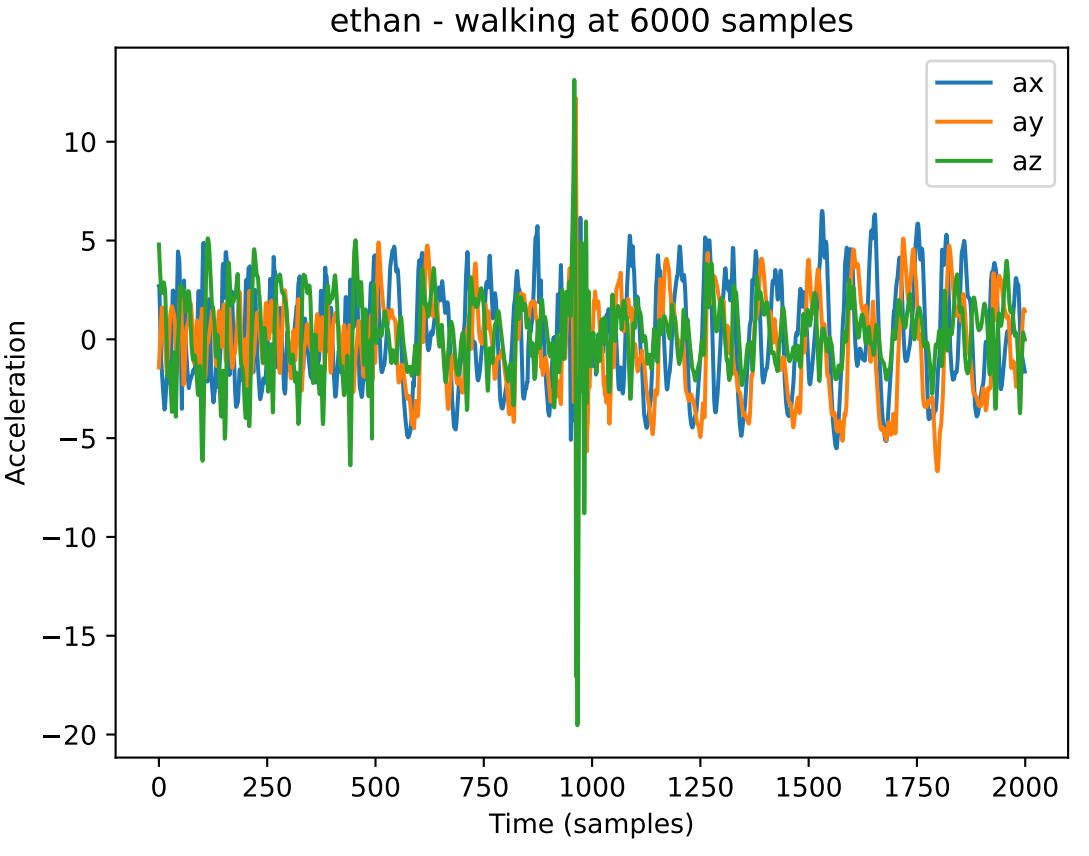

Walking vs Jumping Magnitude - ethan at 10000 samples
walking
jumping
Acceleration Magnitude
Time (samples)
0
20
40
60
80
100
0
250
500
750
1000
1250
1500
1750
2000

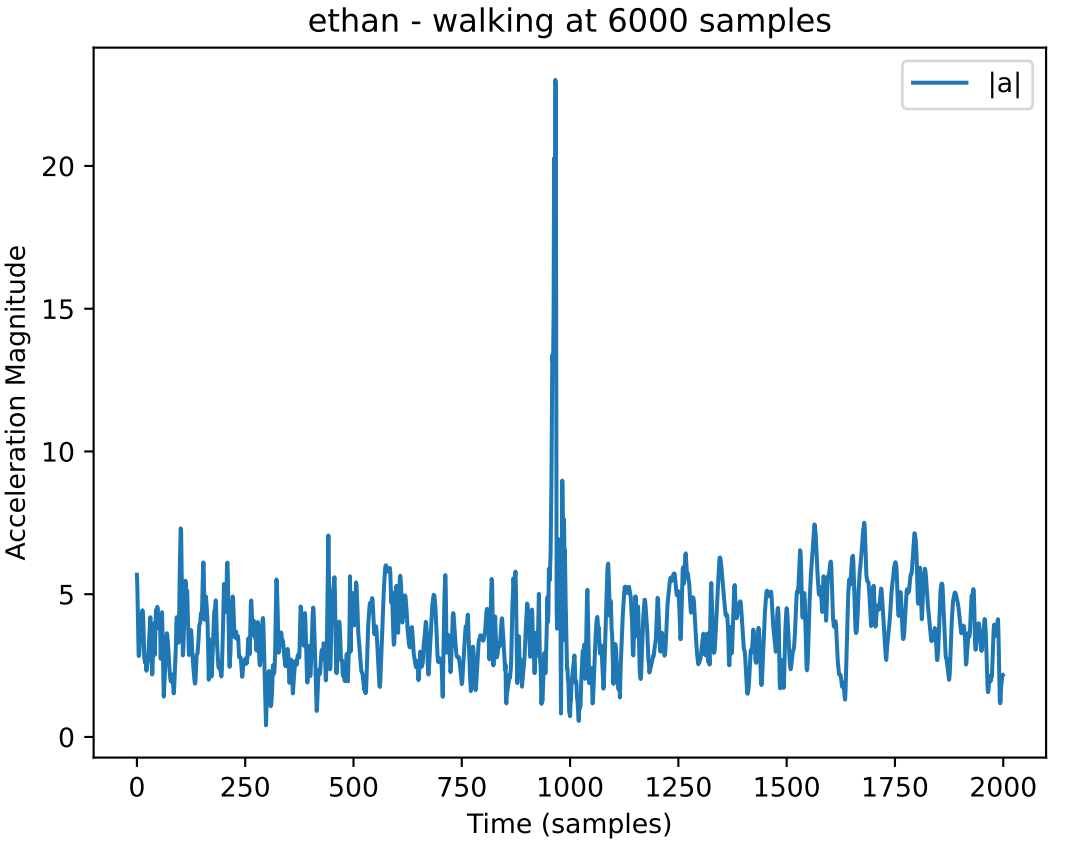

ethan - walking at 1000 samples
Acceleration
Time (samples)
ax
ay
az
6
4
2
0
−2
−4
−6
0
250
500
750
1000
1250
1500
1750
2000

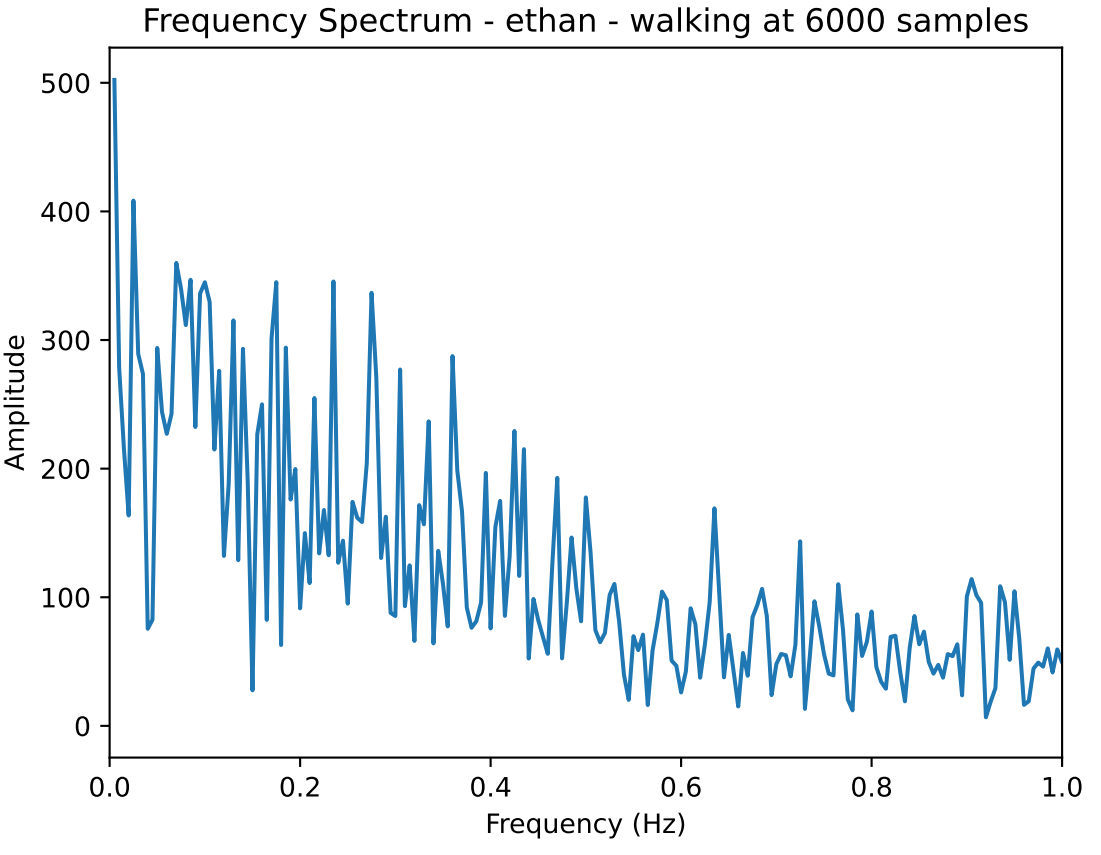

ethan - walking at 1000 samples
|a|
Acceleration Magnitude
Time (samples)
1
2
3
4
5
6
7
0
250
500
750
1000
1250
1500
1750
2000

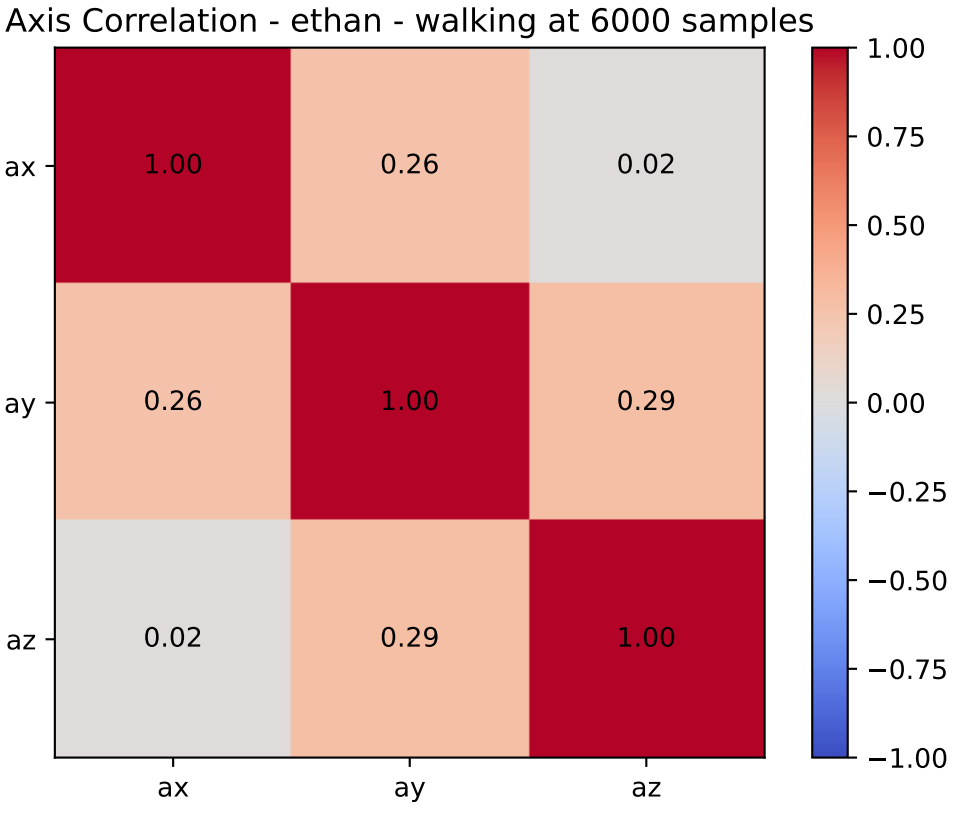

Frequency Spectrum - ethan - walking at 1000 samples
Amplitude
Frequency (Hz)
1000
800
600
400
200
0
0.0
0.2
0.4
0.6
0.8
1.0

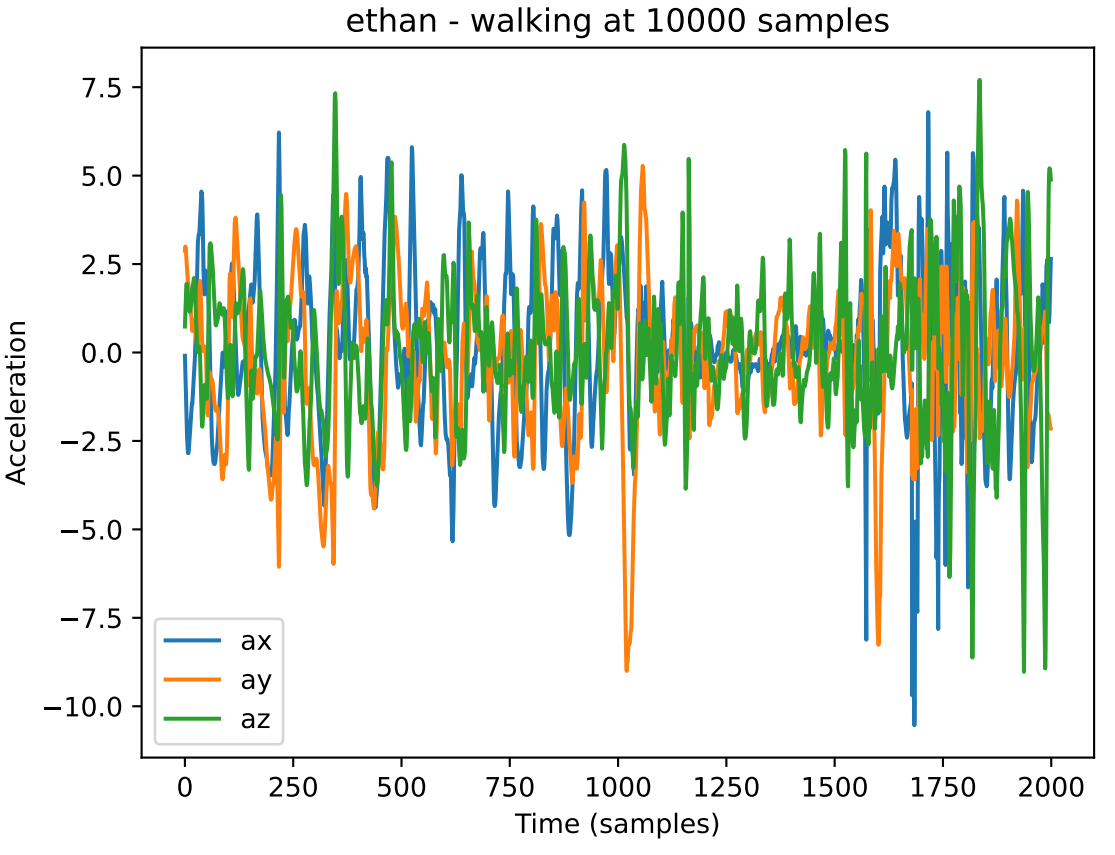

Axis Correlation - ethan - walking at 1000 samples
ax
ay
az
1.00
-0.08
-0.24
-0.08
1.00
0.01
-0.24
0.01
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

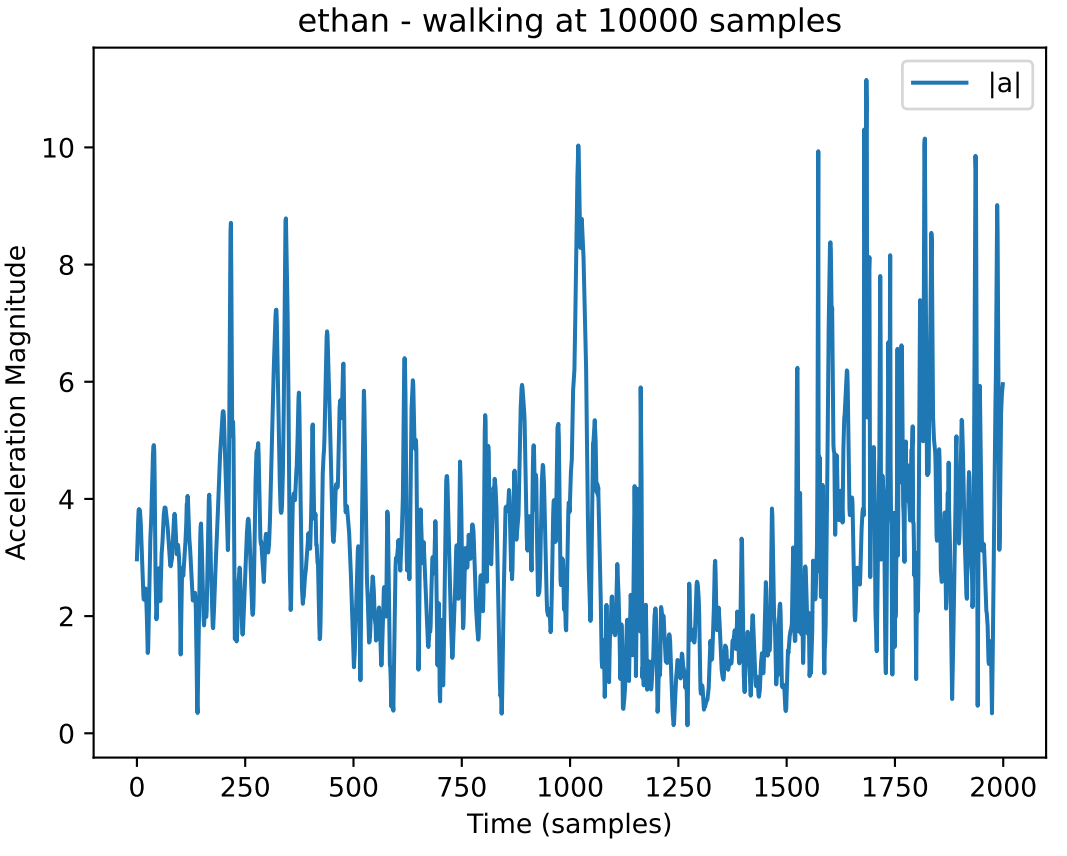

ethan - walking at 6000 samples
ax
ay
az
10
5
0
−5
−10
−15
−20
Acceleration
0
250
500
750
1000
1250
1500
1750
2000
Time (samples)

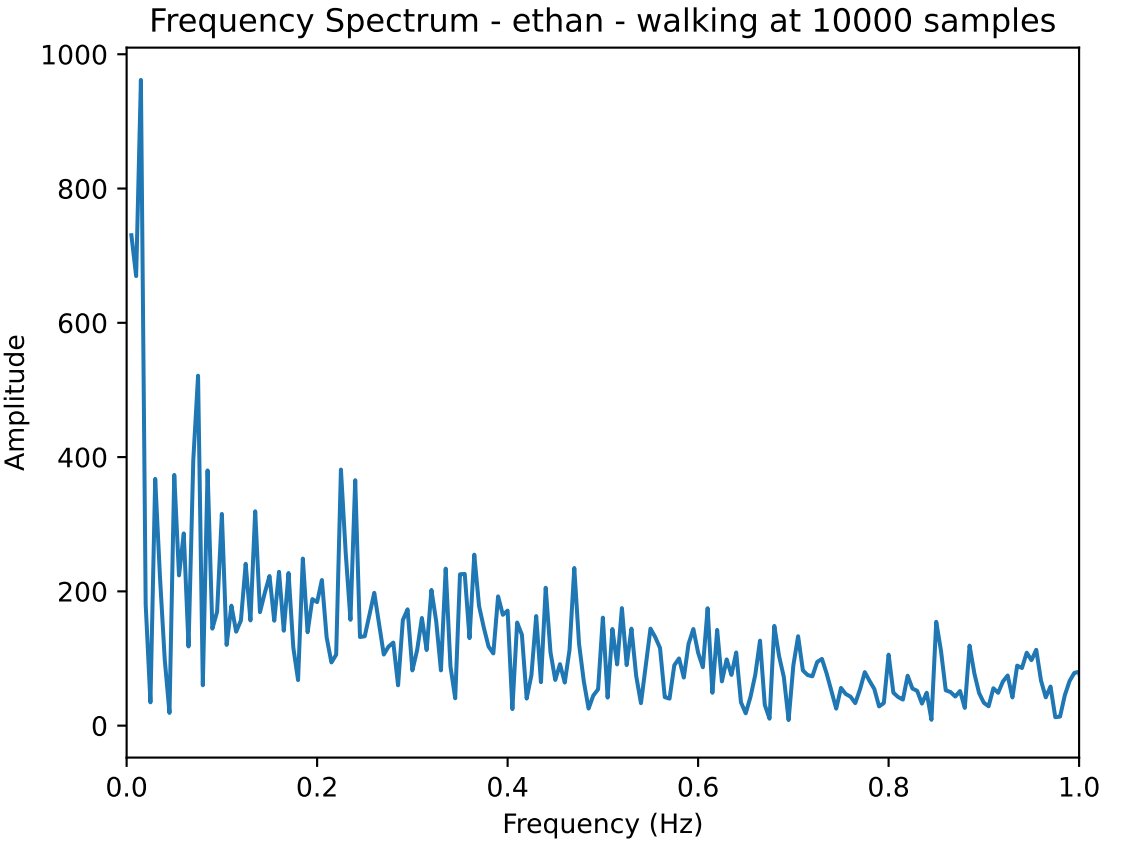

ethan - walking at 6000 samples
|a|
Acceleration Magnitude
0
5
10
15
20
0
250
500
750
1000
1250
1500
1750
2000
Time (samples)

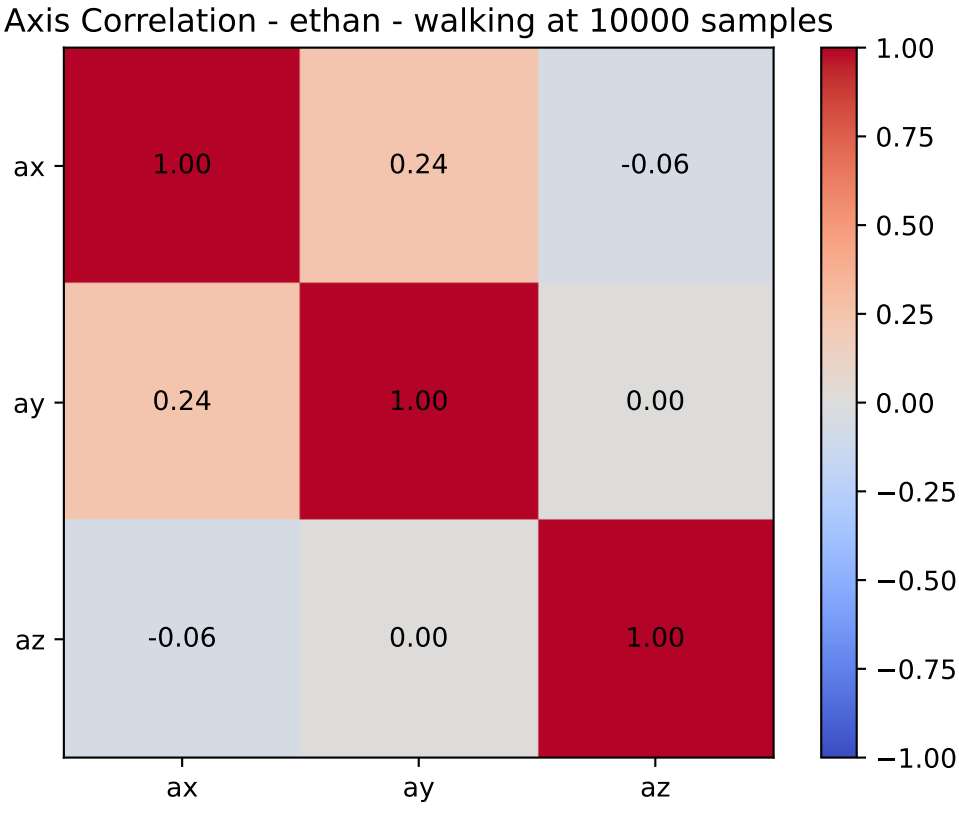

Frequency Spectrum - ethan - walking at 6000 samples
Amplitude
Frequency (Hz)
500
400
300
200
100
0
0.0
0.2
0.4
0.6
0.8
1.0

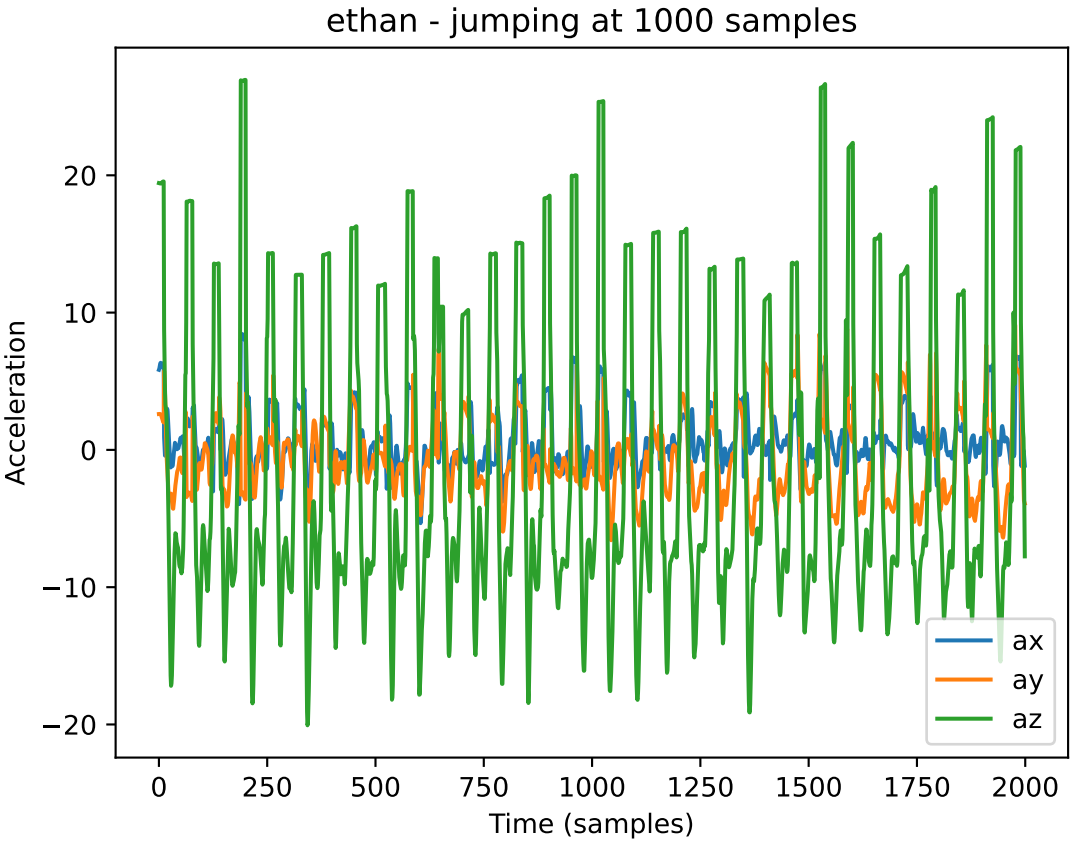

Axis Correlation - ethan - walking at 6000 samples
ax
ay
az
1.00
0.26
0.02
0.26
1.00
0.29
0.02
0.29
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

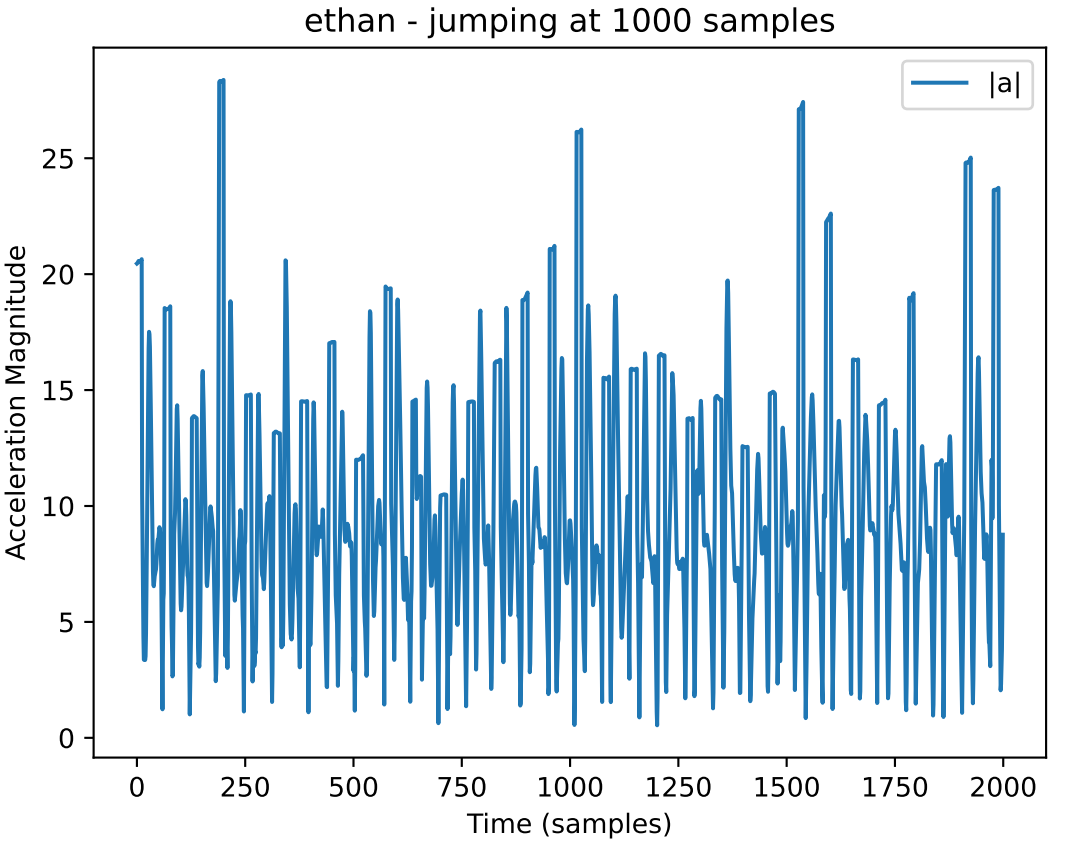

ethan - walking at 10000 samples
Acceleration
Time (samples)
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
0
250
500
750
1000
1250
1500
1750
2000
ax
ay
az

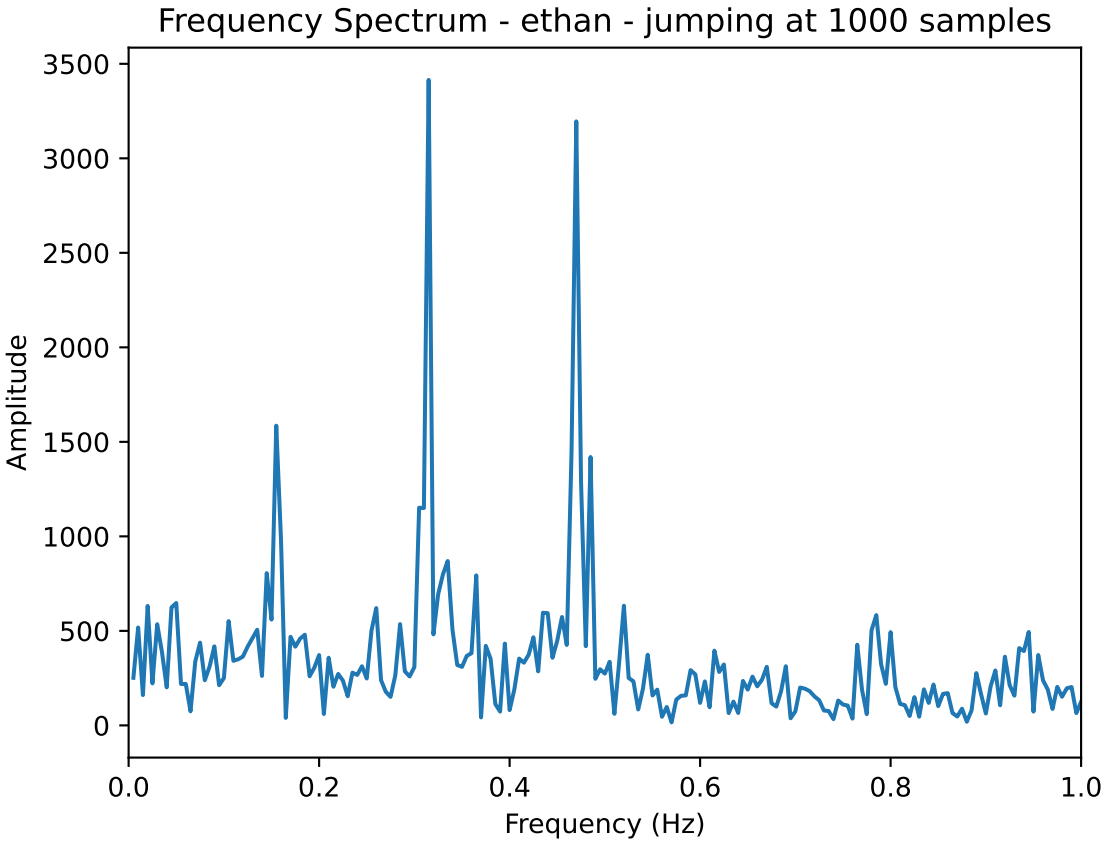

ethan - walking at 10000 samples
|a|
Acceleration Magnitude
Time (samples)
0
2
4
6
8
10
0
250
500
750
1000
1250
1500
1750
2000

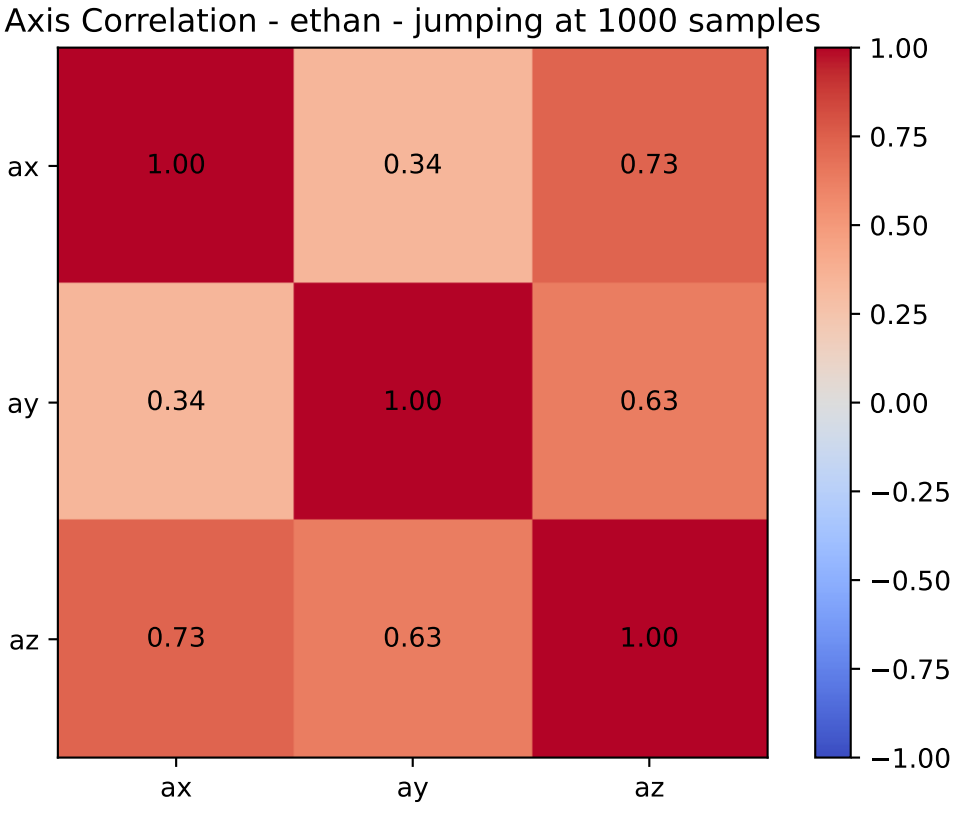

Frequency Spectrum - ethan - walking at 10000 samples
Amplitude
Frequency (Hz)
1000
800
600
400
200
0
0.0
0.2
0.4
0.6
0.8
1.0

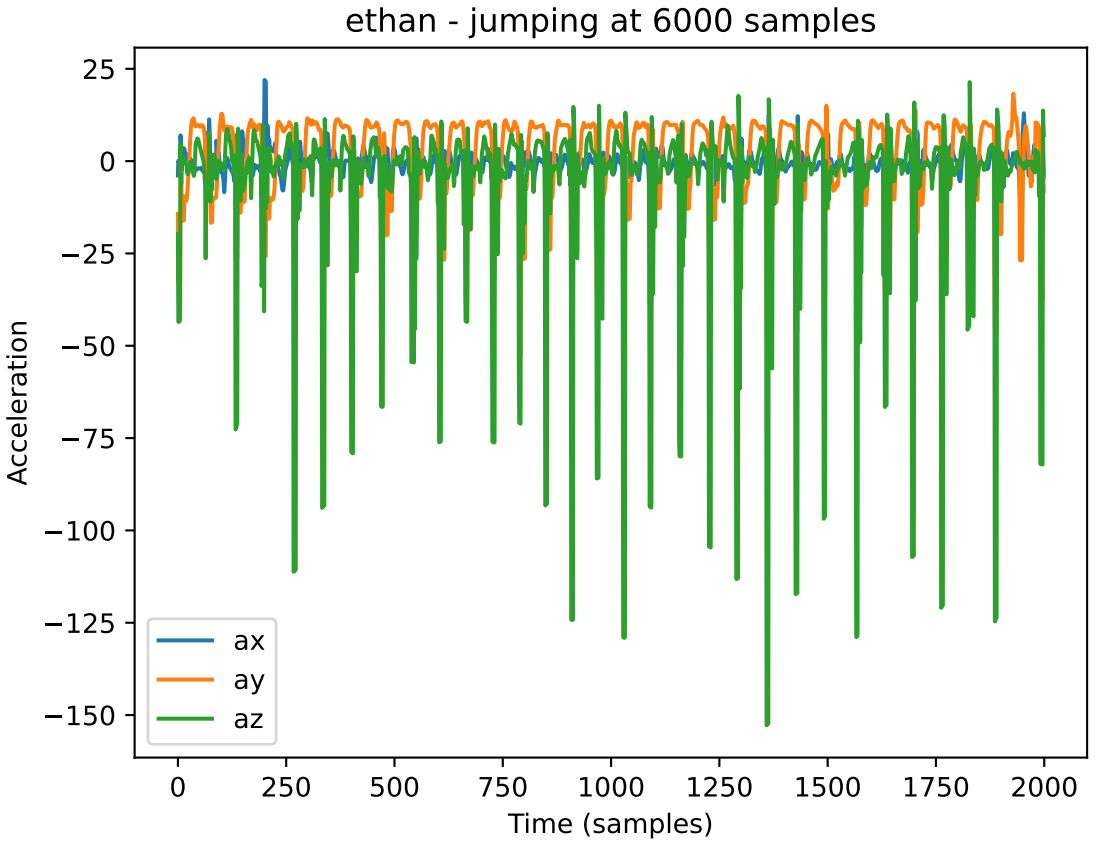

Axis Correlation - ethan - walking at 10000 samples
ax
ay
az
1.00
0.24
-0.06
0.24
1.00
0.00
-0.06
0.00
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

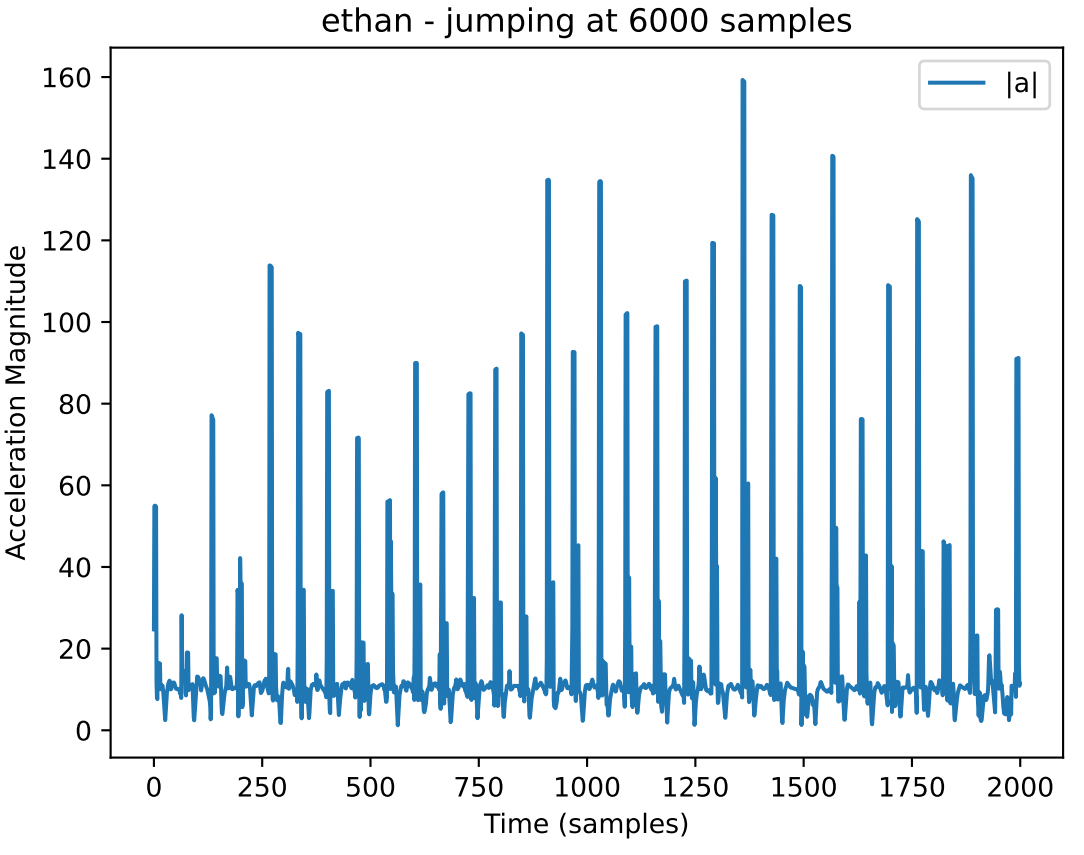

ethan - jumping at 1000 samples
Acceleration
Time (samples)
20
10
0
−10
−20
0
250
500
750
1000
1250
1500
1750
2000
ax
ay
az

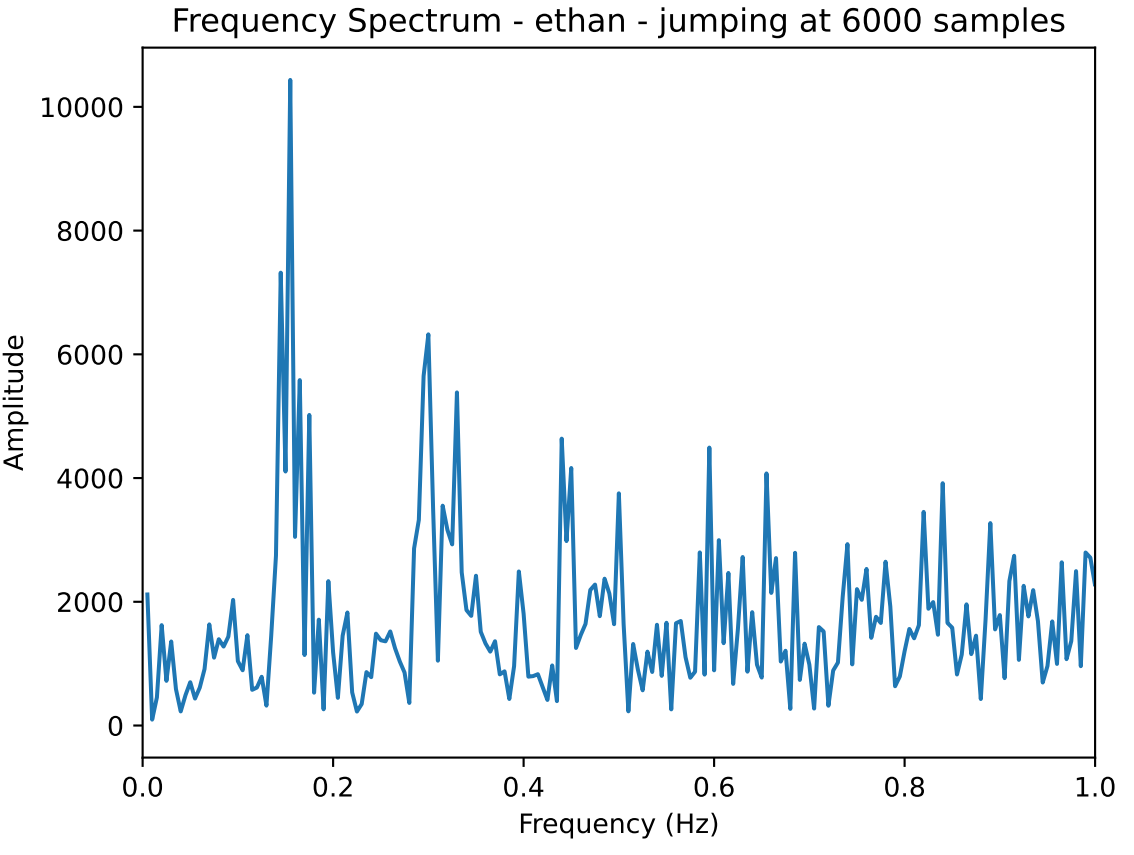

ethan - jumping at 1000 samples
|a|
Acceleration Magnitude
Time (samples)
0
5
10
15
20
25
0
250
500
750
1000
1250
1500
1750
2000

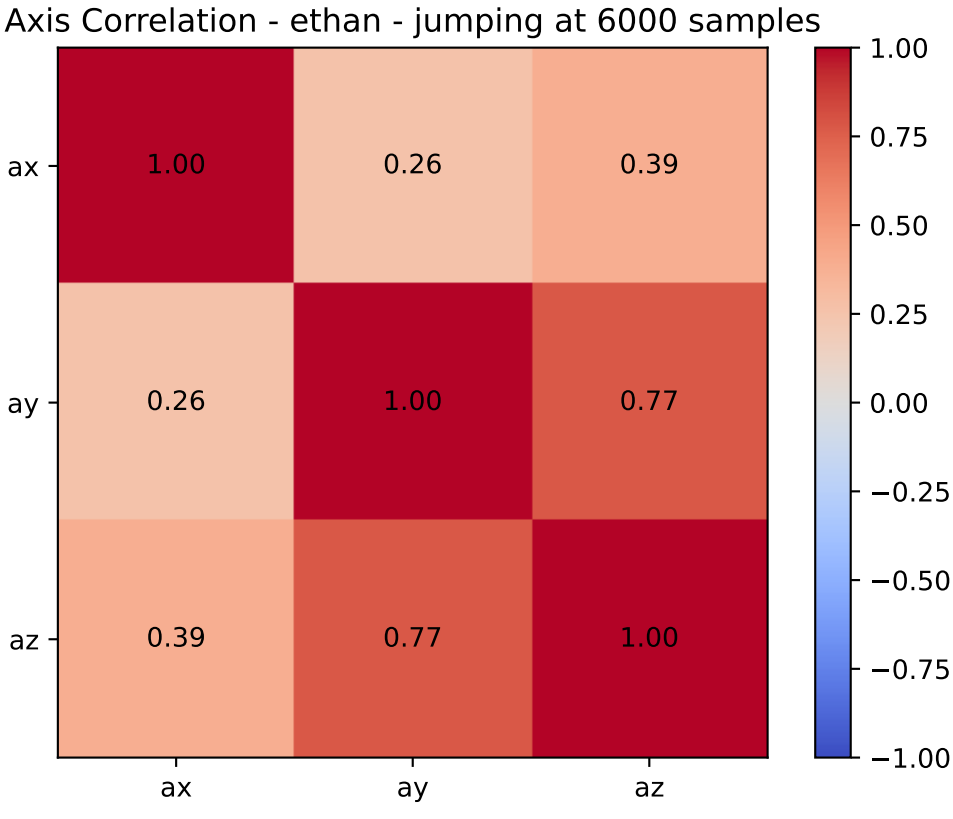

Frequency Spectrum - ethan - jumping at 1000 samples
Amplitude
Frequency (Hz)
3500
3000
2500
2000
1500
1000
500
0
0.0
0.2
0.4
0.6
0.8
1.0

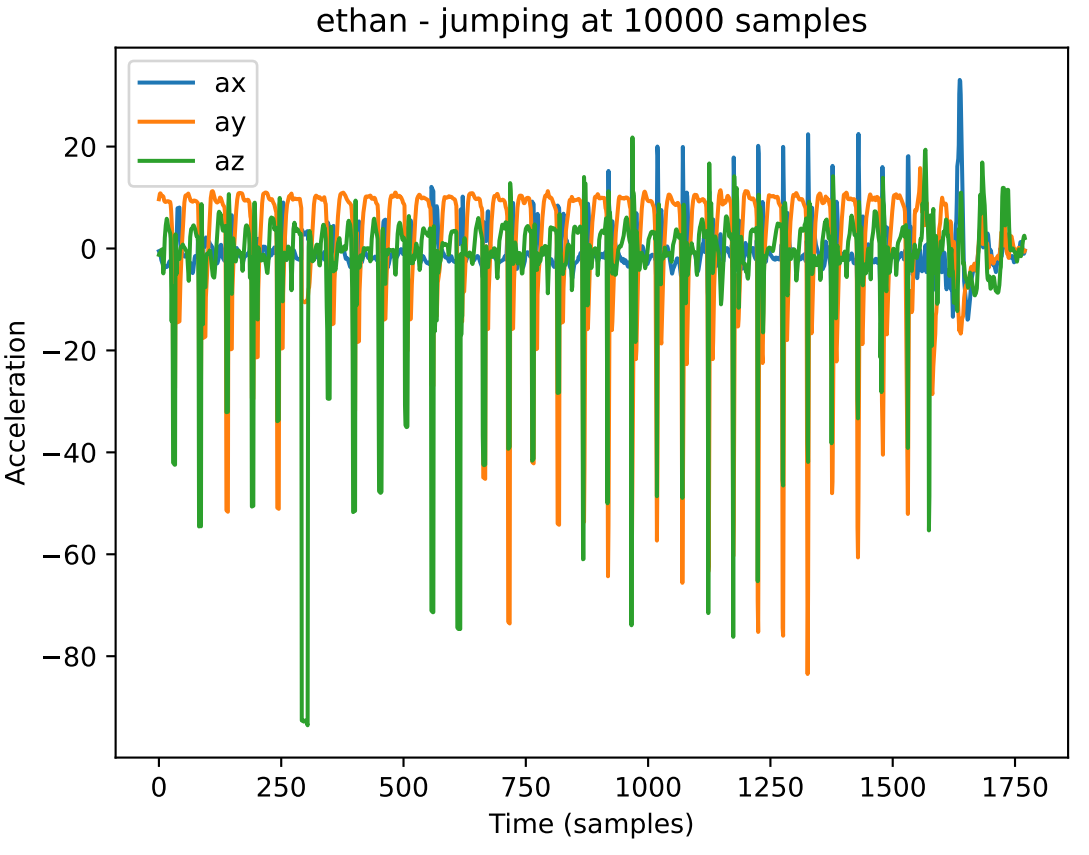

Axis Correlation - ethan - jumping at 1000 samples
ax
ay
az
1.00
0.34
0.73
0.34
1.00
0.63
0.73
0.63
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

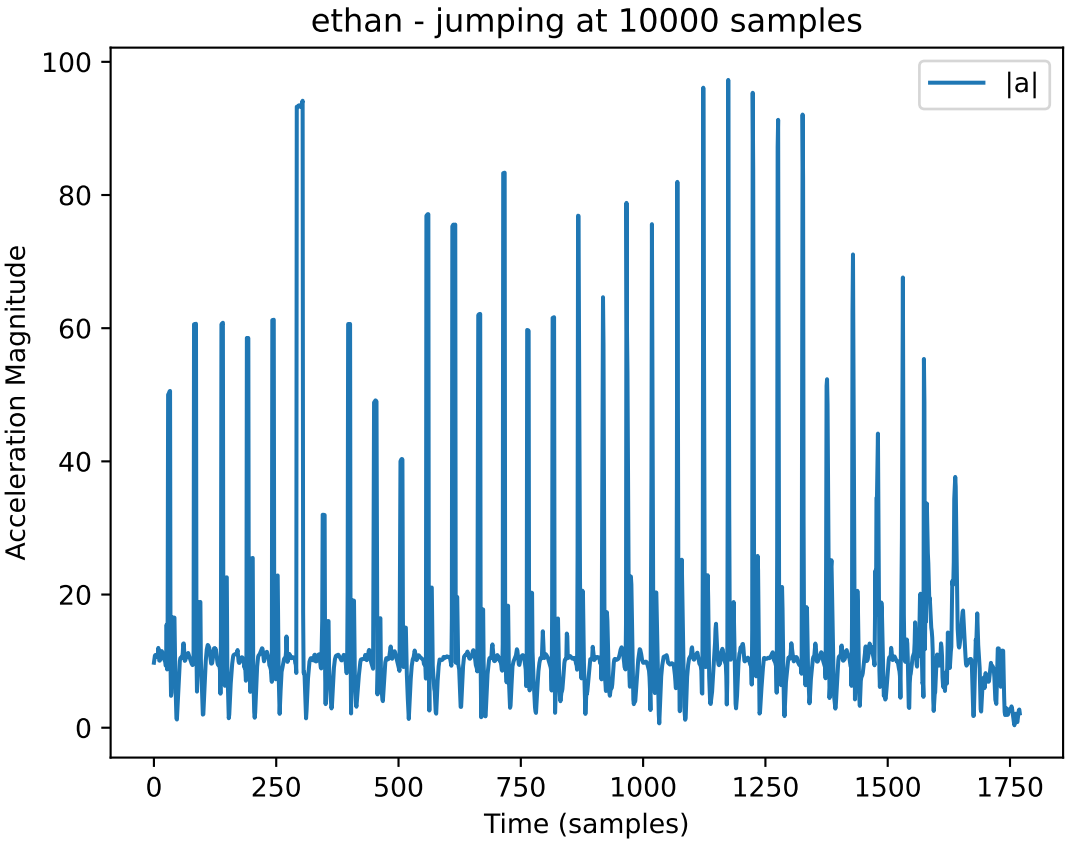

ethan - jumping at 6000 samples
Acceleration
Time (samples)
ax
ay
az
25
0
−25
−50
−75
−100
−125
−150
0
250
500
750
1000
1250
1500
1750
2000

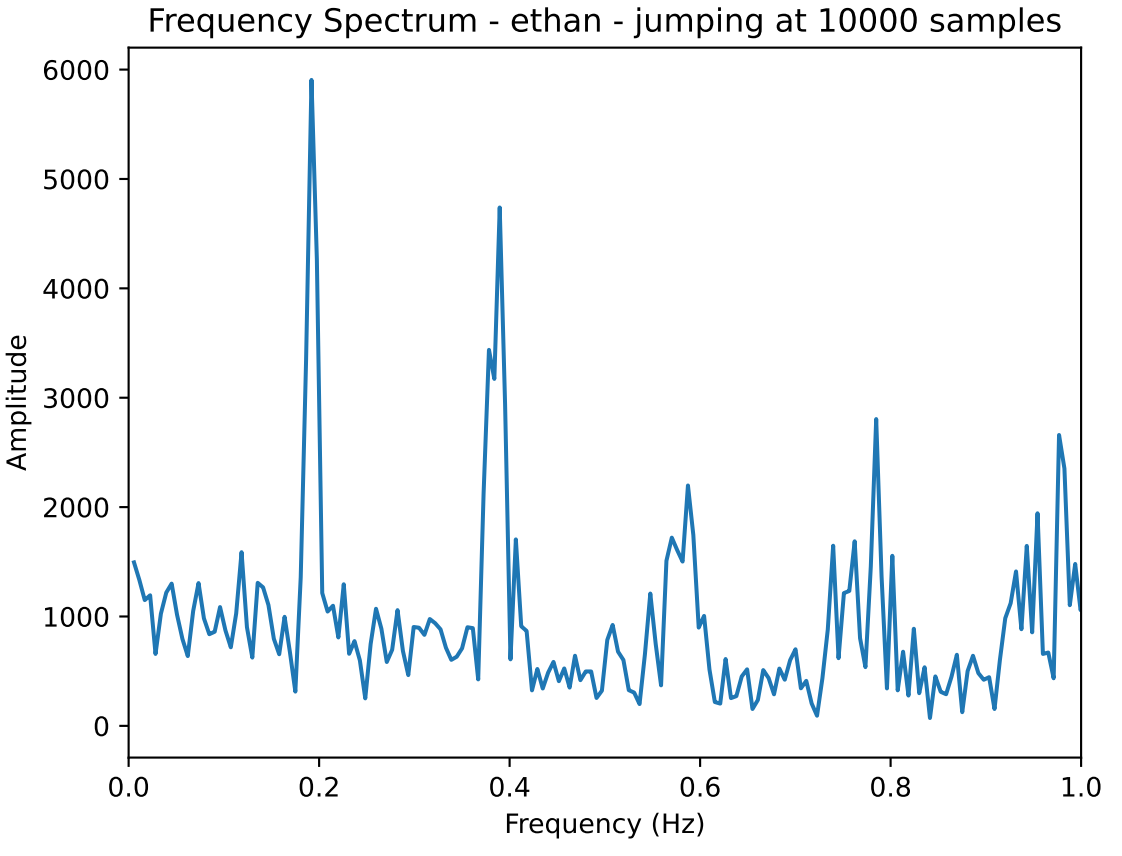

ethan - jumping at 6000 samples
|a|
Acceleration Magnitude
Time (samples)
0
20
40
60
80
100
120
140
160
0
250
500
750
1000
1250
1500
1750
2000

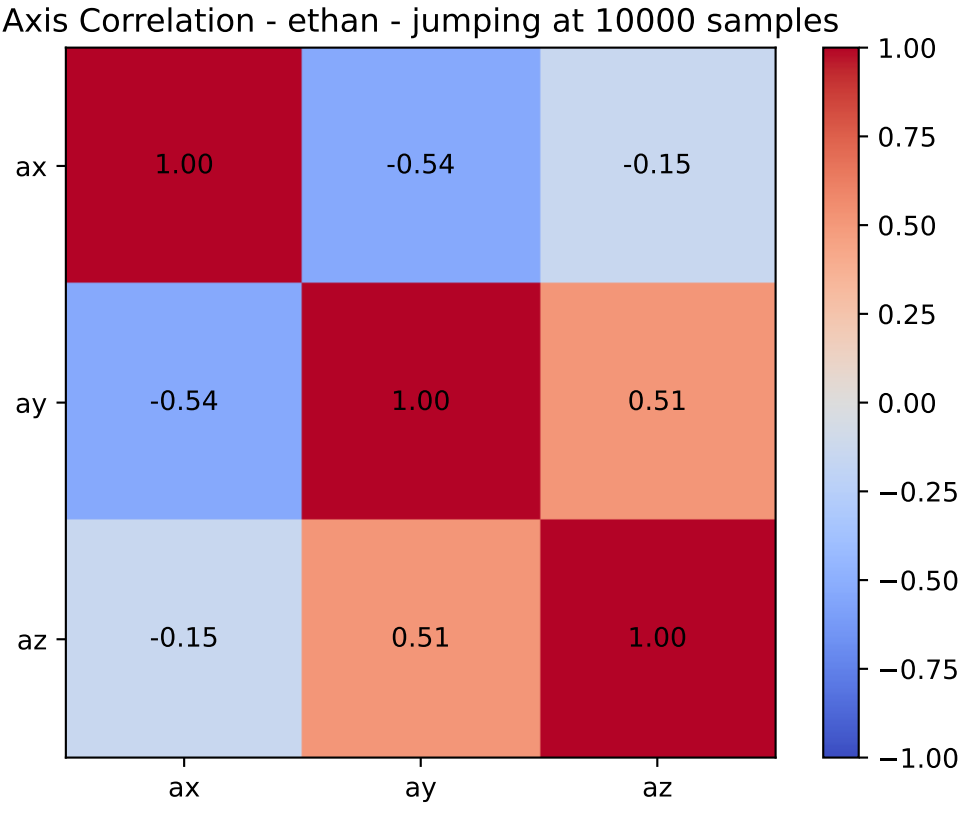

Frequency Spectrum - ethan - jumping at 6000 samples
Amplitude
10000
8000
6000
4000
2000
0
0.0
0.2
0.4
0.6
0.8
1.0
Frequency (Hz)


Axis Correlation - ethan - jumping at 6000 samples
ax
ay
az
1.00
0.26
0.39
0.26
1.00
0.77
0.39
0.77
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00


ethan - jumping at 10000 samples
ax
ay
az
20
0
−20
−40
−60
−80
Acceleration
0
250
500
750
1000
1250
1500
1750
Time (samples)


ethan - jumping at 10000 samples
|a|
Acceleration Magnitude
Time (samples)
0
20
40
60
80
100
0
250
500
750
1000
1250
1500
1750


Frequency Spectrum - ethan - jumping at 10000 samples
Amplitude
Frequency (Hz)
0
1000
2000
3000
4000
5000
6000
0.0
0.2
0.4
0.6
0.8
1.0


Axis Correlation - ethan - jumping at 10000 samples
ax
ay
az
1.00
-0.54
-0.15
-0.54
1.00
0.51
-0.15
0.51
1.00
ax
ay
az
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00